Document

Project: DevOps Implementation

Note: Contents have been deleted. This is not a full version of the document.

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Prepared by

Author

Jean-Philippe LO SIOU

Version: 1.0

DOCUMENT CONTROL

Date	Author	Ver	Comments
11/03/2025	Jean-Philippe LO SIOU	0.1	First Drafts
14/03/2025	Jean-Philippe LO SIOU	0.2	First Release
17/03/2025	Jean-Philippe LO SIOU	0.3	Various Minor Updates (Formatting)
18/03/2025	Jean-Philippe LO SIOU	0.4	Adding challenges
29/03/2025	Jean-Philippe LO SIOU	0.5	Adding Challenges
30/03/2025	Jean-Philippe LO SIOU	0.6	Various Updates
02/04/2025	Jean-Philippe LO SIOU	0.7-1.0	Final Release

Contents

OBJECTIVE

This project consists of hands-on experience in DevOps practices, implementing automation, and creating a fully integrated CI/CD pipeline. By the end, this project is a fully automated system where changes are automatically tested, deployed, and maintained.

Expected Deliverables

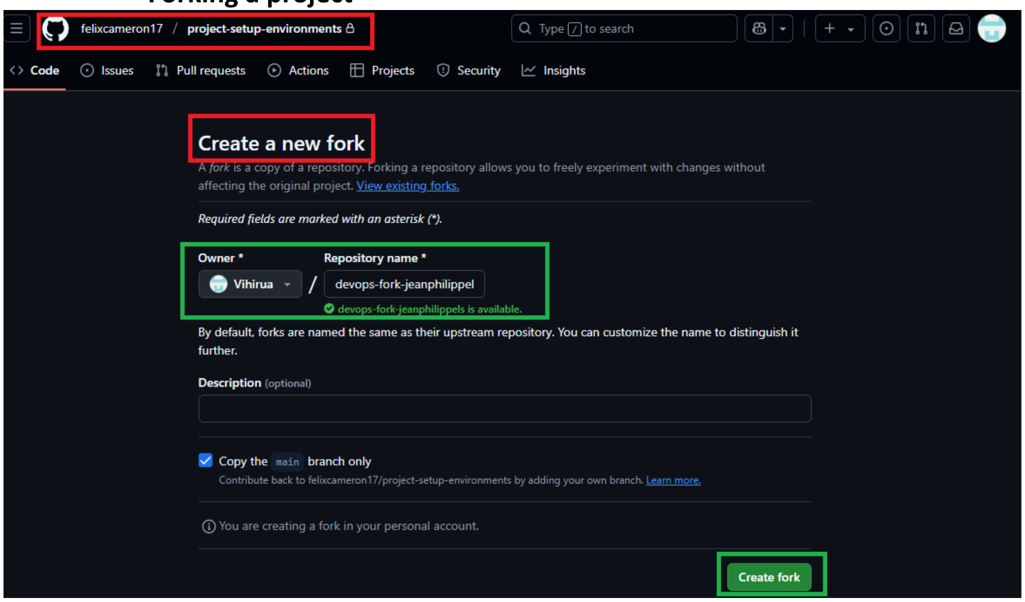
1. A **GitHub repository** containing the forked project with all DevOps implementations.
2. A **fully automated CI/CD pipeline**.
3. An **Infrastructure as Code (IaC) setup**.
4. A **containerized application deployed to the cloud**.
5. A **detailed report explaining the execution, challenges faced, and learnings**.

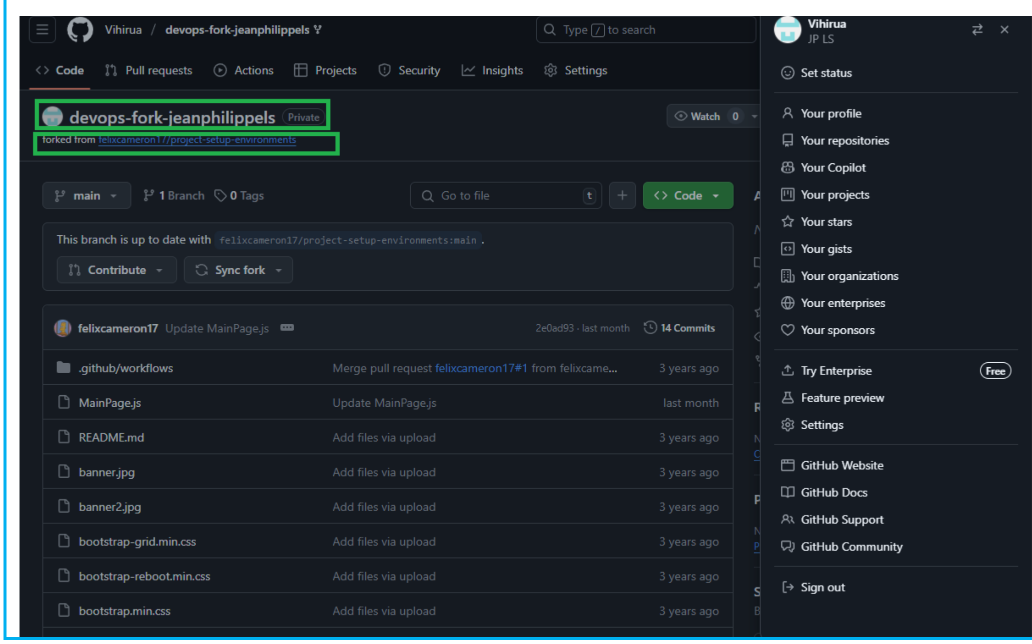
SELECTION & FORKING A REPOSITORY

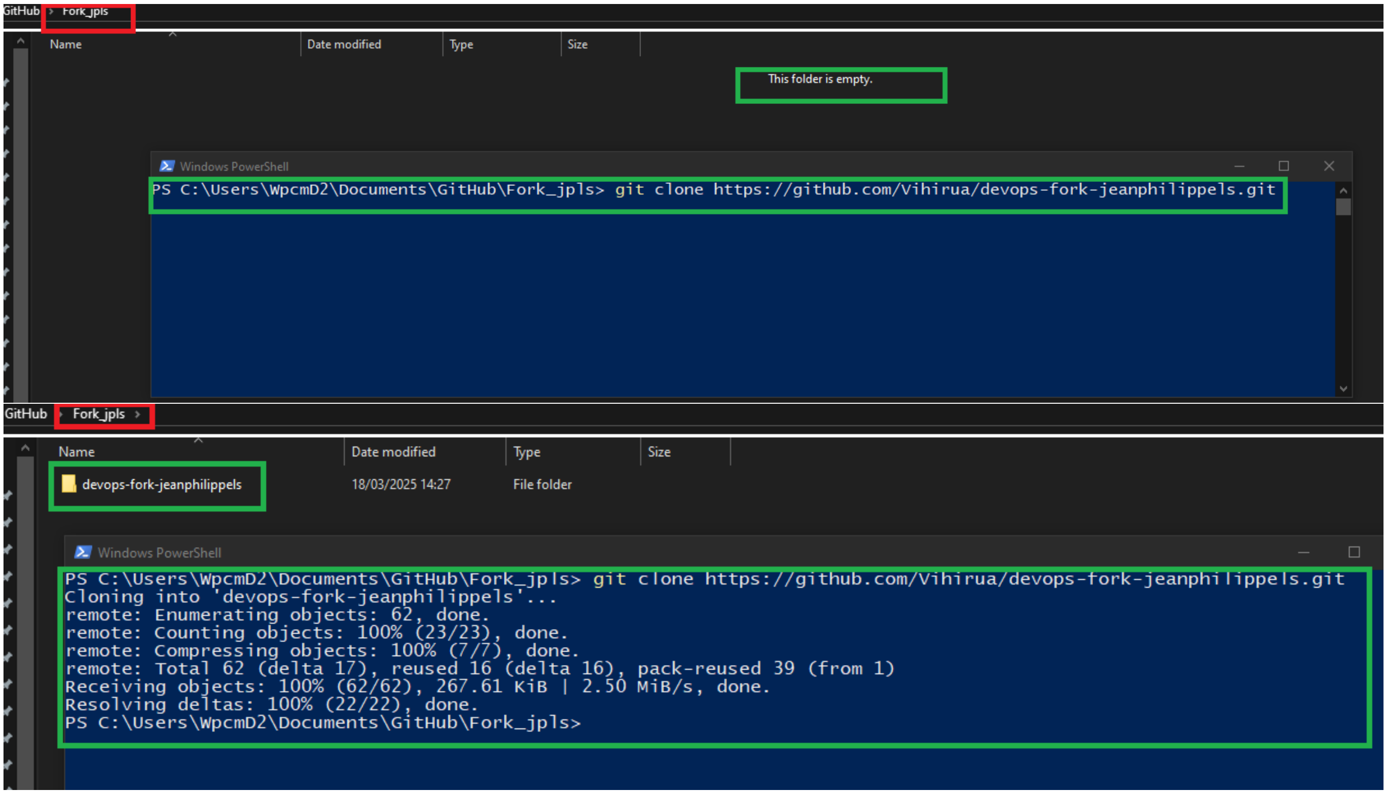
Git & Version Control

- **Forking a project**
- **Create a new branch for DevOps changes**
- **Practice Git Workflow:**
  - **Make changes and commit**
  - **Push changes to GitHub.**
  - **Create a Pull Request (PR) and merge changes.**

- **Forking a project**

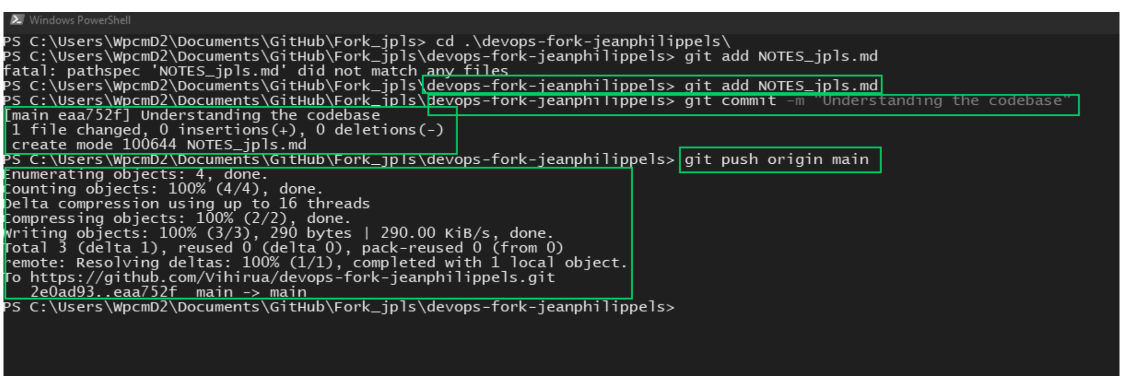






- `Git clone`

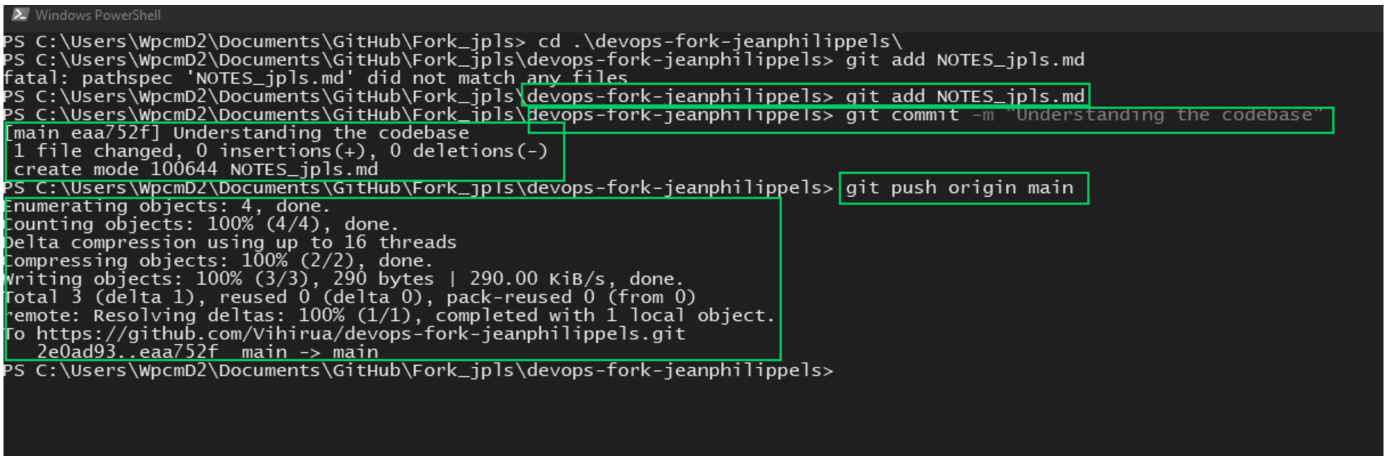
- **Create a new branch for DevOps changes**



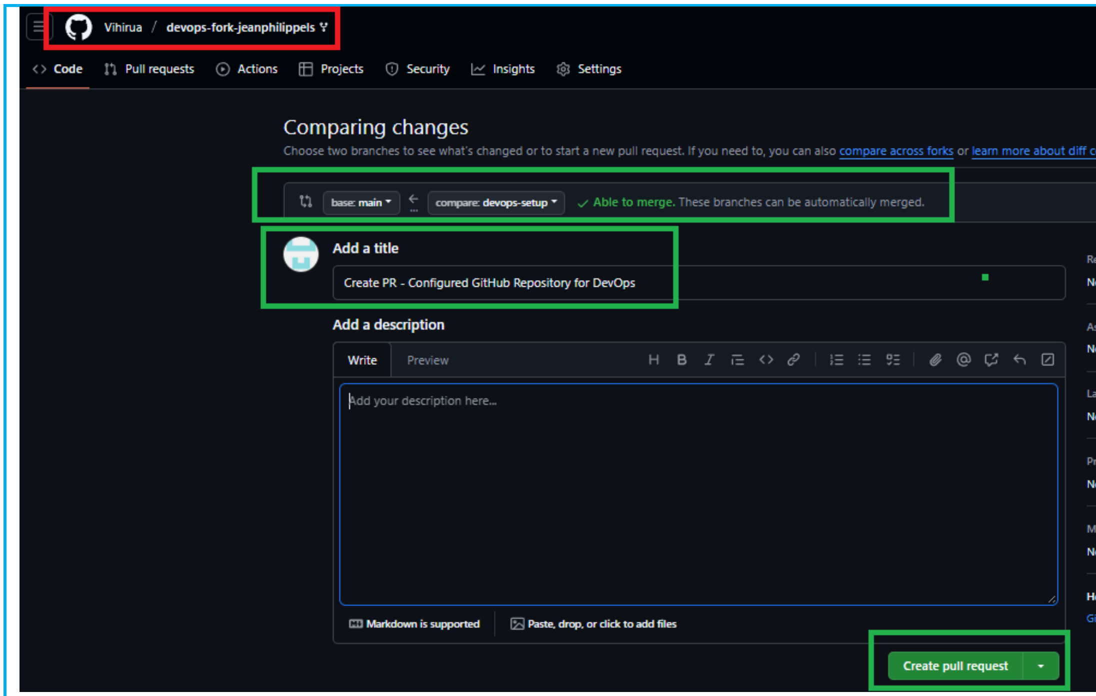
- `Git add NOTES_jpls.md`
- `Git commit -m “Understanding the codebase”`

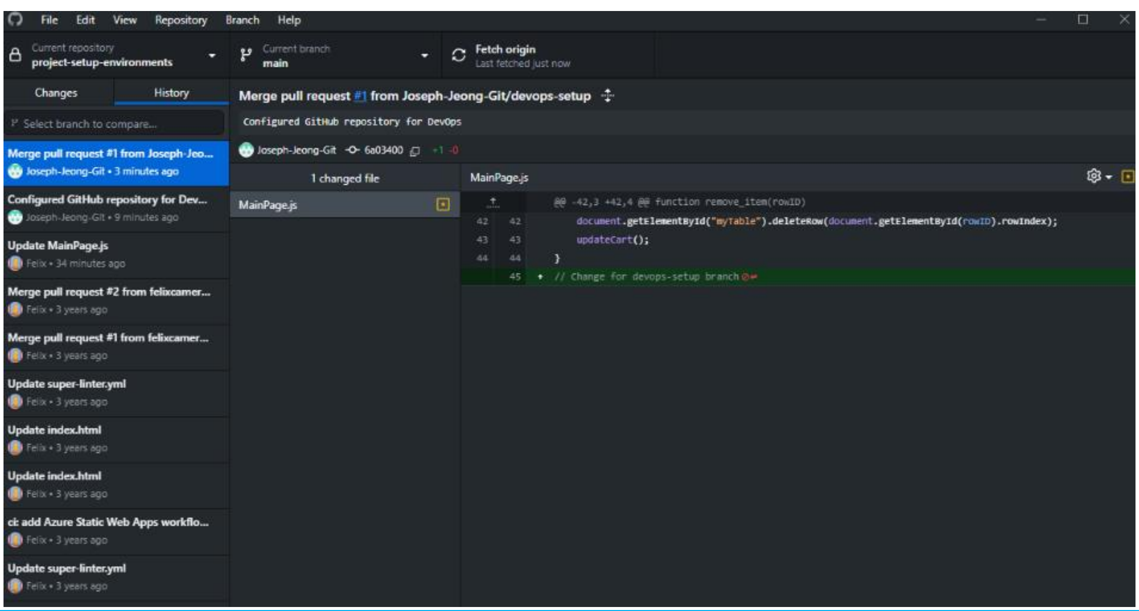
- `Git push origin main`

  - **Push changes to GitHub.**



  - **Create a Pull Request (PR) and merge changes.**


Vihirua / devops-fork-jeanphilippels
Code Pull requests Actions Projects Security Insights Settings
Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff co
base: main compare: devops-setup Able to merge. These branches can be automatically merged.
Add a title
Create PR - Configured GitHub Repository for DevOps
Add a description
Write Preview
Add your description here...
Markdown is supported
Paste, drop, or click to add files
Create pull request


File Edit View Repository Branch Help
Current repository project-setup-environments
Current branch main
Fetch origin Last fetched just now
Changes History
Merge pull request #1 from Joseph-Jeong-Git/devops-setup
Configured GitHub repository for DevOps
Select branch to compare...
1 changed file
MainPage.js
Merge pull request #1 from Joseph-Jeo...
Configured GitHub repository for Dev...
Update MainPage.js
Merge pull request #2 from felixcamer...
Merge pull request #1 from felixcamer...
Update super-linter.yml
Update index.html
Update index.html
ci: add Azure Static Web Apps workflo...
Update super-linter.yml

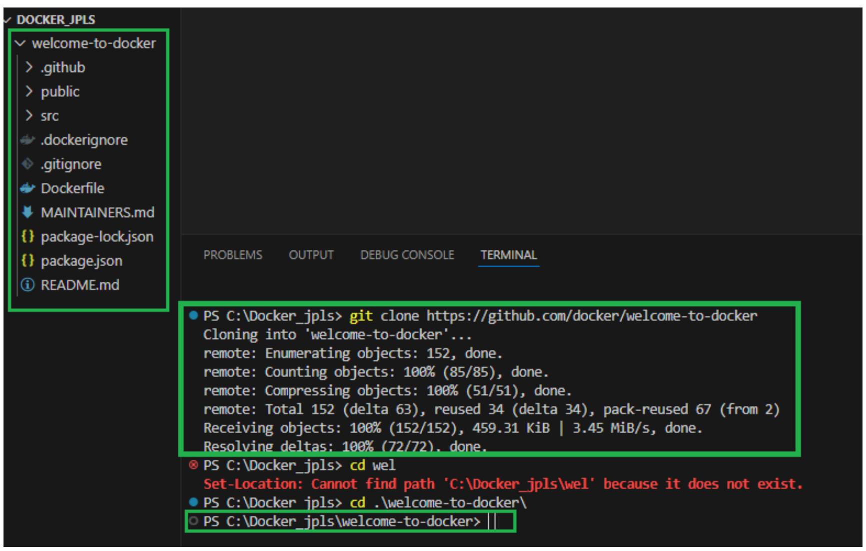
CONTAINERIZATION WITH DOCKER & KUBERNETES

Deployment in Local Docker Container

- **Run a Docker container**
- **Push Docker Image to Docker Hub.**
- **Building DevOps image with Dockerfile**
- **Building container with the DevOps image**
- **Testing container with the DevOps image**
- **Get into the DevOps docker container**
- **Pushing DevOps image to private docker repository**

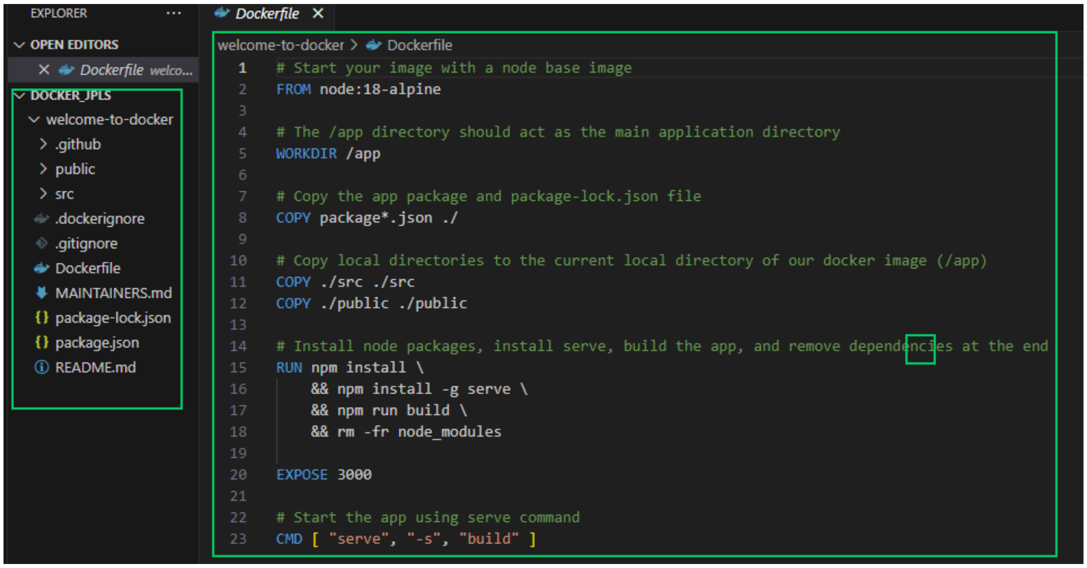
Docker is a software allowing to build, test and deploy applications by using containers. It packages everything in containers that the applications need to run with (Libraries, systems tools, code). It can quickly deploy and scale applications into any environments (any operating Systems)

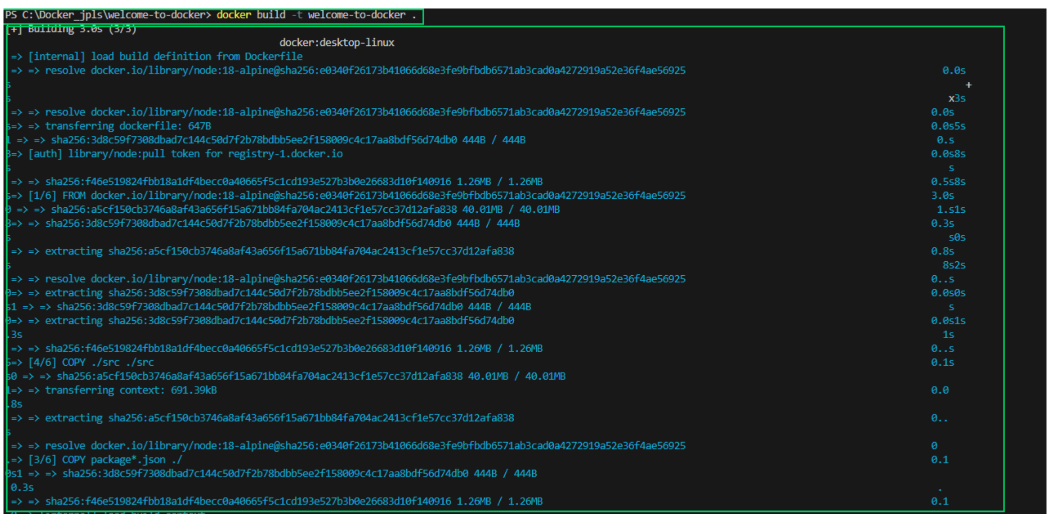
- **Run a Docker container**

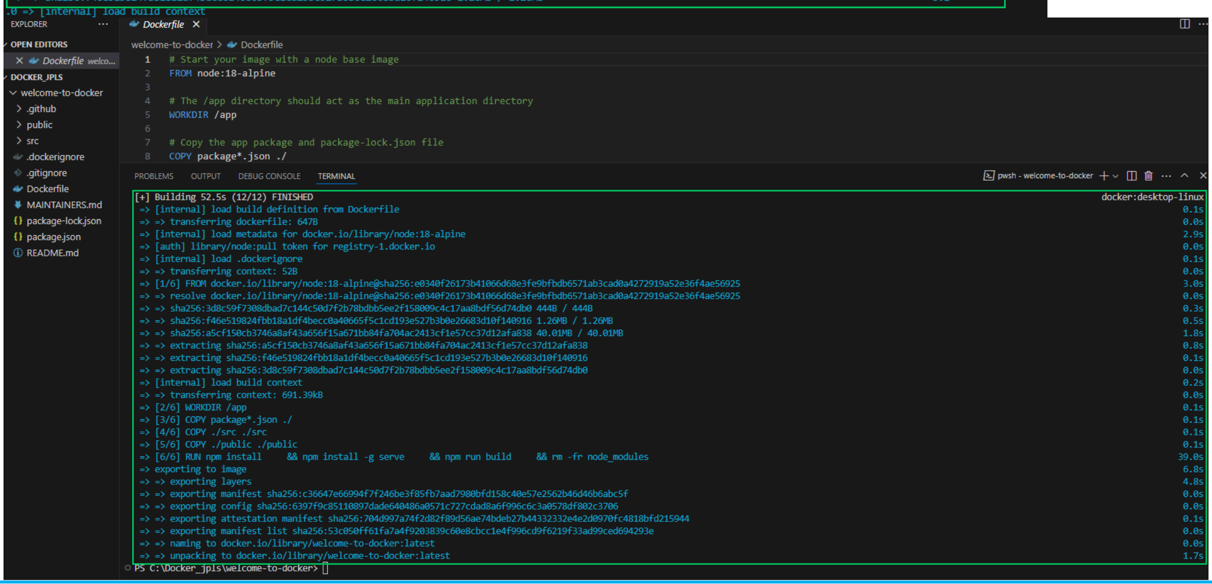


- `Git clone`

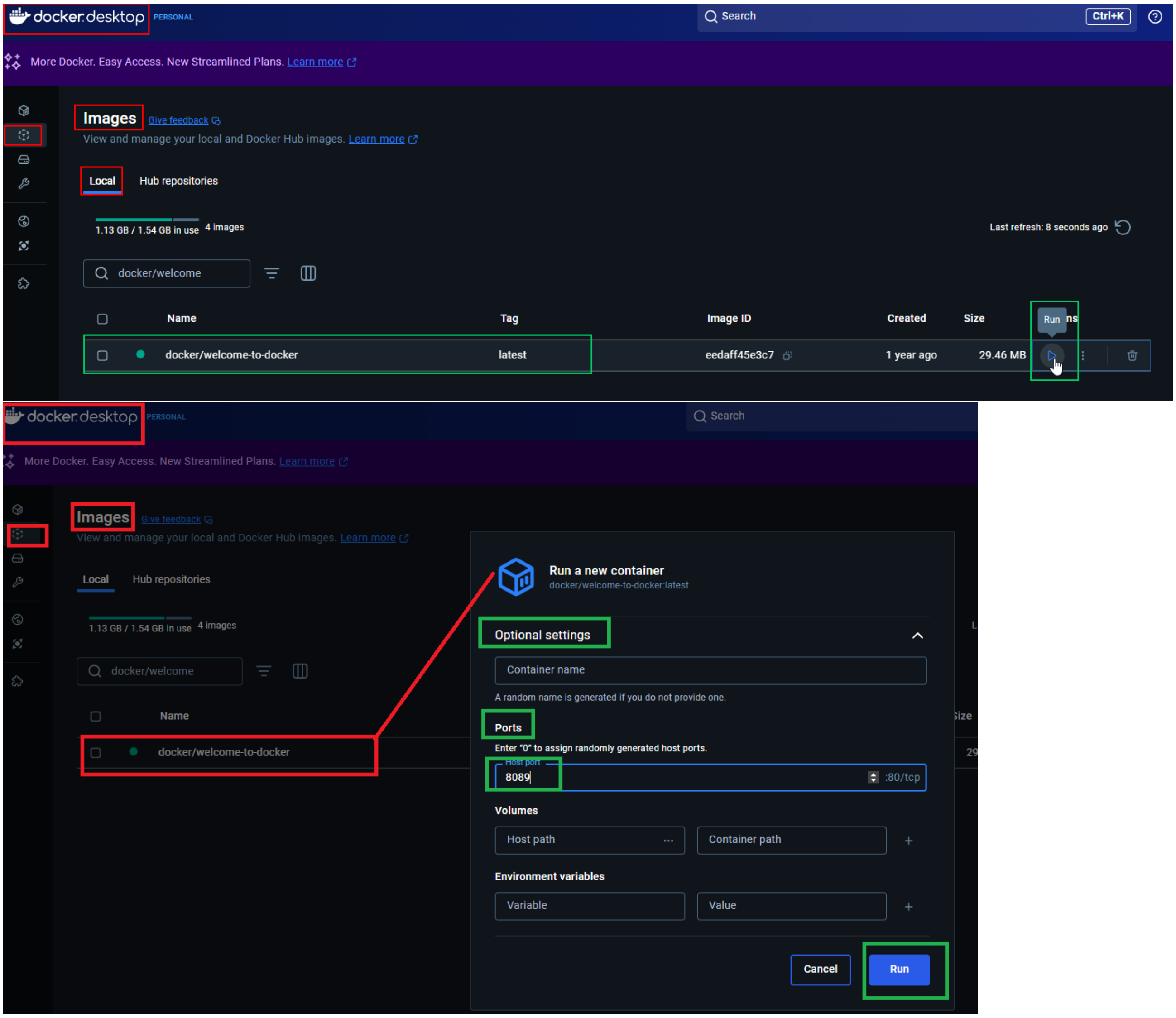
The Dockerfile :

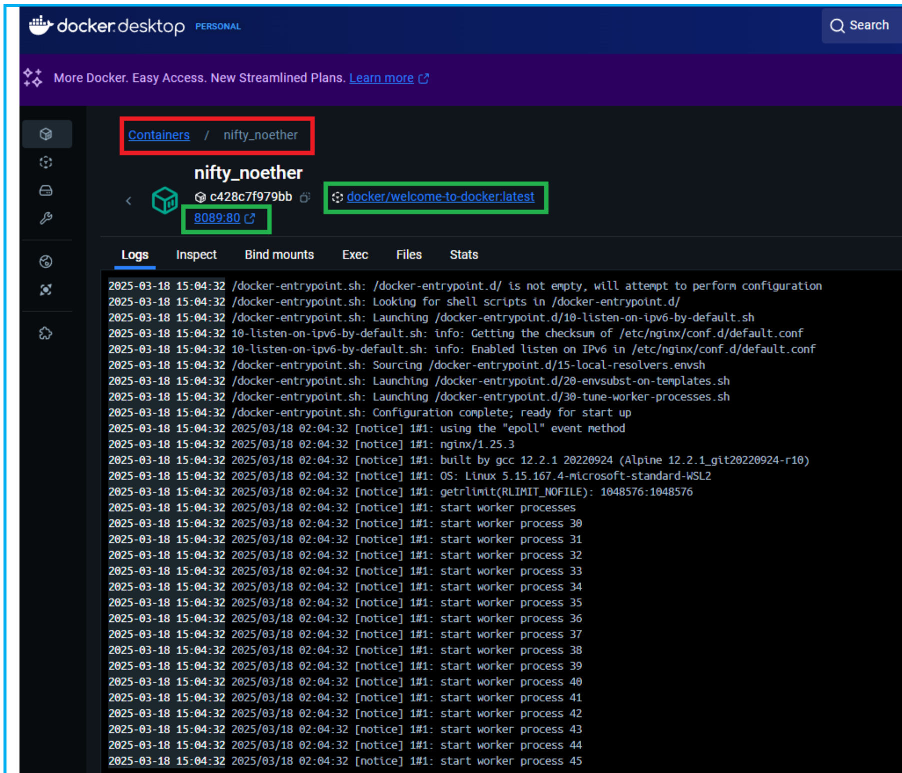


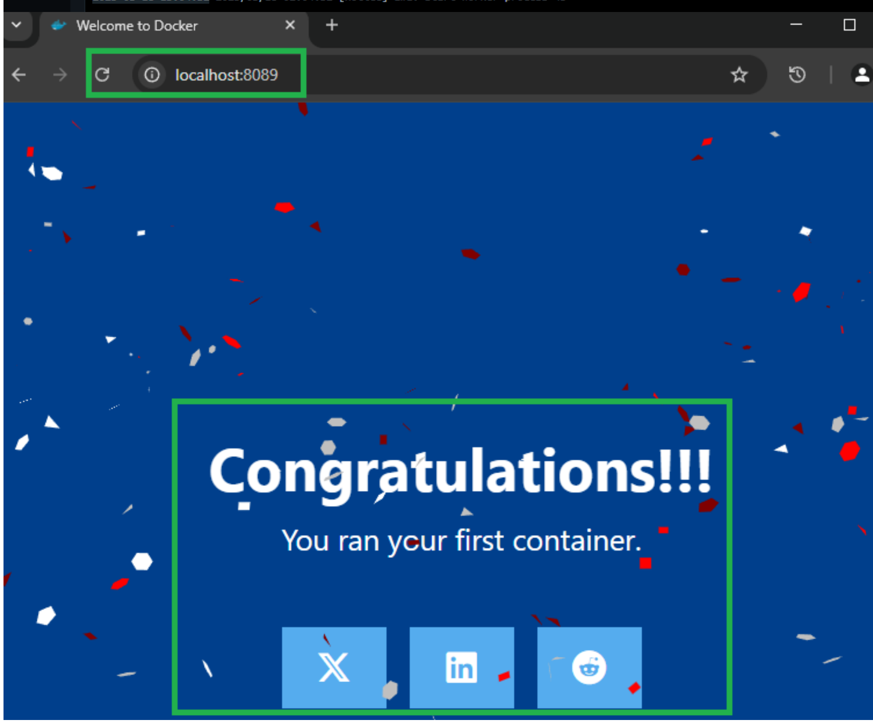




```
•   Docker build -t welcome-to-docker .
```







- **Push Docker Image to Docker Hub.**

```
PS C:\Docker_jpls\welcome-to-docker> docker images --filter=reference=welcome-to-docker
REPOSITORY          TAG       IMAGE ID       CREATED          SIZE
welcome-to-docker   latest    53c050ff61fa   23 minutes ago   383MB
PS C:\Docker_jpls\welcome-to-docker> docker tag welcome-to-docker dockerlspf123/welcome-to-docker-jpls:latest
```

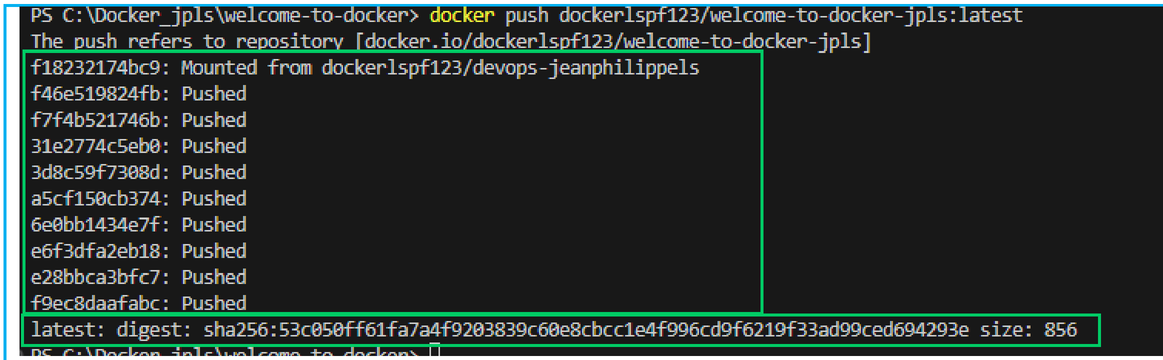
- Docker images
- Docker tag welcome-to-docker dockerlspf123/welcome-to-docker-jpls:latest

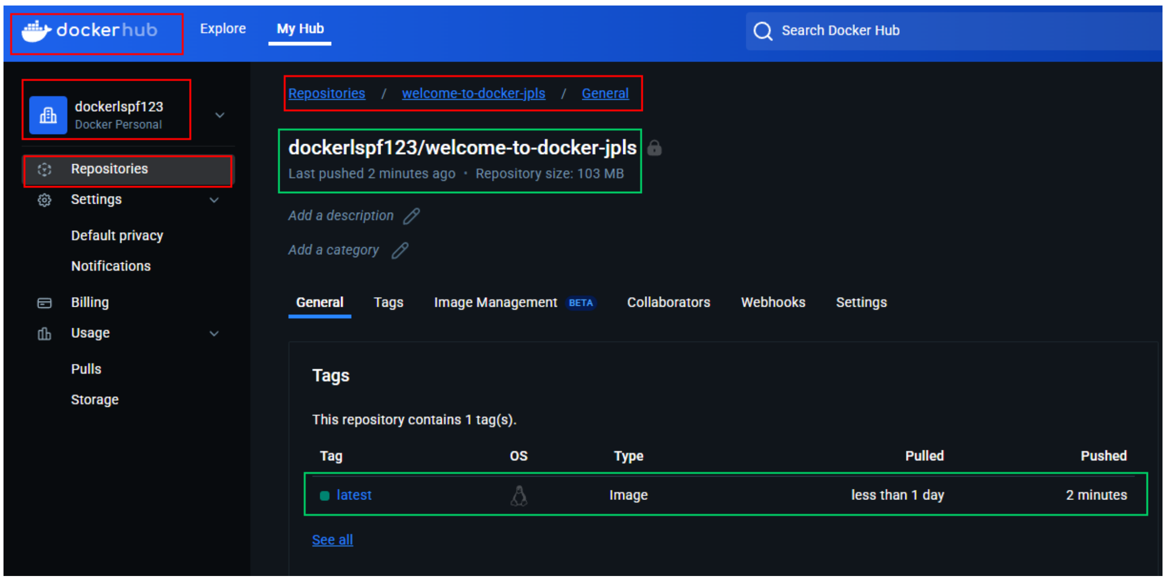
```
PS C:\Docker_jpls\welcome-to-docker> docker images --filter=reference=dockerlspf123/welcome-to-docker-jpls
REPOSITORY                             TAG       IMAGE ID       CREATED          SIZE
dockerlspf123/welcome-to-docker-jpls   latest    53c050ff61fa   24 minutes ago   383MB
PS C:\Docker_jpls\welcome-to-docker>
```

```
PS C:\Docker_jpls\welcome-to-docker> docker login
Authenticating with existing credentials...
Login Succeeded
```

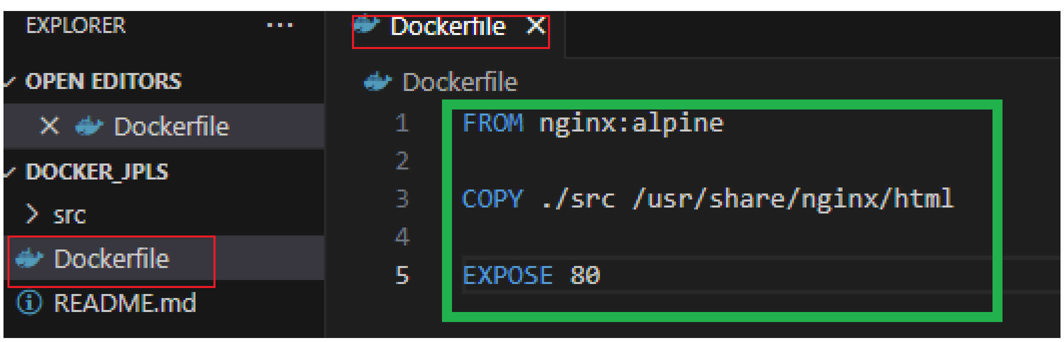
```
PS C:\Docker_jpls\welcome-to-docker> docker push dockerlspf123/welcome-to-docker-jpls:latest
The push refers to repository [docker.io/dockerlspf123/welcome-to-docker-jpls]
f18232174bc9: Mounted from dockerlspf123/devops-jeanphilippels
f46e519824fb: Pushing [==================================================>]  1.261MB/1.261MB
f7f4b521746b: Pushed
31e2774c5eb0: Pushing [===>                                               ]  4.194MB/62.88MB
3d8c59f7308d: Pushed
a5cf150cb374: Pushing [=======>                                           ]  6.291MB/40.01MB
6e0bb1434e7f: Pushed
e6f3dfa2eb18: Pushed
e28bbca3bfc7: Pushed
f9ec8daafabc: Pushed
```

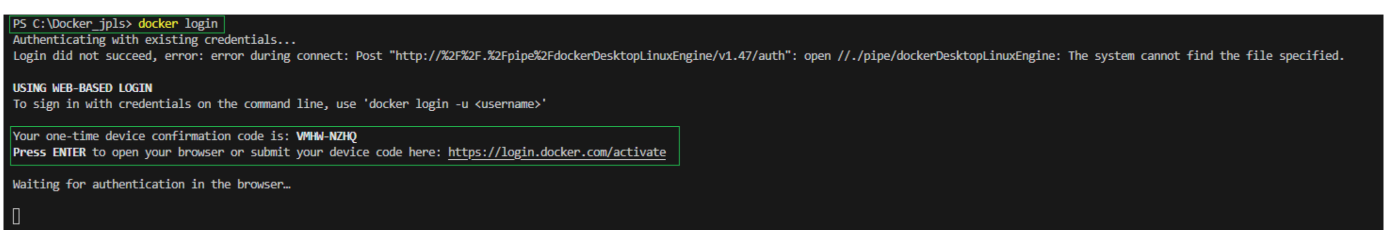
- Docker push dockerlspf123/welcome-to-docker-jpls:latest

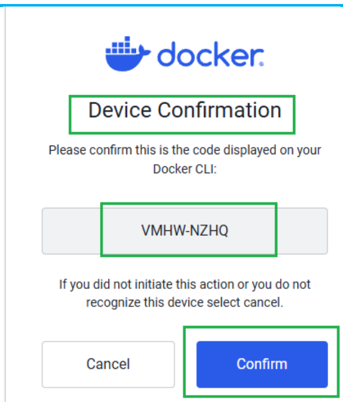


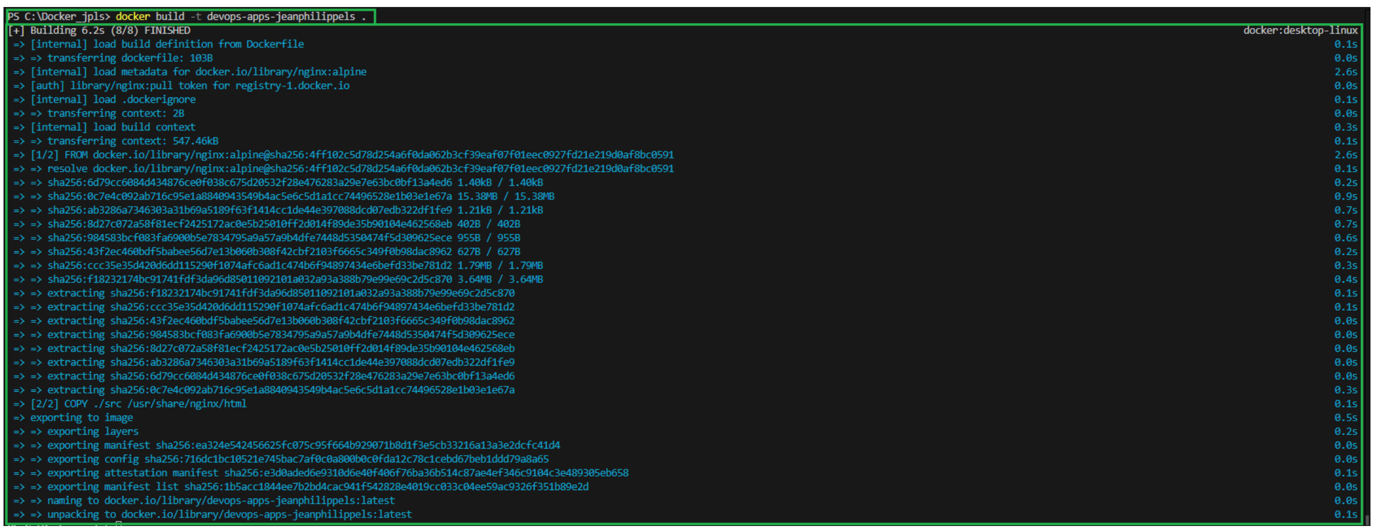


- **Building DevOps image with Dockerfile**

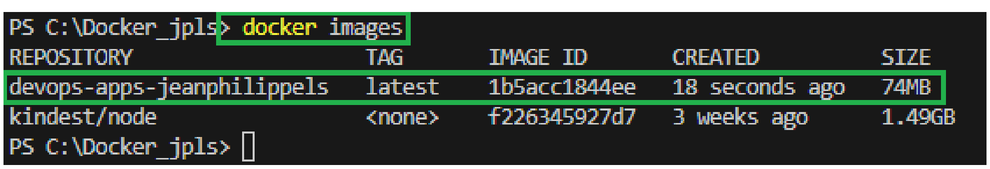






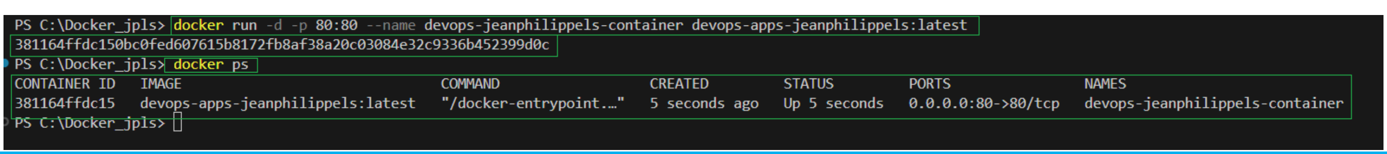



```
docker build -t devops-apps-jeanphilippels .
```



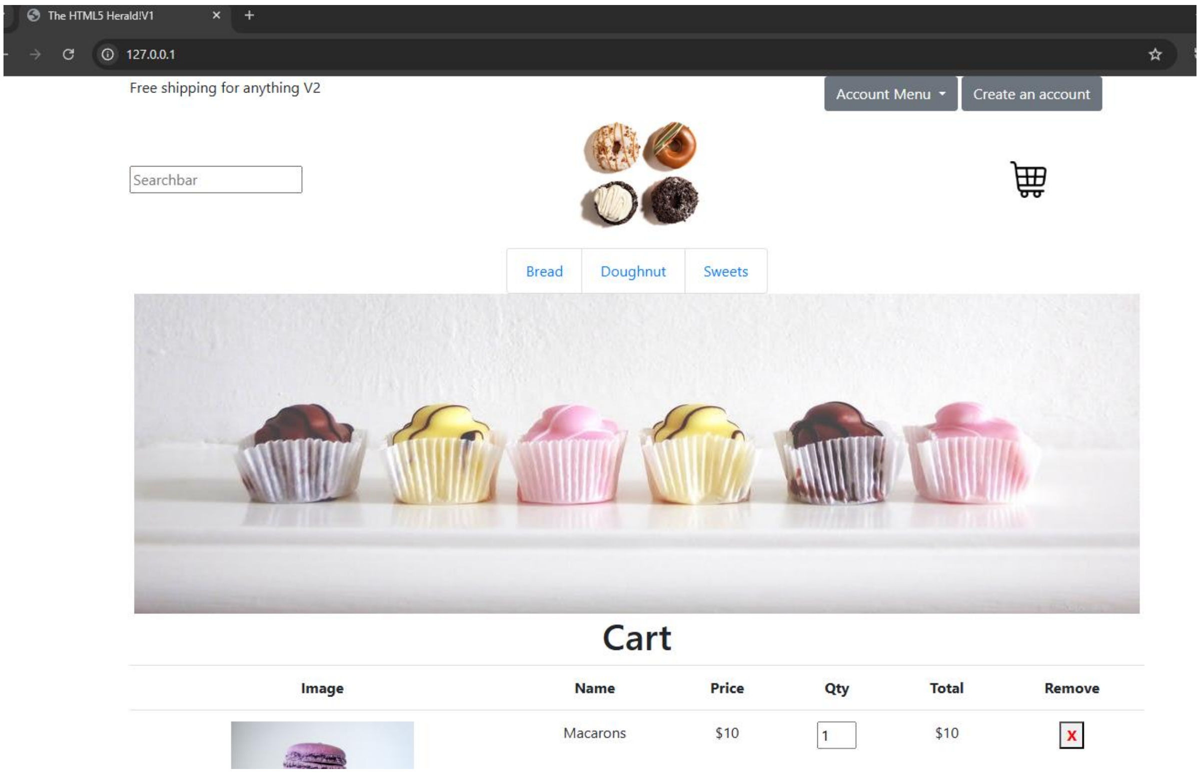

```
docker images
```

- **Building container with the DevOps image**

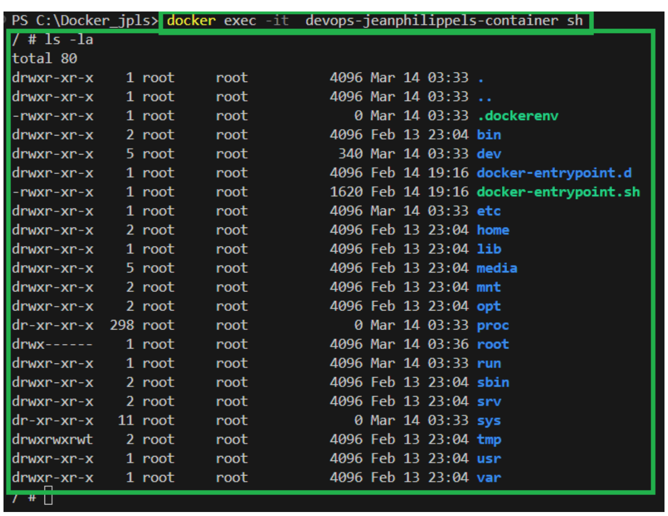


```
•   docker run -d -p 80:80 --name devops-jeanphilippels-container devops-apps
-jeanphilippels:latest

•  docker ps
```

- **Testing container with the DevOps image**

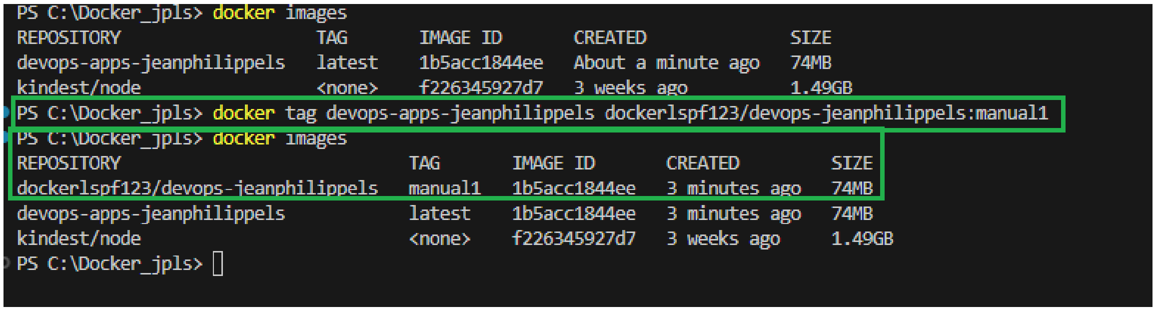


- **Get into the DevOps docker container**



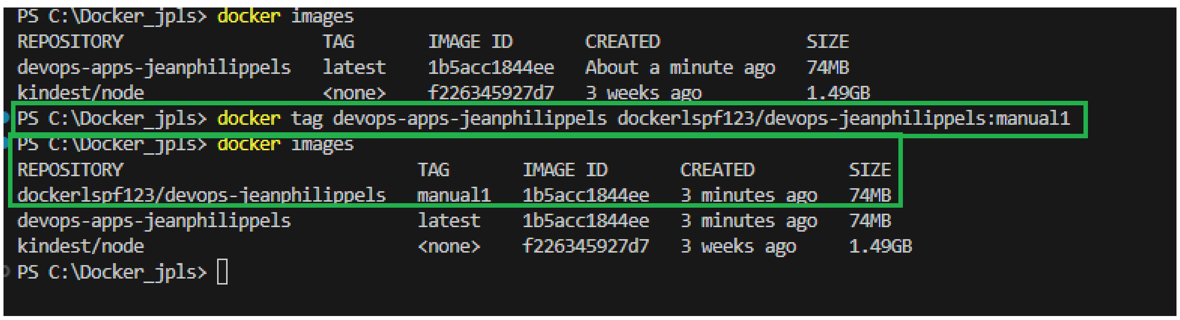
- `docker exec -it  devops-apps-jeanphilippels:latest sh`

- **Pushing DevOps image to private docker repository**



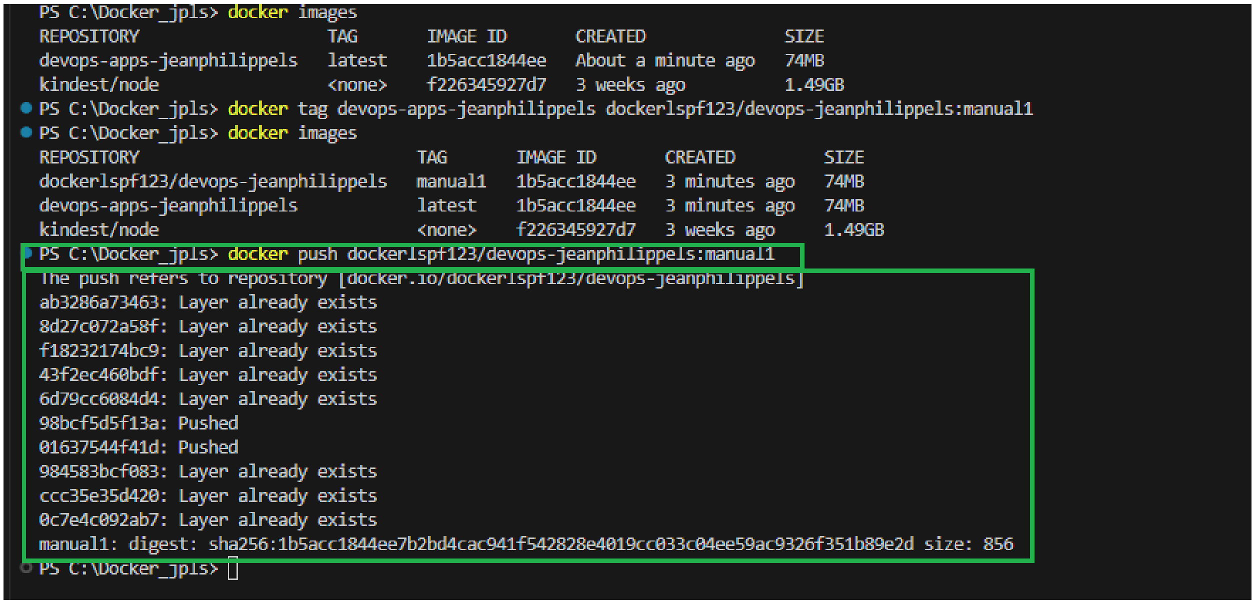
```
PS C:\Docker_jpls> docker images
REPOSITORY                   TAG       IMAGE ID       CREATED          SIZE
devops-apps-jeanphilippels   latest    1b5acc1844ee   About a minute ago   74MB
kindest/node                 <none>    f226345927d7   3 weeks ago      1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY                            TAG       IMAGE ID       CREATED         SIZE
dockerlspf123/devops-jeanphilippels   manual1   1b5acc1844ee   3 minutes ago   74MB
devops-apps-jeanphilippels            latest    1b5acc1844ee   3 minutes ago   74MB
kindest/node                          <none>    f226345927d7   3 weeks ago     1.49GB
PS C:\Docker_jpls>
```

- `docker images`



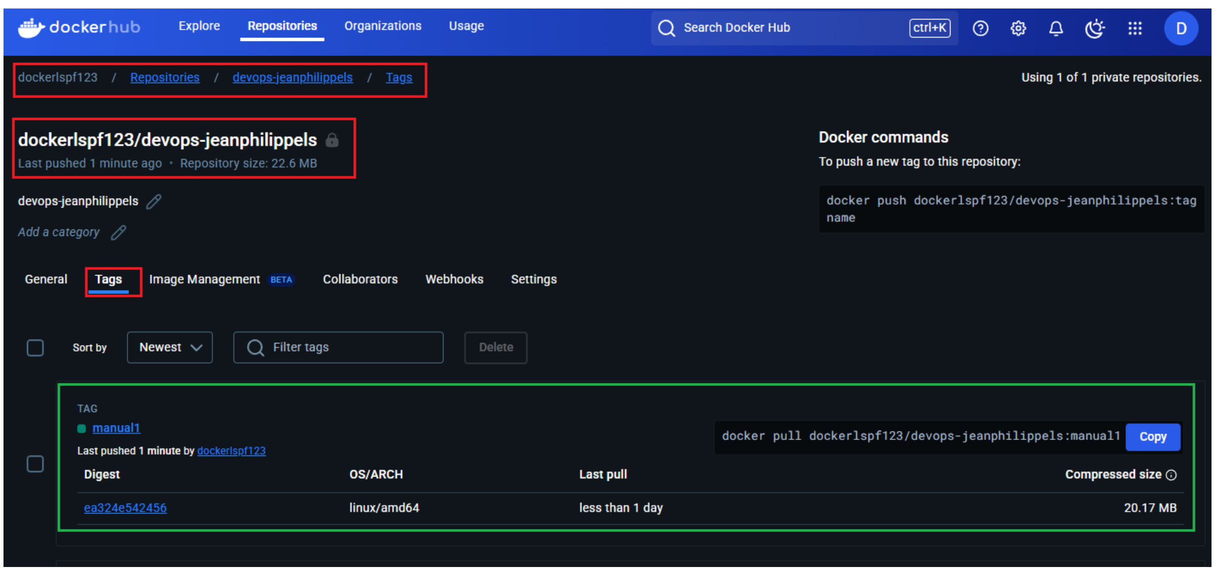
```
PS C:\Docker_jpls> docker images
REPOSITORY                   TAG       IMAGE ID       CREATED          SIZE
devops-apps-jeanphilippels   latest    1b5acc1844ee   About a minute ago   74MB
kindest/node                 <none>    f226345927d7   3 weeks ago      1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY                            TAG       IMAGE ID       CREATED         SIZE
dockerlspf123/devops-jeanphilippels   manual1   1b5acc1844ee   3 minutes ago   74MB
devops-apps-jeanphilippels            latest    1b5acc1844ee   3 minutes ago   74MB
kindest/node                          <none>    f226345927d7   3 weeks ago     1.49GB
PS C:\Docker_jpls>
```

- `docker tags devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1`
- `Docker images`



```
PS C:\Docker_jpls> docker images
REPOSITORY                TAG       IMAGE ID       CREATED              SIZE
devops-apps-jeanphilippels latest   1b5acc1844ee   About a minute ago   74MB
kindest/node              <none>    f226345927d7   3 weeks ago          1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY                             TAG       IMAGE ID       CREATED         SIZE
dockerlspf123/devops-jeanphilippels    manual1   1b5acc1844ee   3 minutes ago   74MB
devops-apps-jeanphilippels             latest    1b5acc1844ee   3 minutes ago   74MB
kindest/node                           <none>    f226345927d7   3 weeks ago     1.49GB
PS C:\Docker_jpls> docker push dockerlspf123/devops-jeanphilippels:manual1
The push refers to repository [docker.io/dockerlspf123/devops-jeanphilippels]
ab3286a73463: Layer already exists
8d27c072a58f: Layer already exists
f18232174bc9: Layer already exists
43f2ec460bdf: Layer already exists
6d79cc6084d4: Layer already exists
98bcf5d5f13a: Pushed
01637544f41d: Pushed
984583bcf083: Layer already exists
ccc35e35d420: Layer already exists
0c7e4c092ab7: Layer already exists
manual1: digest: sha256:1b5acc1844ee7b2bd4cac941f542828e4019cc033c04ee59ac9326f351b89e2d size: 856
PS C:\Docker_jpls>
```

- `docker push dockerlspf123/devops-jeanphilippels:manual1`



Deployment in Local Kubernetes Clusters

- **Deploy the container using Kubernetes / Apply Kubernetes Configuration**
- ***Configure cluster***
- ***Create a namespace***
- ***Create a nginx deployment***
- ***Create a service nginx nodeport***
- ***Create a scale of the nginx deployment***

- **Deploy the container using Kubernetes / Apply Kubernetes Configuration**

- ***Create the custom namespace for devops applications***
  - ***Create the Secrets allowing to use private docker repository***
  - ***Create the service account***
  - ***Create the role cluster for the service account***
  - ***Create the pod's role for the service account***
- ***Bind the role custer and pod's role to the service account***
- ***Create the custom devops web application deployment with secret from a docker hub repository***
- ***Create the service custom devops web application deployment***
- ***Testing the devops web deployment***

Kind (Kubernetes IN Docker) is a tool allowing to setup a local Kubernetes clusters using Docker Containers. Local development can be explored by using this tool.

Kubernetes is a container orchestration platform allows to automates deployment, the scaling, and the management of containerized applications

- **Configure cluster**

```
PS C:\kubernetes> kind create cluster --config .\kind.yaml --name cka-cluster-jeanphilippels
Creating cluster "cka-cluster-jeanphilippels" ...
 • Ensuring node image (kindest/node:v1.32.2) 🖼 ...
 ✓ Ensuring node image (kindest/node:v1.32.2) 🖼
 • Preparing nodes 📦 📦 📦  ...
 ✓ Preparing nodes 📦 📦 📦
 • Writing configuration 📜 ...
 ✓ Writing configuration 📜
 • Starting control-plane 🕹️ ...
 ✓ Starting control-plane 🕹️
 • Installing CNI 🔌 ...
 ✓ Installing CNI 🔌
 • Installing StorageClass 💾 ...
 ✓ Installing StorageClass 💾
 • Joining worker nodes 🚜 ...
 ✓ Joining worker nodes 🚜
Set kubectl context to "kind-cka-cluster-jeanphilippels"
You can now use your cluster with:

kubectl cluster-info --context kind-cka-cluster-jeanphilippels

Have a nice day! 👋
```

```yaml
! kind.yaml
# three node (two workers) cluster config
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
- role: control-plane
  extraPortMappings:
  - containerPort: 30001
    hostPort: 30001
- role: worker
- role: worker
```

```
Kind create cluster -config .\kind.yaml -name cka-cluster-jeanphilippels
```

```
PS C:\kubernetes> kind get clusters
cka-cluster-jeanphilippels
PS C:\kubernetes>
```

- ```
  Kubectl get clusters
  ```

```
PS C:\kubernetes> kubectl get nodes
NAME                                     STATUS   ROLES           AGE   VERSION
cka-cluster-jeanphilippels-control-plane Ready    control-plane   58m   v1.32.2
cka-cluster-jeanphilippels-worker        Ready    <none>          58m   v1.32.2
cka-cluster-jeanphilippels-worker2       Ready    <none>          58m   v1.32.2
PS C:\kubernetes>
```

- **Create a namespace**

Kubernetes namespaces allow to organize partition and organize resources within a Kubernetes cluster. They separate in logical groups of resources within a cluster, making cluster management more efficient.
They can denote multiple environments like dev, staging, or production, or for grouping resources that make up an application.

- **Create a nginx deployment**

```
PS C:\kubernetes> kubectl create namespace nginx-apps-jeanphilippels
namespace/nginx-apps-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\nginx-deploy.yml
deployment.apps/nginx-deploy-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\nginx-nodeport.yml
service/nginx-nodeport-jeanphilippels created
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deploy-jeanphilippels   3/3     3            3           23s
PS C:\kubernetes>
```
```

```
PS C:\kubernetes> kubectl get namespace -n nginx-apps-jeanphilippels
NAME                        STATUS   AGE
default                     Active   5h26m
dev-apps-jeanphilippels     Active   5h20m
kube-node-lease             Active   5h26m
kube-public                 Active   5h26m
kube-system                 Active   5h26m
local-path-storage          Active   5h26m
nginx-apps-jeanphilippels   Active   4h8m
PS C:\kubernetes>
```

- `Kubectl create namespace nginx-apps-jeanphilippels`
- `Kubectl apply -f .\nginx-deploy.yml`
- `Kubectl apply -f .\nginx-nodeport.yml`
- `Kubectl get deploy -n nginx-apps-jeanphilippels`

```
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
NAME                                         READY   STATUS    RESTARTS   AGE
nginx-deploy-jeanphilippels-5f6bd4879d-4cftr   1/1     Running   0          50s
nginx-deploy-jeanphilippels-5f6bd4879d-gxwqn   1/1     Running   0          50s
nginx-deploy-jeanphilippels-5f6bd4879d-mwvss   1/1     Running   0          50s
PS C:\kubernetes>
```

- `Kubectl get pods -n nginx-apps-jeanphilippels`

```
PS C:\kubernetes> kubectl edit deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
```

- **Example of increase replicas with “edit”**

- `Kubectl edit deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels` kubectl.exe-edit-3503174900.yaml - Notepad

File Edit Format View Help

```yaml
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file w
# reopened with the relevant failures.
#
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{},"labe
  creationTimestamp: "2025-03-10T22:16:49Z"
  generation: 1
  labels:
    env: demo
  name: nginx-deploy-jeanphilippels
  namespace: nginx-apps-jeanphilippels
  resourceVersion: "8152"
  uid: 842eea47-ea30-4bcc-8ca0-db536af5f06f
spec:
  progressDeadlineSeconds: 600
  replicas: 3
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      env: demo
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
```

Ln 1, Col 1 | 100% | Windows (CRLF) | UTF-8

kubectl.exe-edit-3503174900.yaml - Notepad

File Edit Format View Help

```yaml
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file w
# reopened with the relevant failures.
#
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{},"labe
  creationTimestamp: "2025-03-10T22:16:49Z"
  generation: 1
  labels:
    env: demo
  name: nginx-deploy-jeanphilippels
  namespace: nginx-apps-jeanphilippels
  resourceVersion: "8152"
  uid: 842eea47-ea30-4bcc-8ca0-db536af5f06f
spec:
  progressDeadlineSeconds: 600
  replicas: 5
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      env: demo
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
```

```
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deploy-jeanphilippels  5/5     5            5           4m40s
PS C:\kubernetes> 
```

- `Kubectl get deploy -n nginx-apps-jeanphilippels`

```
PS C:\kubernetes> kubectl describe deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
Name:                   nginx-deploy-jeanphilippels
Namespace:              nginx-apps-jeanphilippels
CreationTimestamp:      Tue, 11 Mar 2025 11:16:49 +1300
Labels:                 env=demo
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               env=demo
Replicas:               5 desired | 5 updated | 5 total | 5 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  env=demo
  Containers:
   nginx:
    Image:         nginx
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets:  <none>
NewReplicaSet:   nginx-deploy-jeanphilippels-5f6bd4879d (5/5 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  6m3s  deployment-controller  Scaled up replica set nginx-deploy-jeanphilippels-5f6bd4879d from 0 to 3
  Normal  ScalingReplicaSet  92s   deployment-controller  Scaled up replica set nginx-deploy-jeanphilippels-5f6bd4879d from 3 to 5
PS C:\kubernetes>
```

- `Kubectl describe deploy/nginx-deploy-jeanphilippels`

- **Example of increase replicas with “scale”**

```
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deploy-jeanphilippels  5/5     5            5           3h33m
PS C:\kubernetes> kubectl scale deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels --replicas=10
deployment.apps/nginx-deploy-jeanphilippels scaled
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deploy-jeanphilippels  7/10    10           7           3h34m
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deploy-jeanphilippels  10/10   10           10          3h34m
```

- `Kubectl get deploy -n nginx-apps-jeanphilippels`
- `Kubectl scale deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels -replicas=10`

- **`Kubectl get deploy -n nginx-apps-jeanphilippels`**

```
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
NAME                                        READY   STATUS    RESTARTS      AGE
nginx-deploy-jeanphilippels-5f6bd4879d-4cftr   1/1     Running   1 (99s ago)   3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-4zmh4   1/1     Running   0             58s
nginx-deploy-jeanphilippels-5f6bd4879d-7rv42   1/1     Running   0             58s
nginx-deploy-jeanphilippels-5f6bd4879d-g6x8l   1/1     Running   0             58s
nginx-deploy-jeanphilippels-5f6bd4879d-gxwqn   1/1     Running   1 (99s ago)   3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-lqldx   1/1     Running   1 (99s ago)   3h30m
nginx-deploy-jeanphilippels-5f6bd4879d-mwvss   1/1     Running   1 (99s ago)   3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-sndnz   1/1     Running   1 (99s ago)   3h30m
nginx-deploy-jeanphilippels-5f6bd4879d-t6rzf   1/1     Running   0             58s
nginx-deploy-jeanphilippels-5f6bd4879d-vt8cz   1/1     Running   0             58s
PS C:\kubernetes>
```

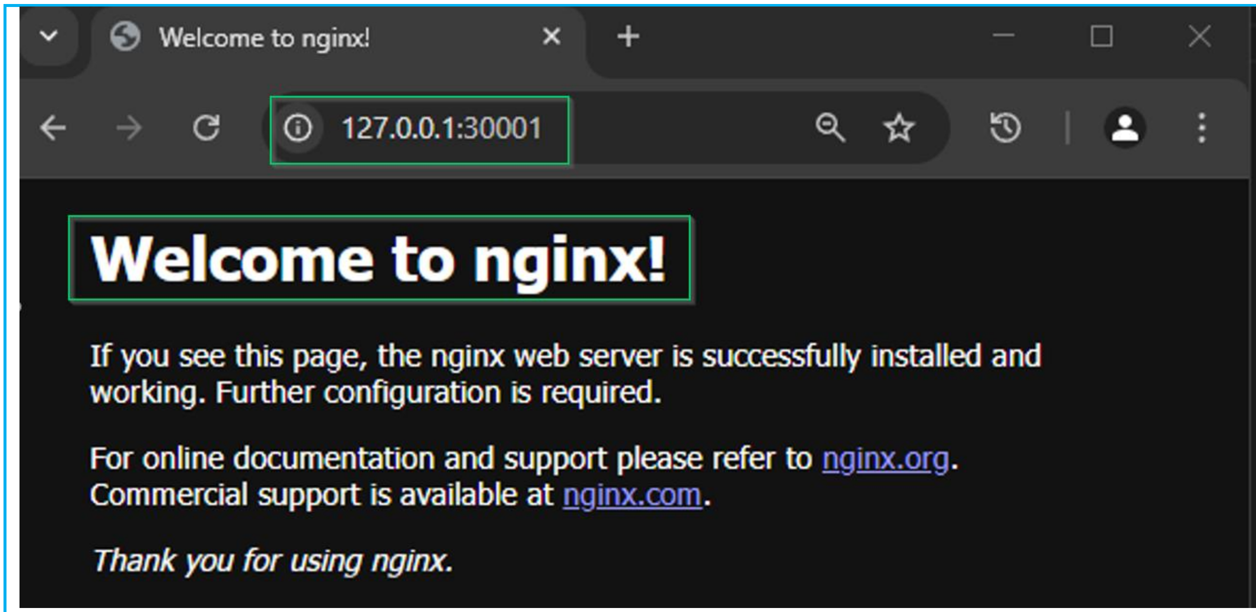
- `Kubectl get pods -n nginx-apps-jeanphilippels`

```
PS C:\kubernetes> kubectl get svc -n nginx-apps-jeanphilippels
NAME                          TYPE       CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
nginx-nodeport-jeanphilippels   NodePort   10.96.81.182   <none>        80:30001/TCP   4h5m
PS C:\kubernetes>
```

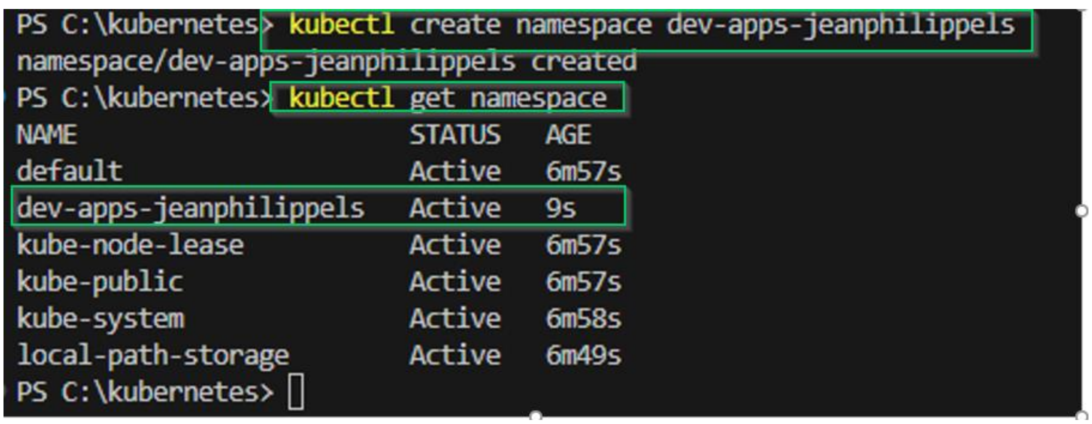
- **Delete a nginx deployment**

```
PS C:\kubernetes> kubectl delete deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
deployment.apps "nginx-deploy-jeanphilippels" deleted
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
No resources found in nginx-apps-jeanphilippels namespace.
PS C:\kubernetes>
```

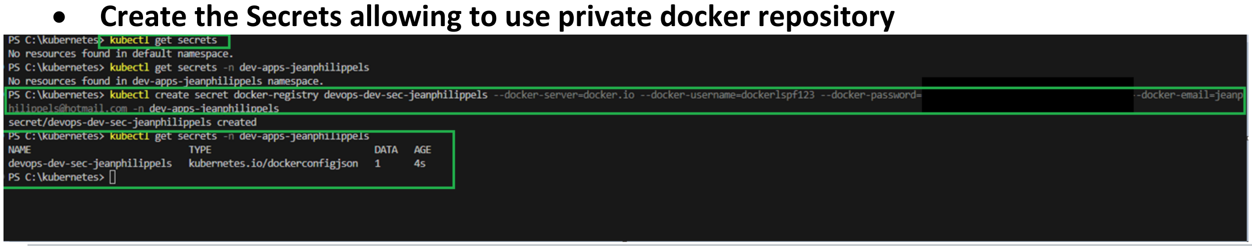
- `Kubectl delete deploy/nginx-deploy-jeanphilippes -n nginx-apps-jeanphilippels`



- **Create the custom namespace for devops apps**

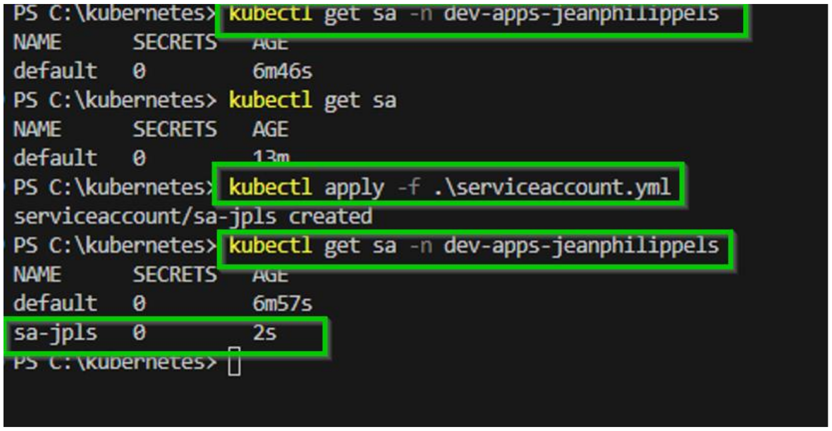


- **Create the Secrets allowing to use private docker repository**




```
• kubectl create secret docker-registry devops-dev-sec-jeanphilippels --docker-serv
er=docker.io --docker-username=dockerlspf123 --docker-password=<PATACCESSTOKEN> --d
ocker-email=jeanphilippels@hotmail.com -n dev-apps-jeanphilippels
```

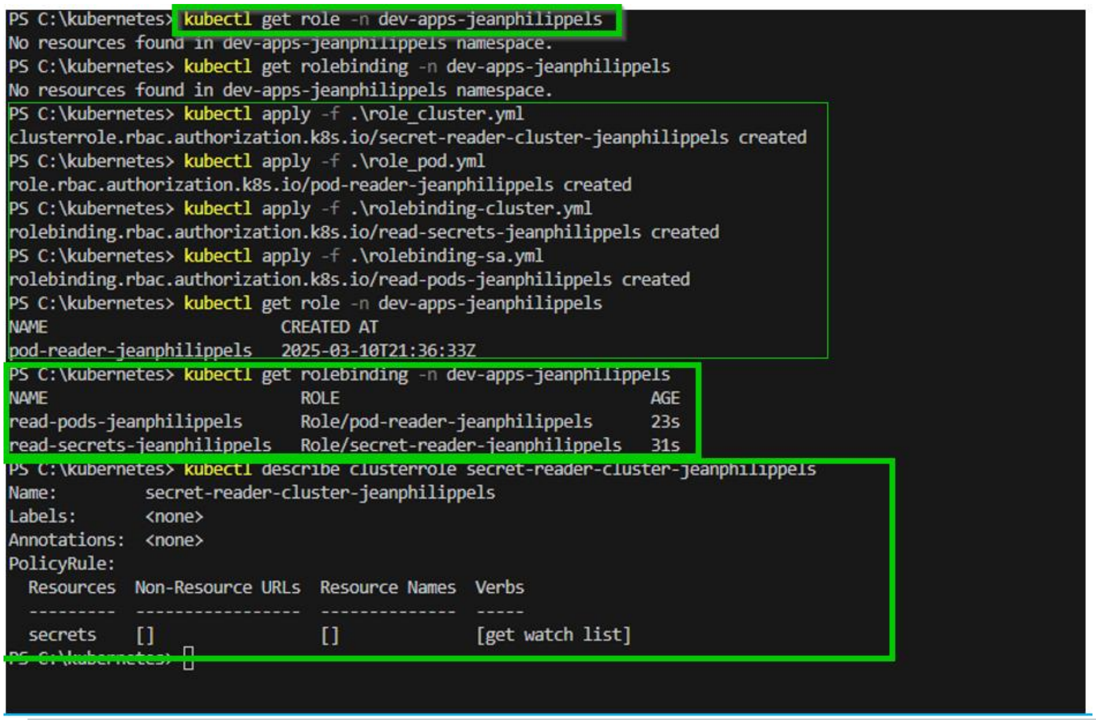
- **Create the service account**



```
PS C:\kubernetes> kubectl get sa -n dev-apps-jeanphilippels
NAME      SECRETS   AGE
default   0         6m46s
PS C:\kubernetes> kubectl get sa
NAME      SECRETS   AGE
default   0         13m
PS C:\kubernetes> kubectl apply -f .\serviceaccount.yml
serviceaccount/sa-jpls created
PS C:\kubernetes> kubectl get sa -n dev-apps-jeanphilippels
NAME      SECRETS   AGE
default   0         6m57s
sa-jpls   0         2s
PS C:\kubernetes>
```

- `kubectl apply -f .\serviceaccount.yml -n dev-apps-jeanphilippels`

- **Create the role cluster for the service account**
- **Create the pod's role for the service account**
- **Bind the role custer and pod's role to the service account**



```
PS C:\kubernetes> kubectl get role -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl get rolebinding -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl apply -f .\role_cluster.yml
clusterrole.rbac.authorization.k8s.io/secret-reader-cluster-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\role_pod.yml
role.rbac.authorization.k8s.io/pod-reader-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\rolebinding-cluster.yml
rolebinding.rbac.authorization.k8s.io/read-secrets-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\rolebinding-sa.yml
rolebinding.rbac.authorization.k8s.io/read-pods-jeanphilippels created
PS C:\kubernetes> kubectl get role -n dev-apps-jeanphilippels
NAME                          CREATED AT
pod-reader-jeanphilippels     2025-03-10T21:36:33Z
PS C:\kubernetes> kubectl get rolebinding -n dev-apps-jeanphilippels
NAME                          ROLE                                AGE
read-pods-jeanphilippels      Role/pod-reader-jeanphilippels      23s
read-secrets-jeanphilippels   Role/secret-reader-jeanphilippels   31s
PS C:\kubernetes> kubectl describe clusterrole secret-reader-cluster-jeanphilippels
Name:         secret-reader-cluster-jeanphilippels
Labels:       <none>
Annotations:  <none>
PolicyRule:
  Resources  Non-Resource URLs  Resource Names  Verbs
  ---------  -----------------  --------------  -----
  secrets    []                 []              [get watch list]
PS C:\kubernetes>
```

- `kubectl apply -f .\role_cluster.yml`

- `kubectl apply -f .\role_pod.yml`
- `kubectl apply -f .\rolebinding-cluster.yml`
- `kubectl apply -f .\rolebinding-sa.yml`
- `kubectl get role -n dev-apps-jeanphilippels`
- `kubectl get rolebinding -n dev-apps-jeanphilippels`
- `kubectl describe clusterrole secret-reader-cluster-jeanphilippels`

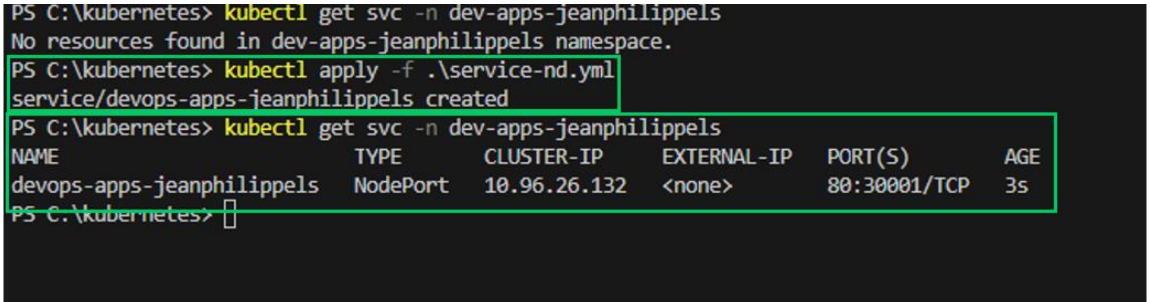
- **Create the custom devops web application deployment with secret from a docker hub repository**

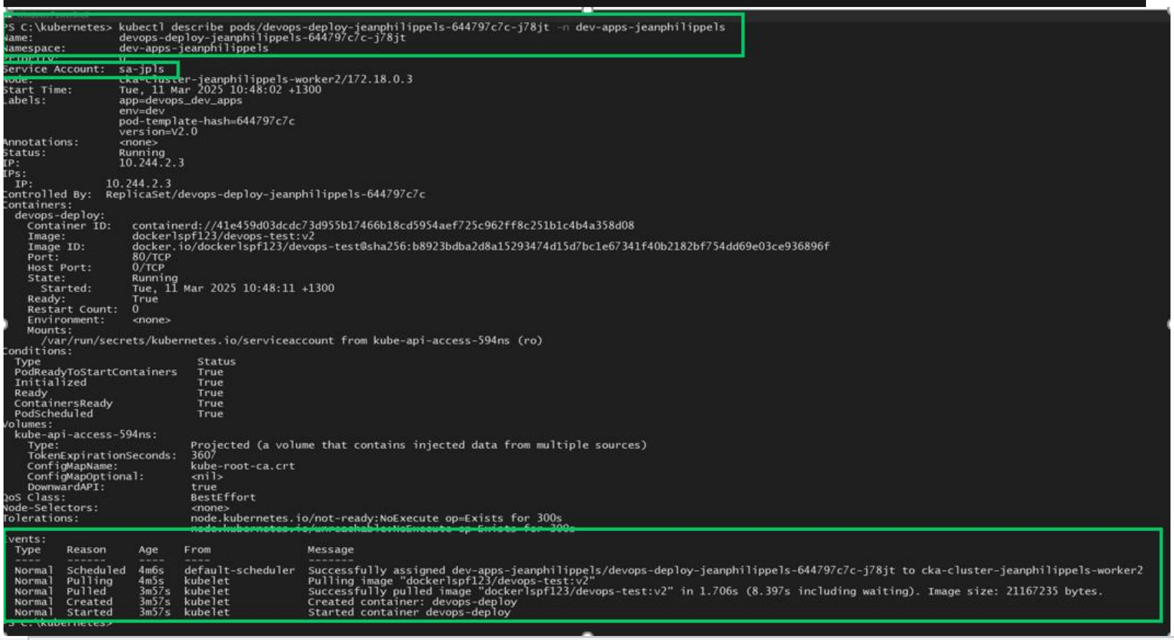
```
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl apply -f .\deploy_secrets.yml
deployment.apps/devops-deploy-jeanphilippels created
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
NAME                                           READY   STATUS              RESTARTS   AGE
devops-deploy-jeanphilippels-644797c7c-j78jt   0/1     ContainerCreating   0          2s
devops-deploy-jeanphilippels-644797c7c-rfphk   0/1     ContainerCreating   0          2s
devops-deploy-jeanphilippels-644797c7c-wh5tn   0/1     ContainerCreating   0          2s
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
NAME                                           READY   STATUS    RESTARTS   AGE
devops-deploy-jeanphilippels-644797c7c-j78jt   1/1     Running   0          14s
devops-deploy-jeanphilippels-644797c7c-rfphk   1/1     Running   0          14s
devops-deploy-jeanphilippels-644797c7c-wh5tn   1/1     Running   0          14s
PS C:\kubernetes>
```

```
PS C:\kubernetes> kubectl get deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels
NAME                           READY   UP-TO-DATE   AVAILABLE   AGE
devops-deploy-jeanphilippels   3/3     3            3           2m1s
PS C:\kubernetes>
```

- `kubectl apply -f .\deploy_secrets.yml`
- `kubectl get pods -n dev-apps-jeanphilippels`
- `kubectl get deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels`

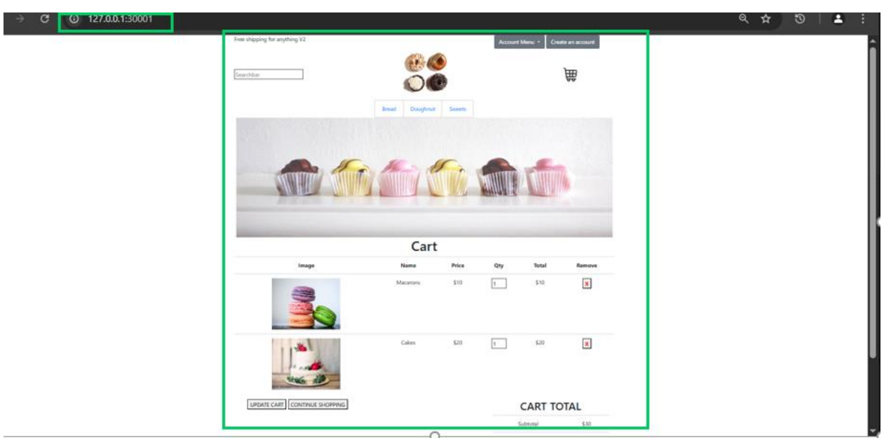
- **Create the service custom devops web application deployment**






```
• kubectl describe pods/ kubectl describe pods/devops-deploy-jeanphilippels-644797c7c-j78jt -n dev-apps-jeanphilippels
```

- **Testing the devops web deployment**



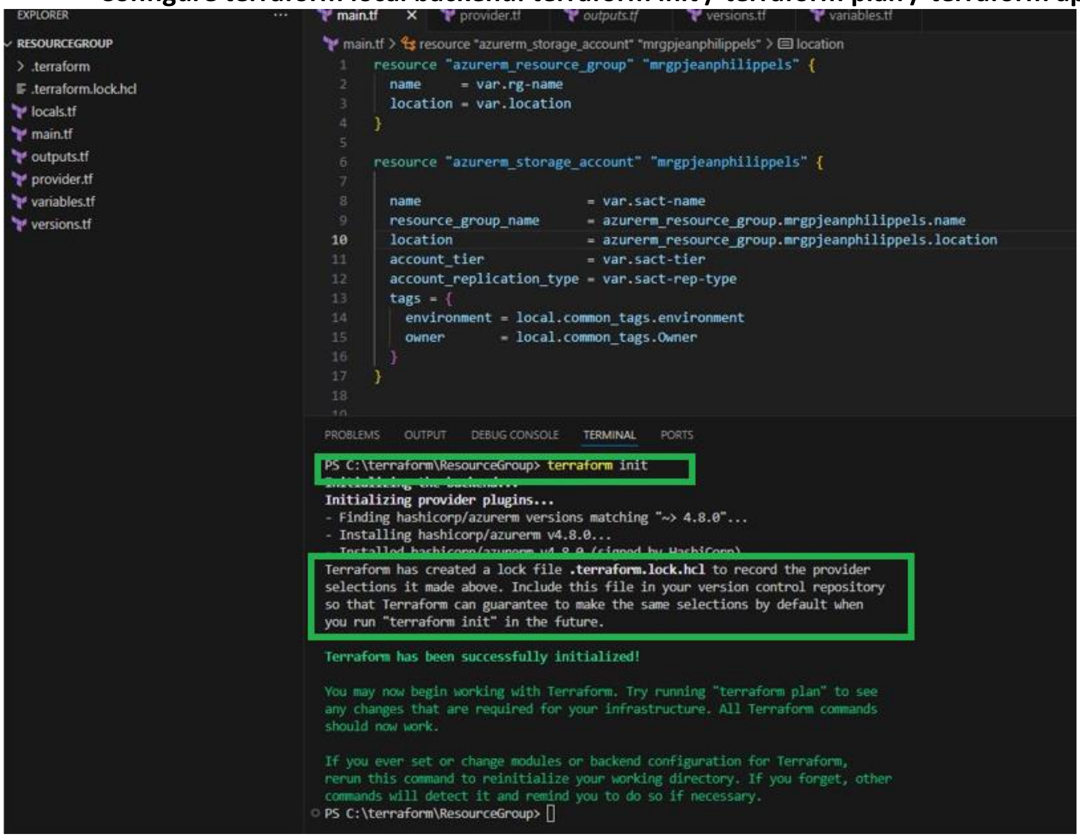
IMPLEMENTING INFRASTRUCTURE AS CODE (IAC)

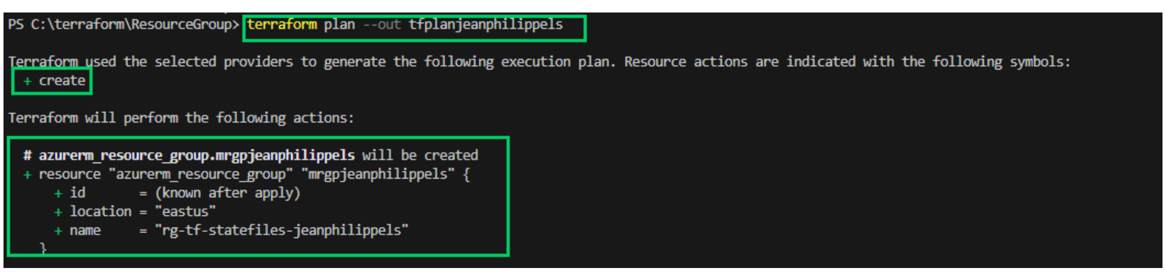
Terraform

- **Configure terraform local backend: terraform init / terraform plan / terraform apply**
- **Destroy resources provisioned with terraform destroy**
- **Configure terraform backend with Microsoft Azure**

Terraform is an Infrastructure as Code tool that allow to provision automatically infrastructures in the cloud and on-premises. It manages resources and configurations across multiple cloud environments.

- **Configure terraform local backend: terraform init / terraform plan / terraform apply**

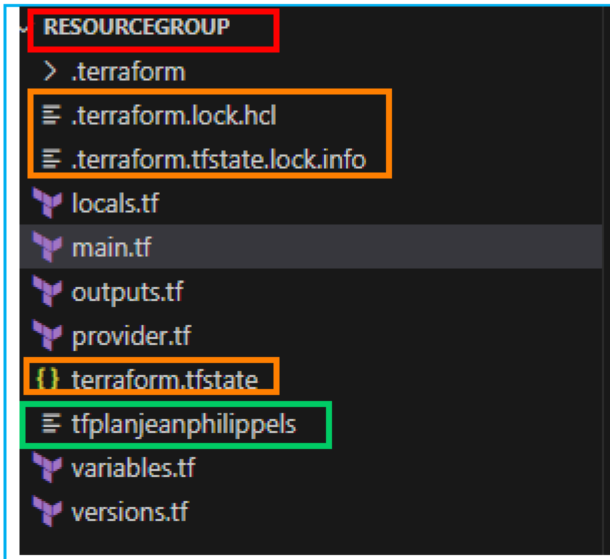




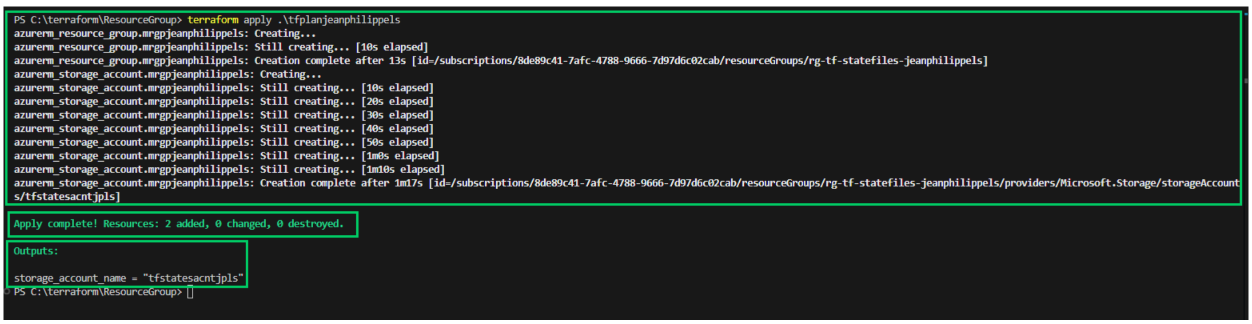
```
# azurerm_storage_account.mrgpjeanphilippels will be created
+ resource "azurerm_storage_account" "mrgpjeanphilippels" {
    + access_tier                       = (known after apply)
    + account_kind                      = "StorageV2"
    + account_replication_type          = "LRS"
    + account_tier                      = "Standard"
    + allow_nested_items_to_be_public   = true
    + cross_tenant_replication_enabled  = false
    + default_to_oauth_authentication   = false
    + dns_endpoint_type                 = "Standard"
    + https_traffic_only_enabled        = true
    + id                                = (known after apply)
    + infrastructure_encryption_enabled = false
    + is_hns_enabled                    = false
    + large_file_share_enabled          = (known after apply)
    + local_user_enabled                = true
    + location                          = "eastus"
    + min_tls_version                   = "TLS1_2"
    + name                              = "tfstatesacntjpls"
    + nfsv3_enabled                     = false
    + primary_access_key                = (sensitive value)
    + primary_blob_connection_string    = (sensitive value)
    + primary_blob_endpoint             = (known after apply)
    + primary_blob_host                 = (known after apply)
    + primary_blob_internet_endpoint    = (known after apply)
    + primary_blob_internet_host        = (known after apply)
    + primary_blob_microsoft_endpoint   = (known after apply)
    + primary_blob_microsoft_host       = (known after apply)
    + primary_connection_string         = (sensitive value)
    + primary_dfs_endpoint              = (known after apply)
    + primary_dfs_host                  = (known after apply)
    + primary_dfs_internet_endpoint     = (known after apply)
    + primary_dfs_internet_host         = (known after apply)
    + primary_dfs_microsoft_endpoint    = (known after apply)
    + primary_dfs_microsoft_host        = (known after apply)
    + primary_file_endpoint             = (known after apply)
    + primary_file_host                 = (known after apply)
    + primary_file_internet_endpoint    = (known after apply)
    + primary_file_internet_host        = (known after apply)
    + primary_file_microsoft_endpoint   = (known after apply)
    + primary_file_microsoft_host       = (known after apply)
    + primary_location                  = (known after apply)
    + primary_queue_endpoint            = (known after apply)
    + primary_queue_host                = (known after apply)
    + primary_queue_microsoft_endpoint  = (known after apply)
    + primary_queue_microsoft_host      = (known after apply)
    + primary_table_endpoint            = (known after apply)
    + primary_table_host                = (known after apply)
    + primary_table_microsoft_endpoint  = (known after apply)
```

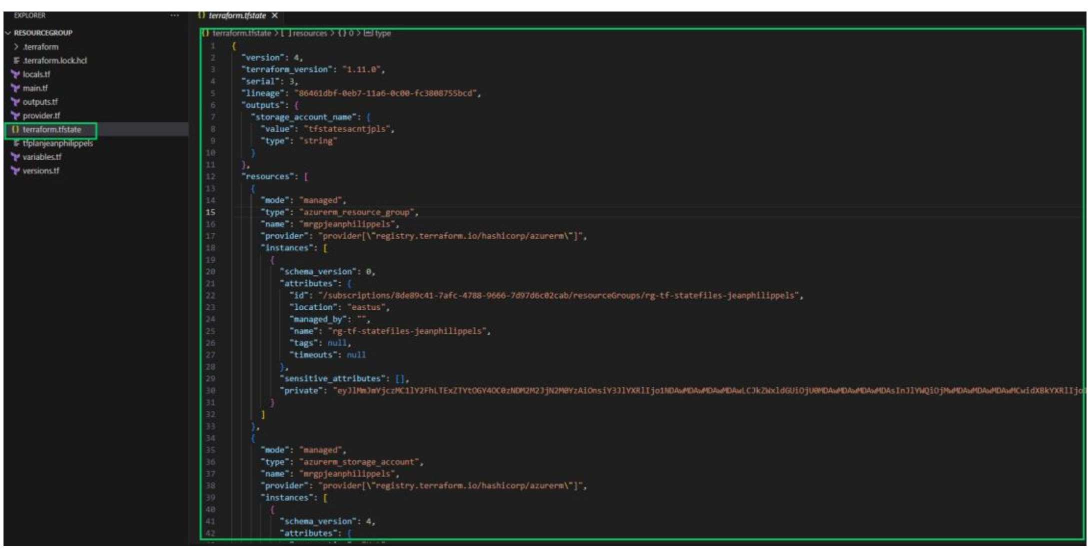
```
      + secondary_file_host              = (known after apply)
      + secondary_file_internet_endpoint  = (known after apply)
      + secondary_file_internet_host      = (known after apply)
      + secondary_file_microsoft_endpoint = (known after apply)
      + secondary_file_microsoft_host     = (known after apply)
      + secondary_location                = (known after apply)
      + secondary_queue_endpoint          = (known after apply)
      + secondary_queue_host              = (known after apply)
      + secondary_queue_microsoft_endpoint = (known after apply)
      + secondary_queue_microsoft_host    = (known after apply)
      + secondary_table_endpoint          = (known after apply)
      + secondary_table_host              = (known after apply)
      + secondary_table_microsoft_endpoint = (known after apply)
      + secondary_table_microsoft_host    = (known after apply)
      + secondary_web_endpoint            = (known after apply)
      + secondary_web_host                = (known after apply)
      + secondary_web_internet_endpoint   = (known after apply)
      + secondary_web_internet_host       = (known after apply)
      + secondary_web_microsoft_endpoint  = (known after apply)
      + secondary_web_microsoft_host      = (known after apply)
      + sftp_enabled                      = false
      + shared_access_key_enabled         = true
      + table_encryption_key_type         = "Service"
      + tags                              = {
          + "environment" = "dev"
          + "owner"       = "IT"
        }

      + blob_properties (known after apply)

      + network_rules (known after apply)

      + queue_properties (known after apply)

      + routing (known after apply)

      + share_properties (known after apply)
    }

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + storage_account_name = "tfstatesacntjpls"
```

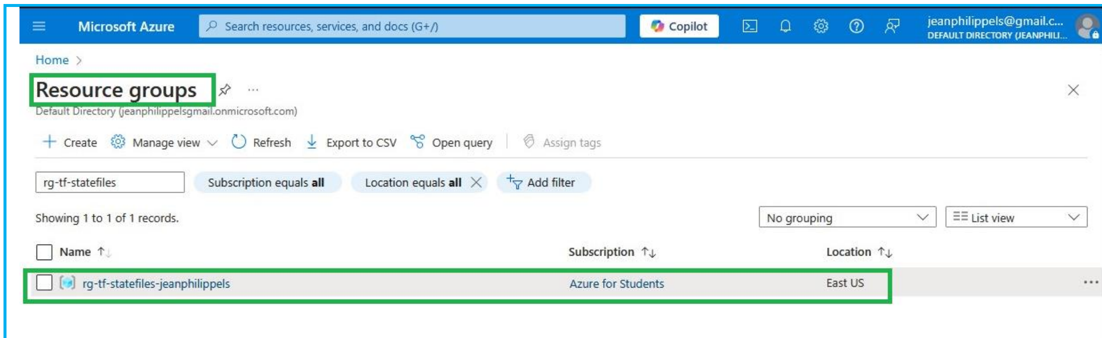
```
Saved the plan to: tfplanjeanphilippels

To perform exactly these actions, run the following command to apply:
    terraform apply "tfplanjeanphilippels"
```

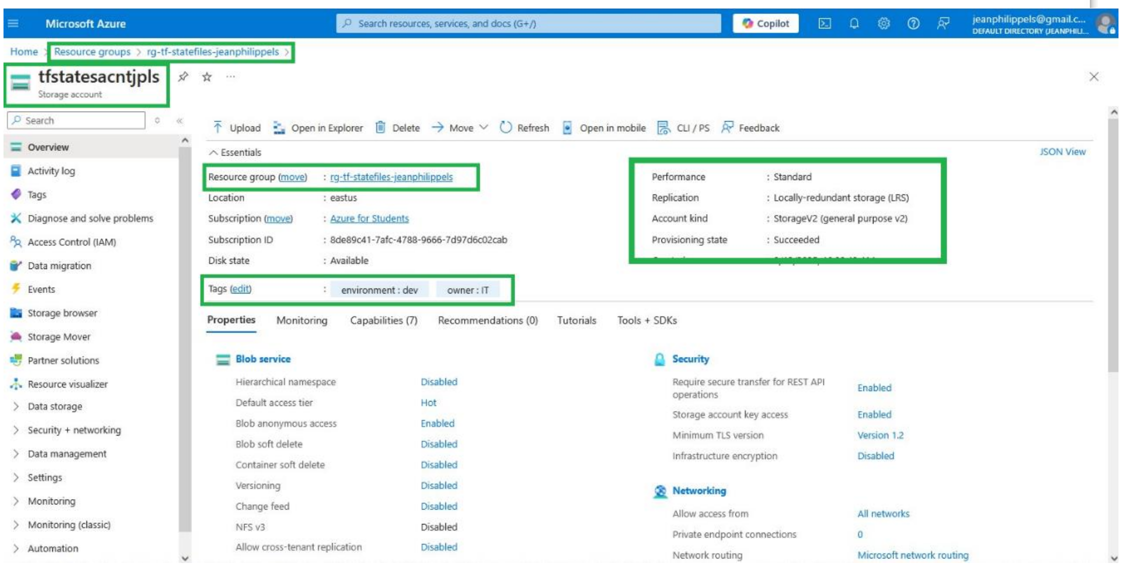


Terraform keeps track of the infrastructure that is deploying in a *state file* called *terraform.tfstate*. When running Terraform, it compares the infrastructure code to the actual resources in Azure.

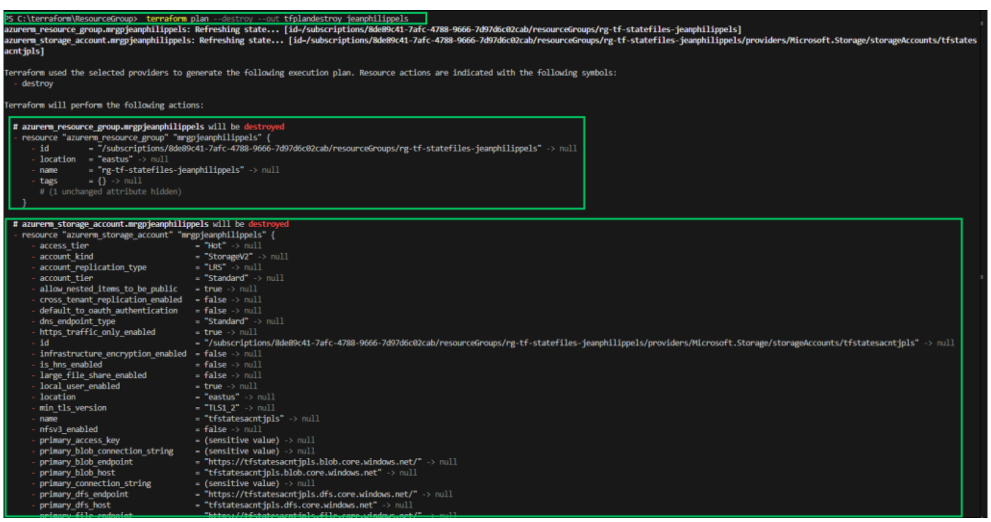




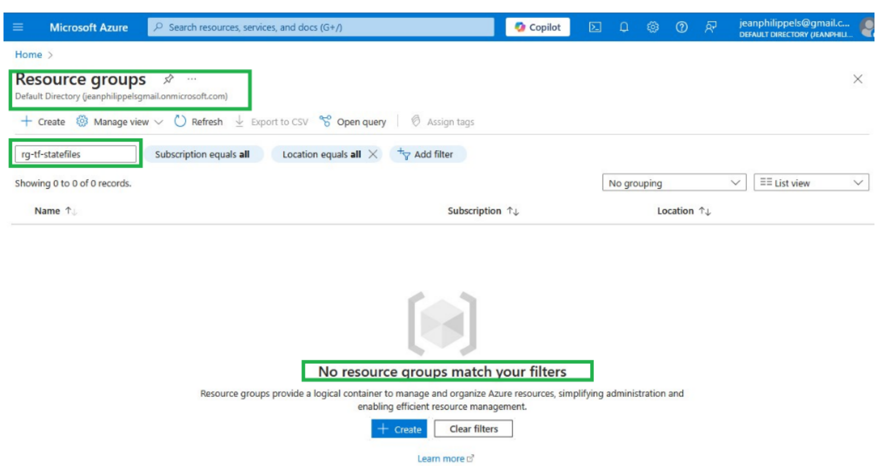




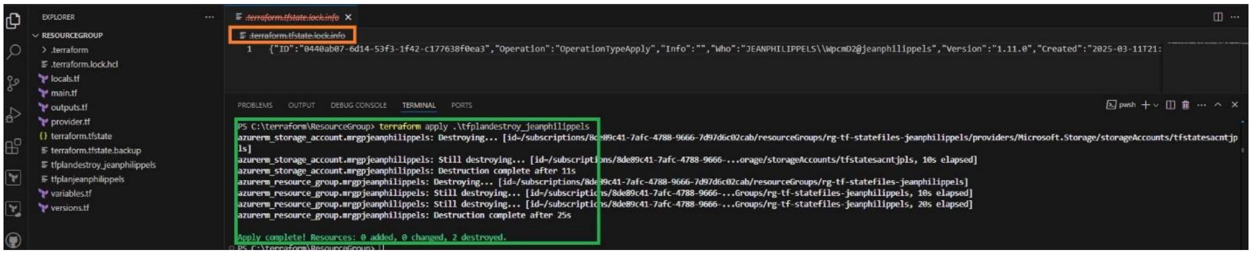
- **Destroy resources provisioned with "terraform destroy"**



```
      - blob_properties {
          - change_feed_enabled           = false -> null
          - change_feed_retention_in_days = 0 -> null
          - last_access_time_enabled      = false -> null
          - versioning_enabled            = false -> null
            # (1 unchanged attribute hidden)
        }

      - queue_properties {
          - hour_metrics {
              - enabled               = false -> null
              - include_apis          = false -> null
              - retention_policy_days = 0 -> null
              - version               = "1.0" -> null
            }
          - logging {
              - delete                = false -> null
              - read                  = false -> null
              - retention_policy_days = 0 -> null
              - version               = "1.0" -> null
              - write                 = false -> null
            }
          - minute_metrics {
              - enabled               = false -> null
              - include_apis          = false -> null
              - retention_policy_days = 0 -> null
              - version               = "1.0" -> null
            }
        }

      - share_properties {
          - retention_policy {
              - days = 7 -> null
            }
        }
    }

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:
  - storage_account_name = "tfstatesacntjpls" -> null

Saved the plan to: tfplandestroy_jeanphilippels

To perform exactly these actions, run the following command to apply:
    terraform apply "tfplandestroy_jeanphilippels"
PS C:\terraform\ResourceGroup>
```



When a terraform apply is executing a *.terraform.tfstate.lock.info* is created in order to lock the tfstate so no one can execute at the same time a terraform command and corrupt the infrastructure.



- **Configure terraform backend with Microsoft Azure**

State file can contain secrets, keeping it in Azure storage enhance security.
This will ensure that the latest version of the state file is used, while locking other user to deploy.

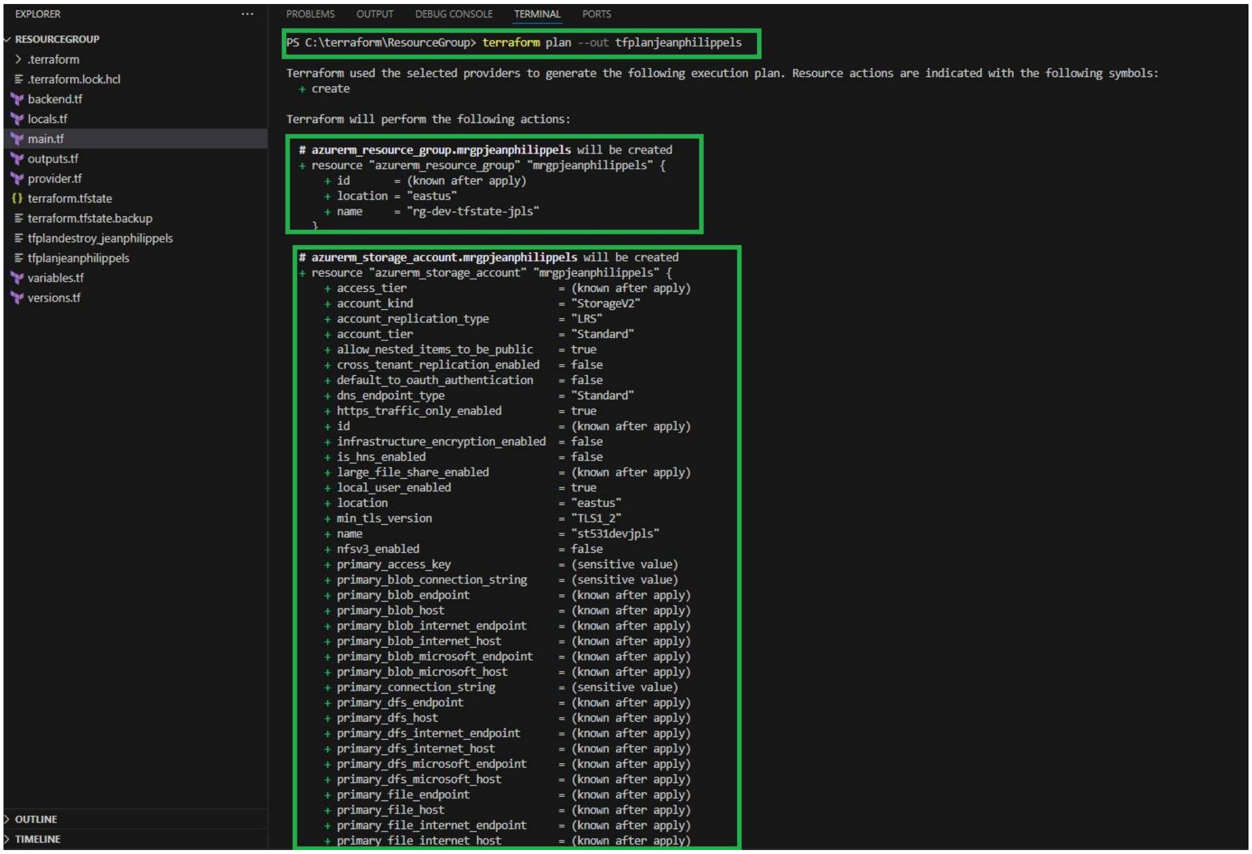
Example of lock state :

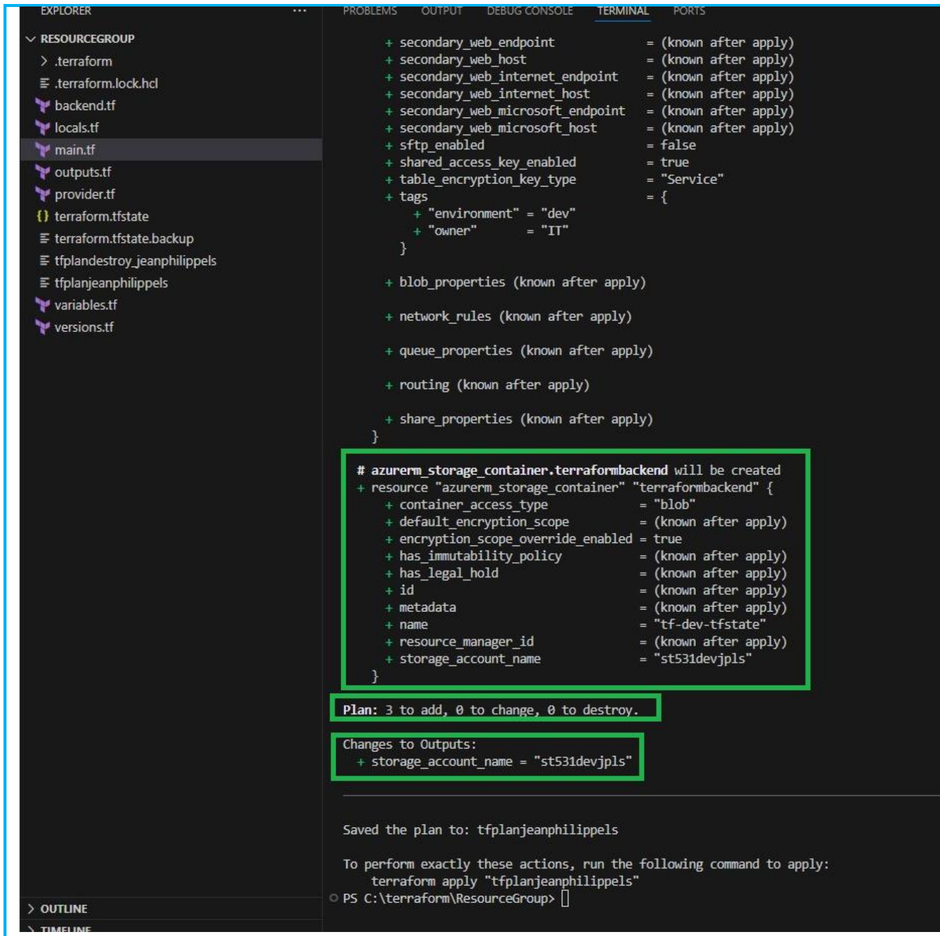
```
PS C:\terraform\Kubernetes> terraform plan --out tfplanjeanphilippels
Acquiring state lock. This may take a few moments...

  Error: Error acquiring the state lock

  Error message: state blob is already locked
  Lock Info:
    ID:        02d75dac-dabb-b390-262d-252dd5f330a9
    Path:      tf-dev-tfstate/tf.dev.tfstate
    Operation: OperationTypeApply
    Who:       runner@fv-az1286-706
    Version:   1.6.0
    Created:   2025-03-13 00:00:34.656203913 +0000 UTC
    Info:


  Terraform acquires a state lock to protect the state from being written
  by multiple users at the same time. Please resolve the issue above and try
  again. For most commands, you can disable locking with the "-lock=false"
  flag, but this is not recommended.

PS C:\terraform\Kubernetes> kubectl force-unlock 02d75dac-dabb-b390-262d-252dd5f330a9
error: unknown command "force-unlock" for "kubectl"
PS C:\terraform\Kubernetes> terraform force-unlock 02d75dac-dabb-b390-262d-252dd5f330a9
Do you really want to force-unlock?
  Terraform will remove the lock on the remote state.
  This will allow local Terraform commands to modify this state, even though it
  may still be in use. Only 'yes' will be accepted to confirm.

  Enter a value: yes

Terraform state has been successfully unlocked!

The state has been unlocked, and Terraform commands should now be able to
obtain a new lock on the remote state.
PS C:\terraform\Kubernetes>
```

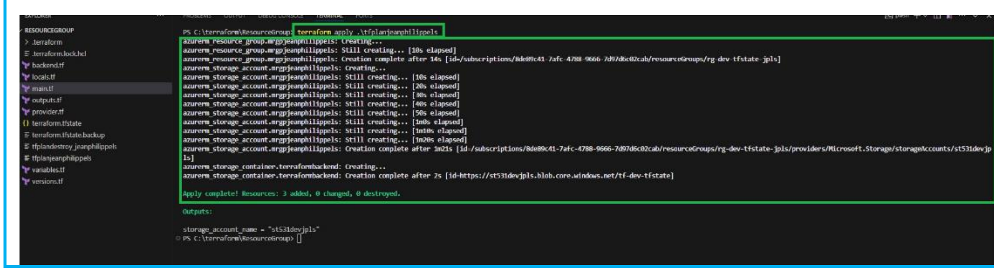
We provision the backend with terraform on Microsoft Azure.

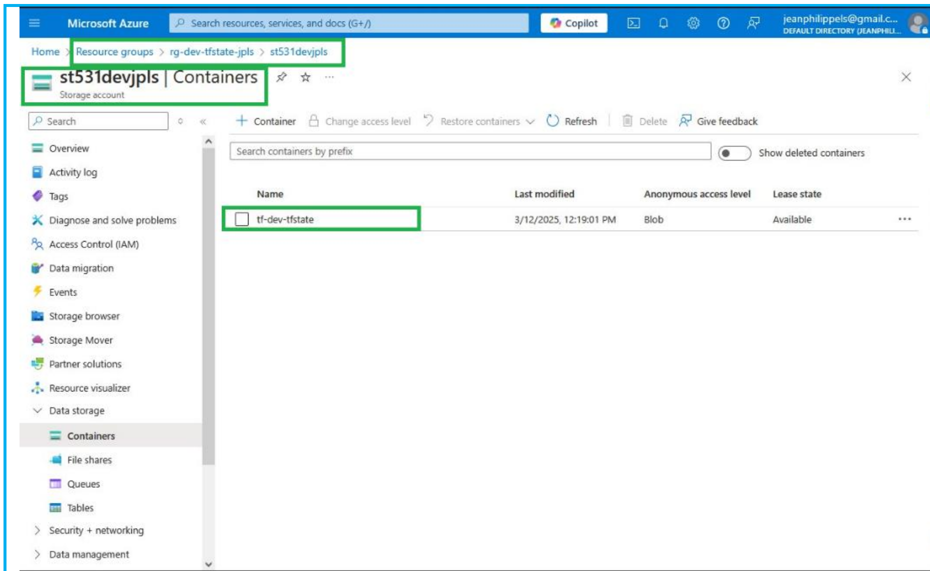
- Provision storage account and container without backend block = terraform.tfstate locally

- After provisioning: add the backend block with references to the resources just created
- Run terraform init.
- Terraform will ask to migrate the local state file into Microsoft Azure Storage

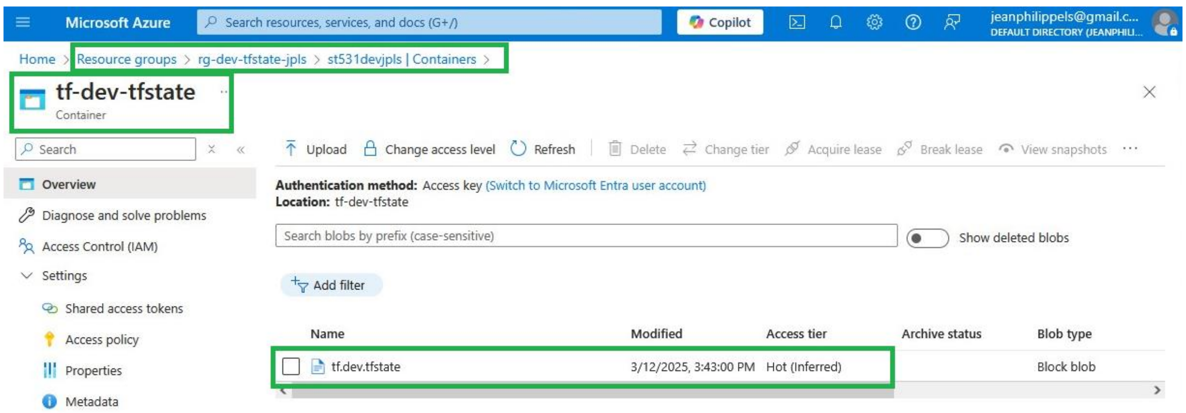



EXPLORER
RESOURCEGROUP
.terraform
.terraform.lock.hcl
backend.tf
locals.tf
main.tf
outputs.tf
provider.tf
terraform.tfstate
terraform.tfstate.backup
tfplandestroy_jeanphilippels
tfplanjeanphilippels
variables.tf
versions.tf
OUTLINE
TIMELINE
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
+ secondary_web_endpoint = (known after apply)
+ secondary_web_host = (known after apply)
+ secondary_web_internet_endpoint = (known after apply)
+ secondary_web_internet_host = (known after apply)
+ secondary_web_microsoft_endpoint = (known after apply)
+ secondary_web_microsoft_host = (known after apply)
+ sftp_enabled = false
+ shared_access_key_enabled = true
+ table_encryption_key_type = "Service"
+ tags = {
+ "environment" = "dev"
+ "owner" = "IT"
}
+ blob_properties (known after apply)
+ network_rules (known after apply)
+ queue_properties (known after apply)
+ routing (known after apply)
+ share_properties (known after apply)
}
# azurerm_storage_container.terraformbackend will be created
+ resource "azurerm_storage_container" "terraformbackend" {
+ container_access_type = "blob"
+ default_encryption_scope = (known after apply)
+ encryption_scope_override_enabled = true
+ has_immutability_policy = (known after apply)
+ has_legal_hold = (known after apply)
+ id = (known after apply)
+ metadata = (known after apply)
+ name = "tf-dev-tfstate"
+ resource_manager_id = (known after apply)
+ storage_account_name = "st531devjpls"
}
Plan: 3 to add, 0 to change, 0 to destroy.
Changes to Outputs:
+ storage_account_name = "st531devjpls"
Saved the plan to: tfplanjeanphilippels
To perform exactly these actions, run the following command to apply:
terraform apply "tfplanjeanphilippels"
PS C:\terraform\ResourceGroup>

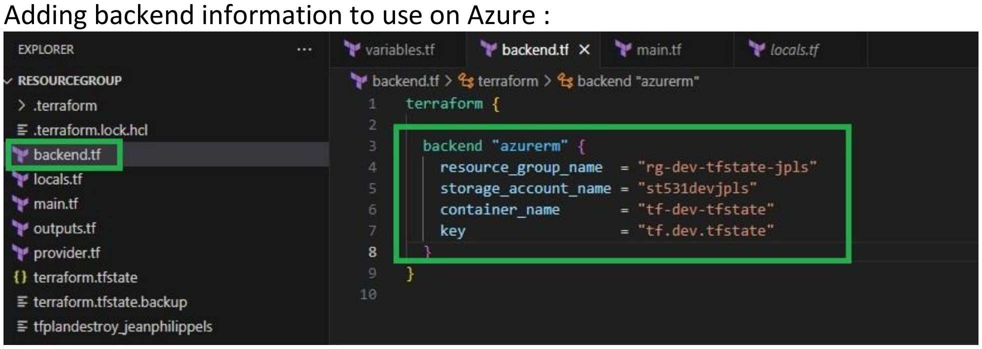

PS C:\terraform\ResourceGroup> terraform apply .\tfplanjeanphilippels
azurerm_resource_group.mrgpjeanphilippels: Creating...
azurerm_resource_group.mrgpjeanphilippels: Still creating... [10s elapsed]
azurerm_resource_group.mrgpjeanphilippels: Creation complete after 14s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-tfstate-jpls]
azurerm_storage_account.mrgpjeanphilippels: Creating...
azurerm_storage_account.mrgpjeanphilippels: Still creating... [10s elapsed]
azurerm_storage_account.mrgpjeanphilippels: Still creating... [20s elapsed]
azurerm_storage_account.mrgpjeanphilippels: Still creating... [30s elapsed]
azurerm_storage_account.mrgpjeanphilippels: Still creating... [40s elapsed]
azurerm_storage_account.mrgpjeanphilippels: Still creating... [50s elapsed]
azurerm_storage_account.mrgpjeanphilippels: Still creating... [1m0s elapsed]
azurerm_storage_account.mrgpjeanphilippels: Still creating... [1m10s elapsed]
azurerm_storage_account.mrgpjeanphilippels: Still creating... [1m20s elapsed]
azurerm_storage_account.mrgpjeanphilippels: Creation complete after 1m21s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-tfstate-jpls/providers/Microsoft.Storage/storageAccounts/st531devjpls]
azurerm_storage_container.terraformbackend: Creating...
azurerm_storage_container.terraformbackend: Creation complete after 2s [id=https://st531devjpls.blob.core.windows.net/tf-dev-tfstate]
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
Outputs:
storage_account_name = "st531devjpls"
PS C:\terraform\ResourceGroup>

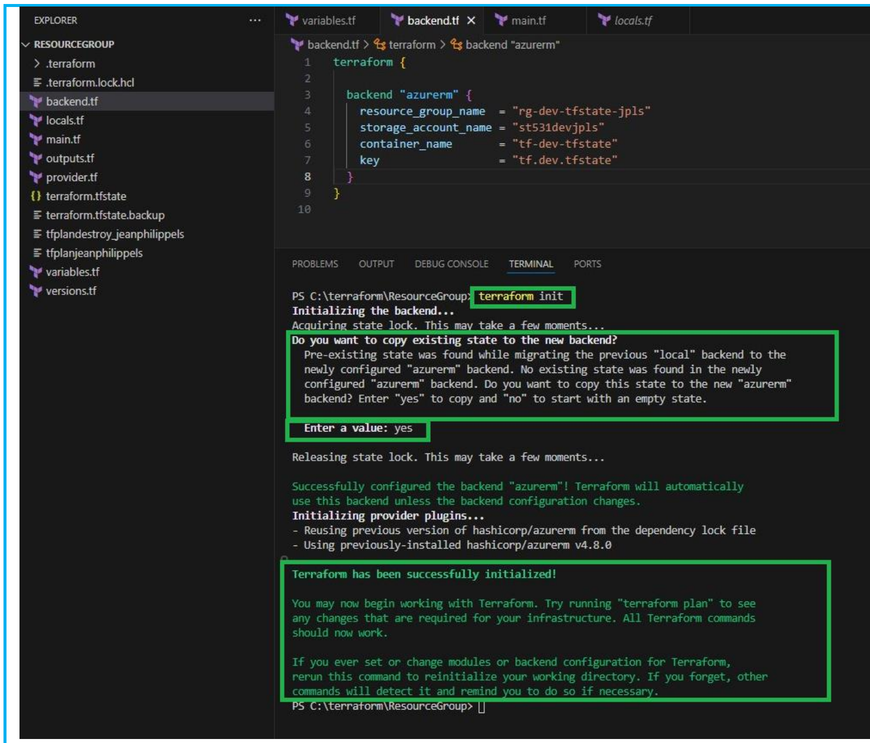


Tf State file is stored in the Azure container with the "tf.dev.tfstate" name.

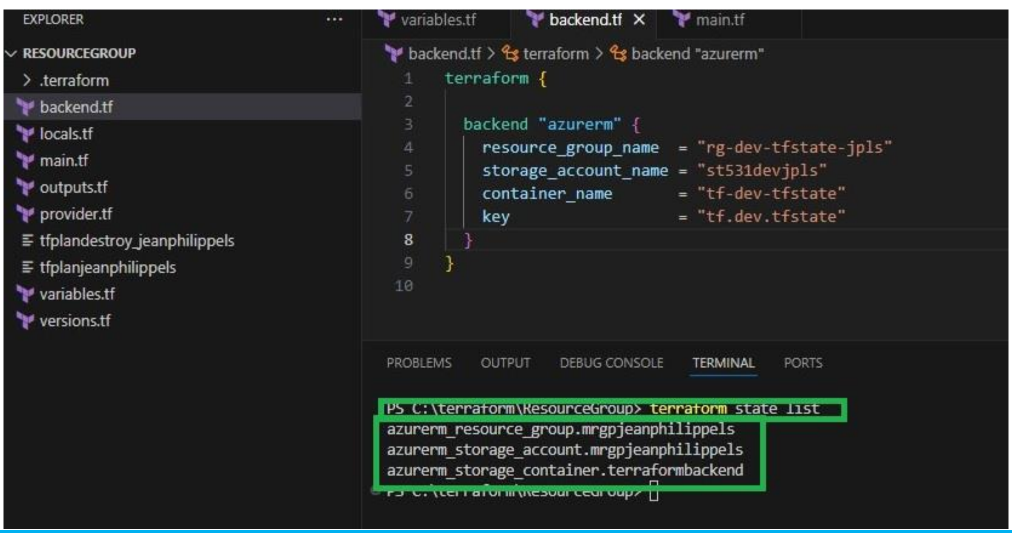


Adding backend information to use on Azure :





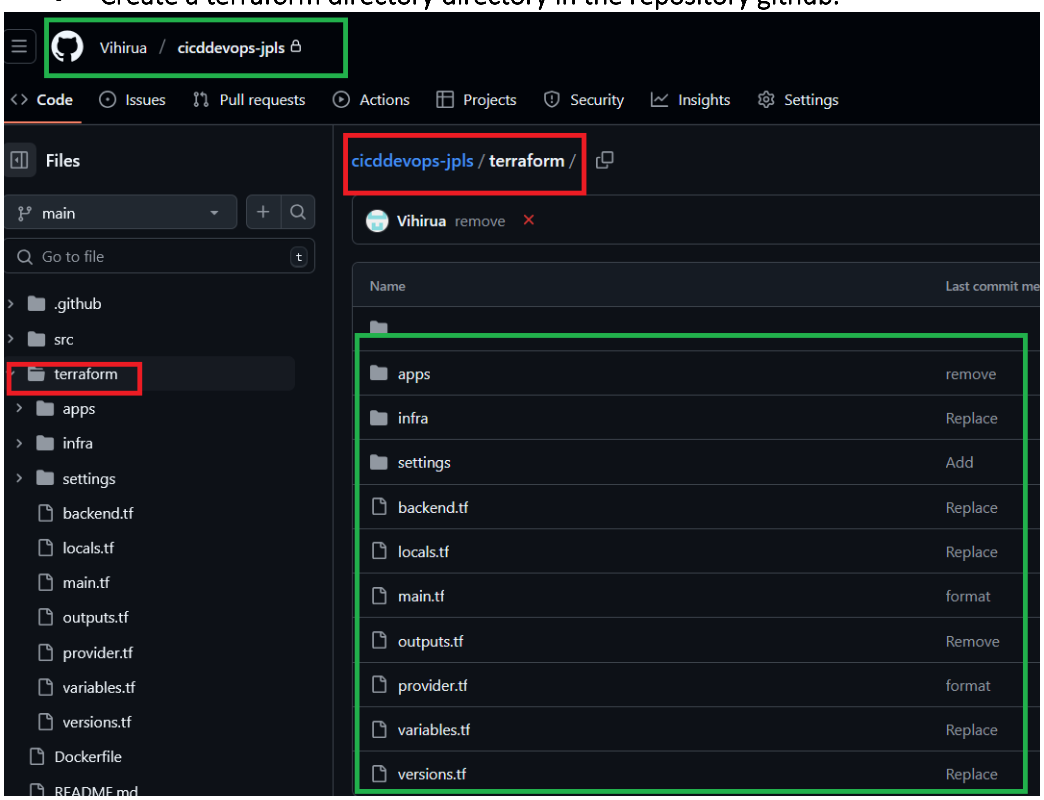
To verify the setup, the local instance of **terraform.tfstate** file has been deleted .
We run command terraform state list. The output is as follow:



Deployment in Azure Kubernetes Cluster through Terraform

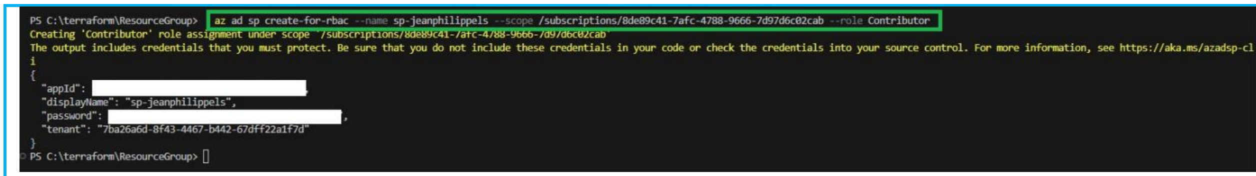
- **Deploy the container using Kubernetes / Apply Kubernetes Configuration**
  - **Create a terraform directory directory in the repository github.**
  - ***Configure Azure Kubernetes Services***
- ***Provision the Azure Kubernetes Services***
- ***Create the custom namespace for devops applications***
- ***Create the Secrets allowing to use private docker repository***
- ***Create the Service account***
- ***Create the role cluster for the service account***
- ***Bind the role cluster to the service account***
- **Create the custom devops web application deployment with secret from a docker hub repository**
- **Create the service custom devops web application deployment**
- ***Provision the Apps in Azure Kubernetes Services***
- **Testing the devops web deployment**

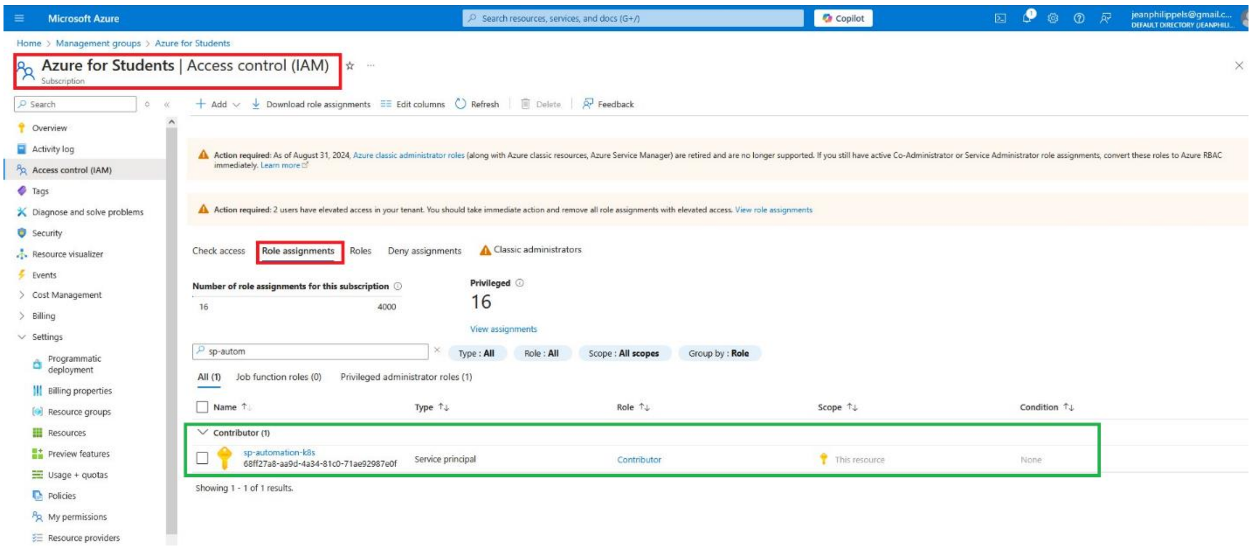
- Create a terraform directory directory in the repository github.



- ***Configure Azure Kubernetes Services***

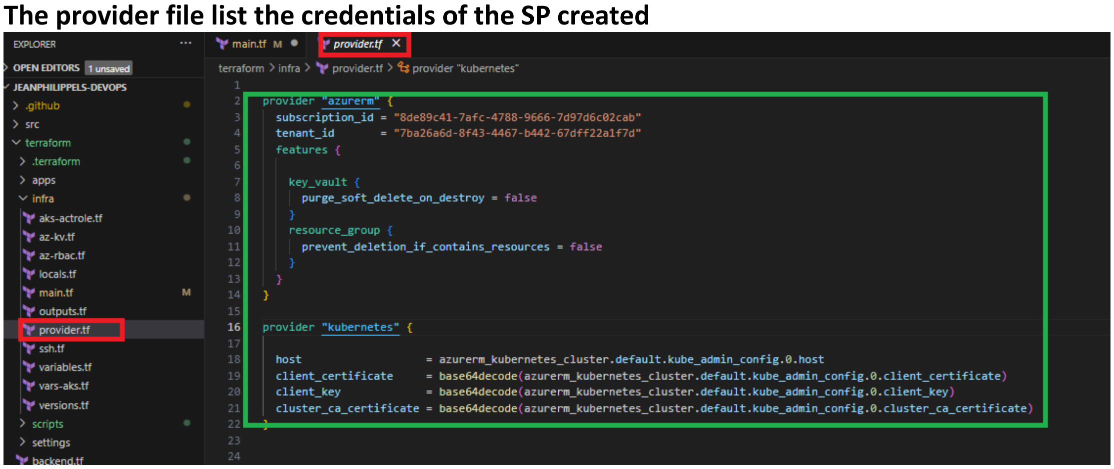
In order terraform to access Microsoft Azure, a Service Principal Name has to be created.
Contributor Role is giving to the Service Principal Name.



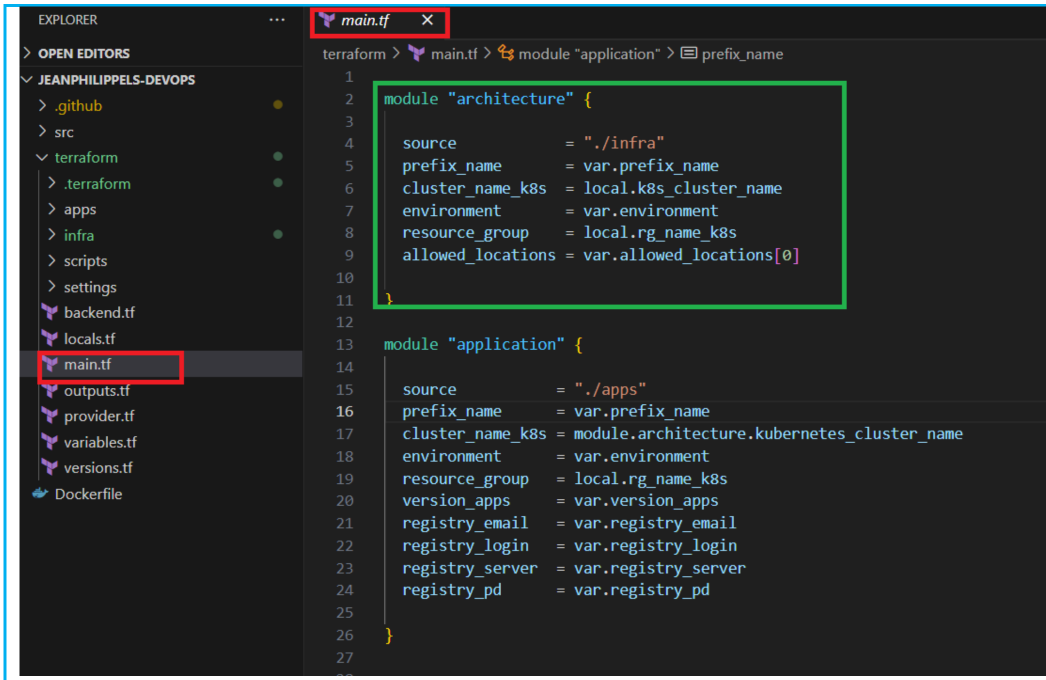


- **Provision the Azure Kubernetes Services**

The provider file list the credentials of the SP created



The main file in the root directory listed the modules use in order to provision the architecture:

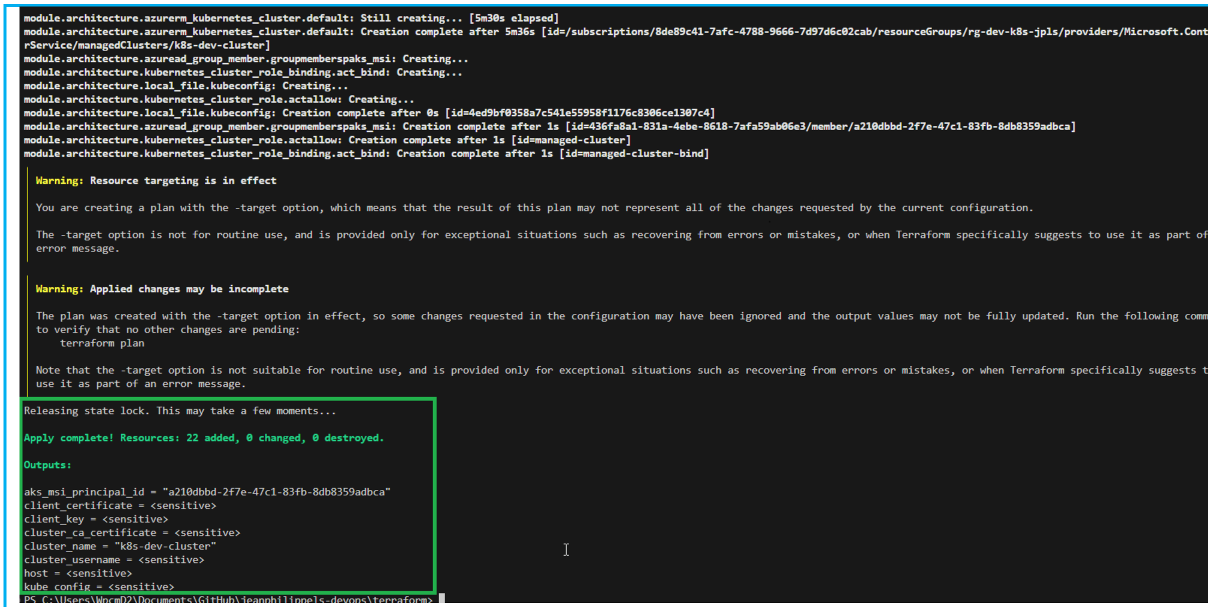


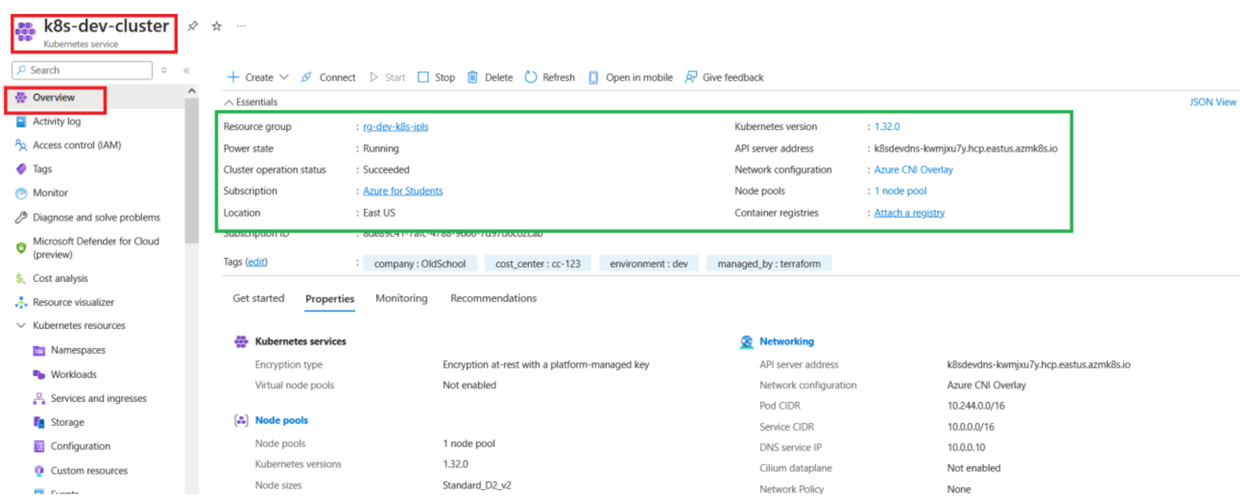
- **Module architecture:**

The variables of the module are the parameters that need to be fill up when calling in the main root directory of terraform :

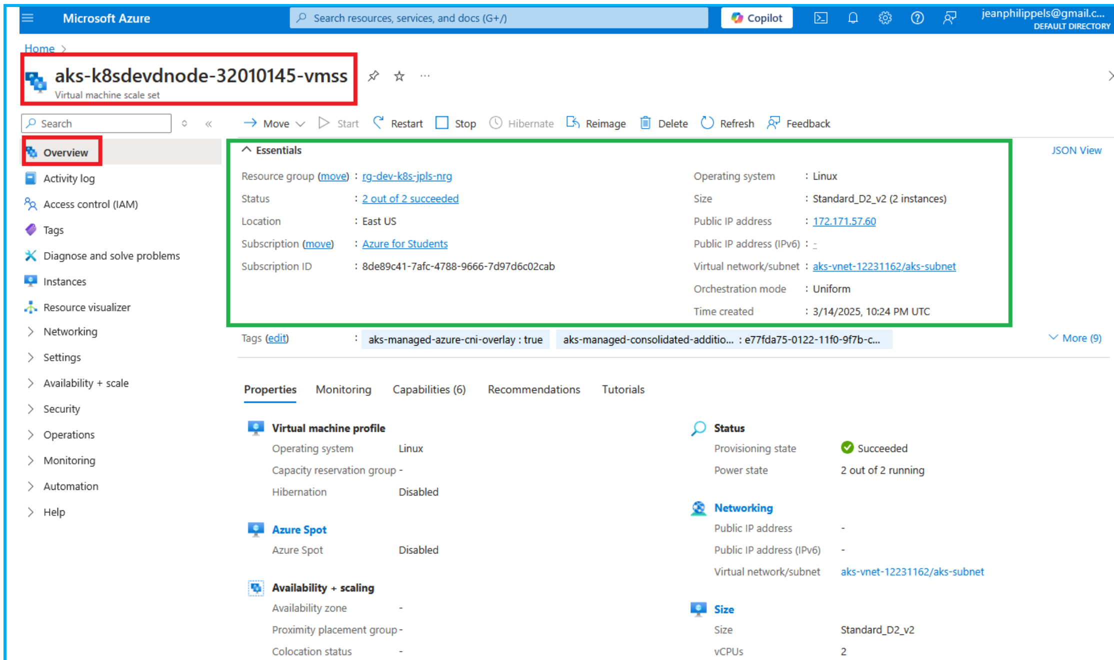
```
PS C:\Users\WpcmD2\Documents\GitHub\jeanphilippels-devops\terraform> terraform apply --target=module.architecture --var-file=settings/dev/variables.tfvars --auto-approve
Acquiring state lock. This may take a few moments...
module.architecture.data.azuread_service_principal.spjplsact: Reading...
module.architecture.data.azuread_client_config.currentad: Reading...
module.architecture.data.azuread_service_principal.spjpls: Reading...
module.architecture.data.azuread_group.azgroupdevops: Reading...
module.architecture.data.azuread_client_config.currentad: Read complete after 0s [id=7ba26a6d-8f43-4467-b442-67dff22a1f7d-49642172-7e15-4440-a15e-bd3fe84ca293-aeb41d91-48fc-4f8c-b320-33346dd37471]
module.architecture.data.azuread_service_principal.spjplsact: Read complete after 1s [id=/servicePrincipals/aeb41d91-48fc-4f8c-b320-33346dd37471]
module.architecture.data.azuread_service_principal.spjpls: Read complete after 1s [id=/servicePrincipals/b6ab327a-b37c-4f32-870f-f6114b902745]
module.architecture.data.azuread_group.azgroupdevops: Read complete after 1s [id=/groups/0658a60b-0195-46b3-b0c2-bbebff1afe2e]
module.architecture.data.azurerm_client_config.currents: Reading...
module.architecture.data.azurerm_client_config.current: Reading...
module.architecture.data.azurerm_client_config.currentsw: Reading...
module.architecture.data.azurerm_role_definition.aksadminrole: Reading...
module.architecture.data.azurerm_role_definition.builtin: Reading...
module.architecture.data.azurerm_subscription.primary: Reading...
module.architecture.data.azurerm_client_config.currents: Read complete after 0s [id=Y2xpZW50Q29uZmlncy9jbGllbnRJZD00OTY0MjE3Mi03ZTE1LTQ0NDAtYTE1ZS1iZDNmZTg0Y2EyOTM7b2JqZWN0SWQ9YWViNDFkOTEtNDhmYy00Zjh L
WIzMjAtMzMzNDZkZDM3NDcxO3N1YnNjcmlwdGlvbklkPThkZTg5YzQxLTdhZmMtNDc4OC05NjY2LTdkOTdkNmMwMmNhYjt0ZW5hbnRJZD03YmEyNmE2ZC04ZjQzLTQ0NjctYjQ0Mi02N2RmZjIyYTFmN2Q=]
module.architecture.data.azurerm_client_config.current: Read complete after 0s [id=Y2xpZW50Q29uZmlncy9jbGllbnRJZD00OTY0MjE3Mi03ZTE1LTQ0NDAtYTE1ZS1iZDNmZTg0Y2EyOTM7b2JqZWN0SWQ9YWViNDFkOTEtNDhmYy00Zjhj W
IzMjAtMzMzNDZkZDM3NDcxO3N1YnNjcmlwdGlvbklkPThkZTg5YzQxLTdhZmMtNDc4OC05NjY2LTdkOTdkNmMwMmNhYjt0ZW5hbnRJZD03YmEyNmE2ZC04ZjQzLTQ0NjctYjQ0Mi02N2RmZjIyYTFmN2Q=]
module.architecture.data.azurerm_client_config.currentsw: Read complete after 0s [id=Y2xpZW50Q29uZmlncy9jbGllbnRJZD00OTY0MjE3Mi03ZTE1LTQ0NDAtYTE1ZS1iZDNmZTg0Y2EyOTM7b2JqZWN0SWQ9YWViNDFkOTEtNDhmYy00Zj j
LWIzMjAtMzMzNDZkZDM3NDcxO3N1YnNjcmlwdGlvbklkPThkZTg5YzQxLTdhZmMtNDc4OC05NjY2LTdkOTdkNmMwMmNhYjt0ZW5hbnRJZD03YmEyNmE2ZC04ZjQzLTQ0NjctYjQ0Mi02N2RmZjIyYTFmN2Q=]
module.architecture.data.azurerm_subscription.primary: Read complete after 0s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab]
module.architecture.data.azurerm_role_definition.builtin: Read complete after 0s [id=/providers/Microsoft.Authorization/roleDefinitions/acdd72a7-3385-48ef-bd42-f606fba81ae7]
module.architecture.data.azurerm_role_definition.aksadminrole: Read complete after 1s [id=/providers/Microsoft.Authorization/roleDefinitions/b1ff04bb-8a4e-4dc4-8eb5-8693973ce19b]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create
 <= read (data resources)

Terraform will perform the following actions:

  # module.architecture.data.azuread_group.aksgroup will be read during apply
  # (depends on a resource or a module with changes pending)
 <= data "azuread_group" "aksgroup" {
      + assignable_to_role          = (known after apply)
      + auto_subscribe_new_members  = (known after apply)
      + behaviors                   = (known after apply)
      + description                 = (known after apply)
      + display_name                = "k8s_admin"
      + dynamic_membership          = (known after apply)
      + external_senders_allowed    = (known after apply)
      + hide_from_address_lists     = (known after apply)
      + hide_from_outlook_clients   = (known after apply)
```

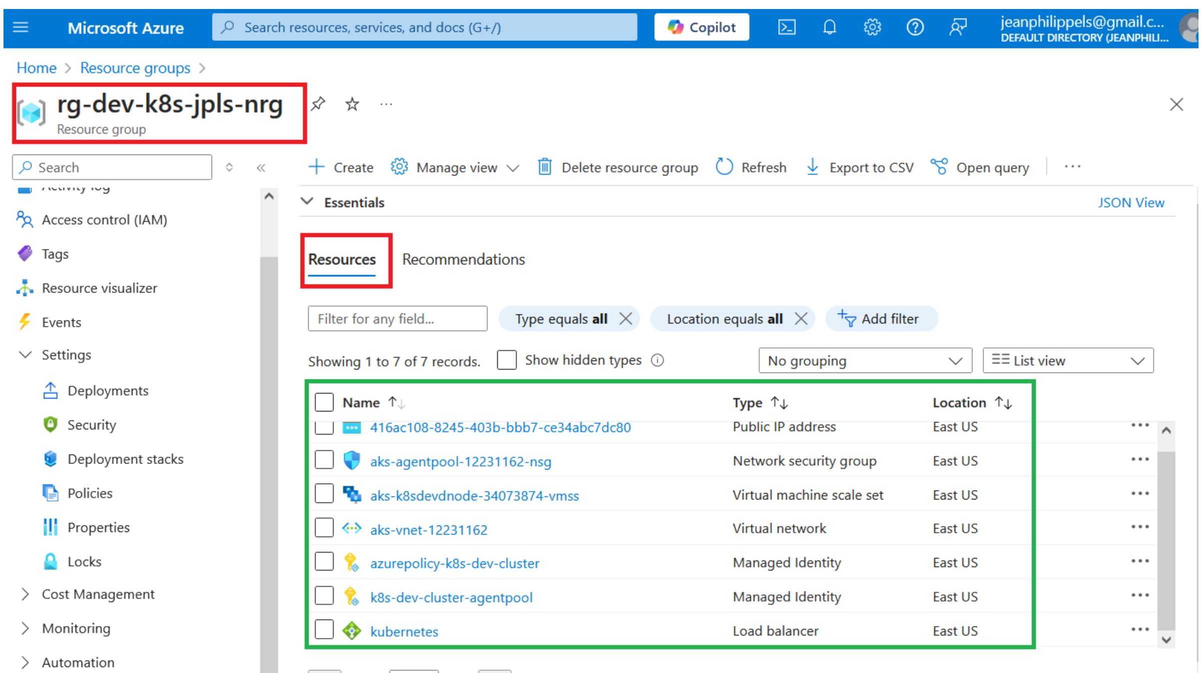
```
      + content_md5          = (known after apply)
      + content_sha1         = (known after apply)
      + content_sha256       = (known after apply)
      + content_sha512       = (known after apply)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "./kubeconfig"
      + id                   = (known after apply)
    }

  # module.architecture.random_pet.ssh_key_name will be created
  + resource "random_pet" "ssh_key_name" {
      + id        = (known after apply)
      + length    = 2
      + prefix    = "ssh"
        # (1 unchanged attribute hidden)
    }

Plan: 22 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + aks_msi_principal_id   = (known after apply)
  + client_certificate     = (sensitive value)
  + client_key             = (sensitive value)
  + cluster_ca_certificate = (sensitive value)
  + cluster_name           = "k8s-dev-cluster"
  + cluster_username       = (sensitive value)
  + host                   = (sensitive value)
  + kube_config            = (sensitive value)
module.architecture.random_pet.ssh_key_name: Creating...
module.architecture.random_pet.ssh_key_name: Creation complete after 0s [id=sshhumbleshrimp]
module.architecture.azurerm_role_assignment.aks_sp_role_assignment: Creating...
module.architecture.azurerm_resource_group.rgaks: Creating...
module.architecture.azuread_group.aks_user: Creating...
module.architecture.azuread_group.aks_admin: Creating...
module.architecture.azurerm_resource_group.rgaks: Still creating... [10s elapsed]
module.architecture.azurerm_resource_group.rgaks: Creation complete after 15s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-k8s-jpls]
module.architecture.azapi_resource.ssh_public_key: Creating...
```



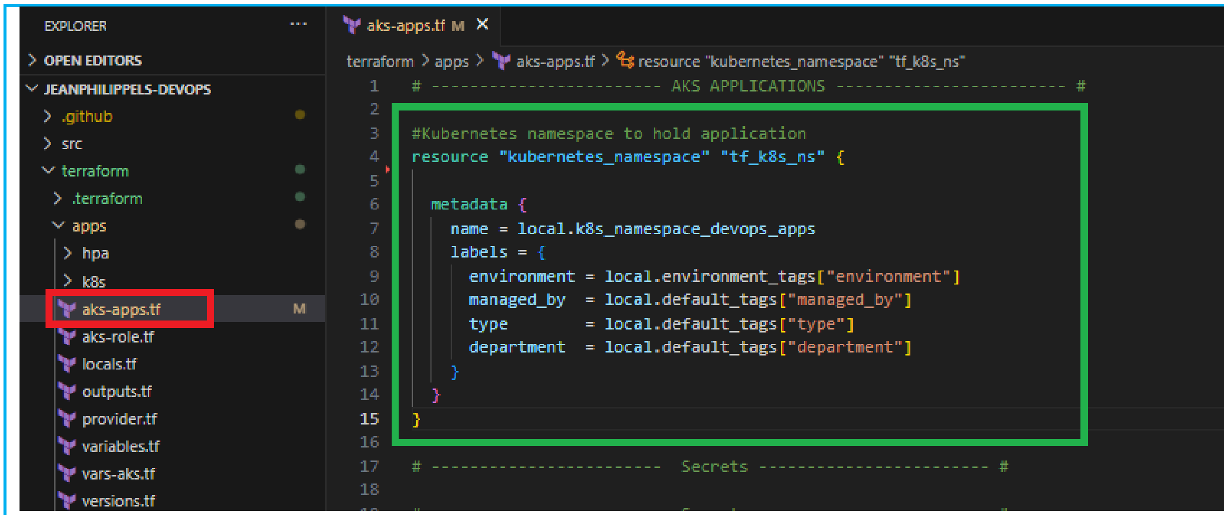


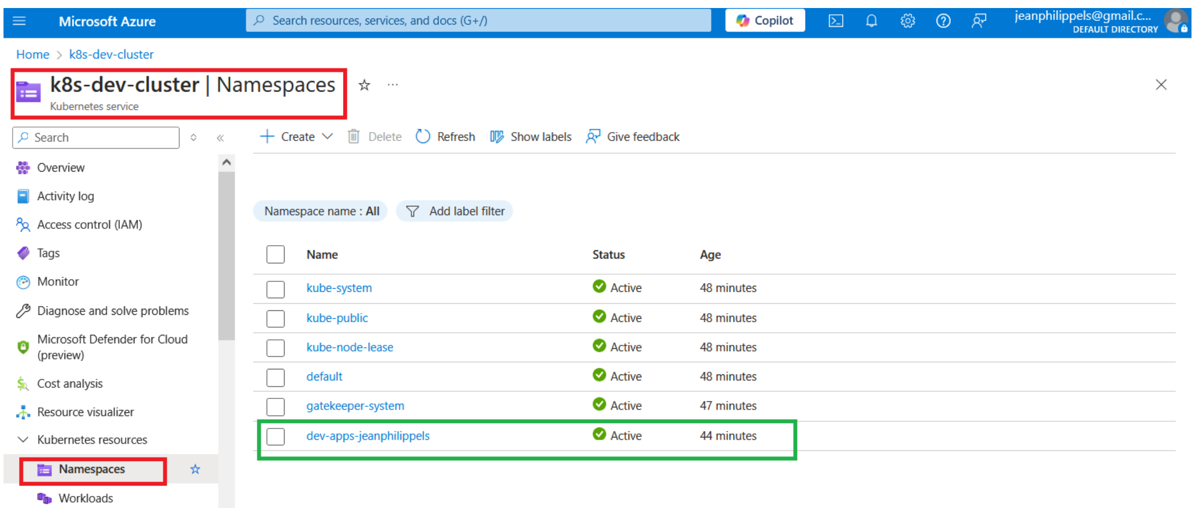
Azure Kubernetes Services relies on Virtual Machine Scale Set which a group of load balanced Virtual Machines.



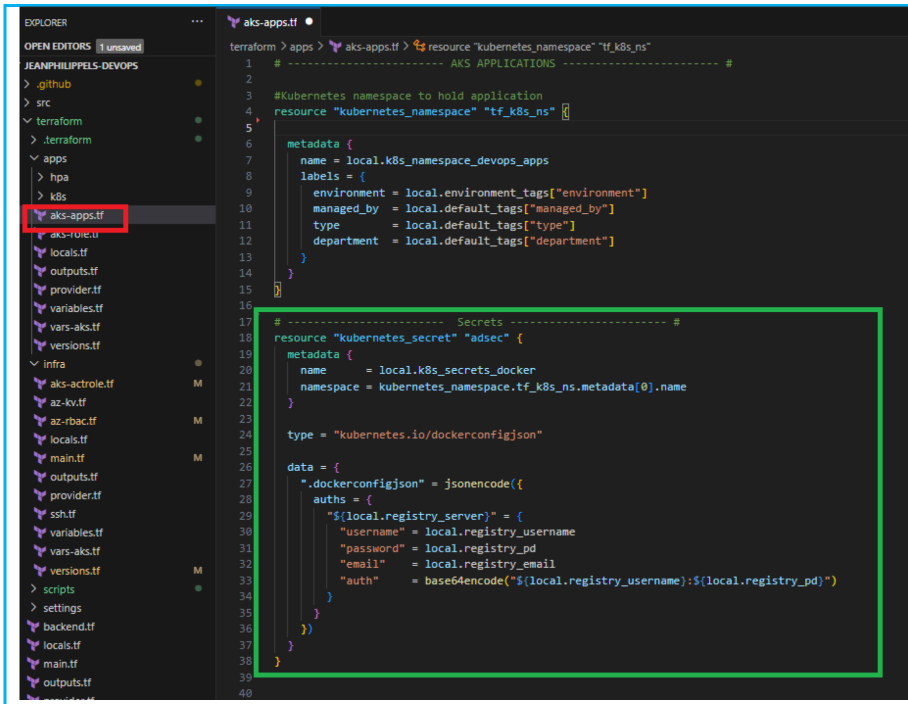


- ***Create the custom namespace for devops applications***

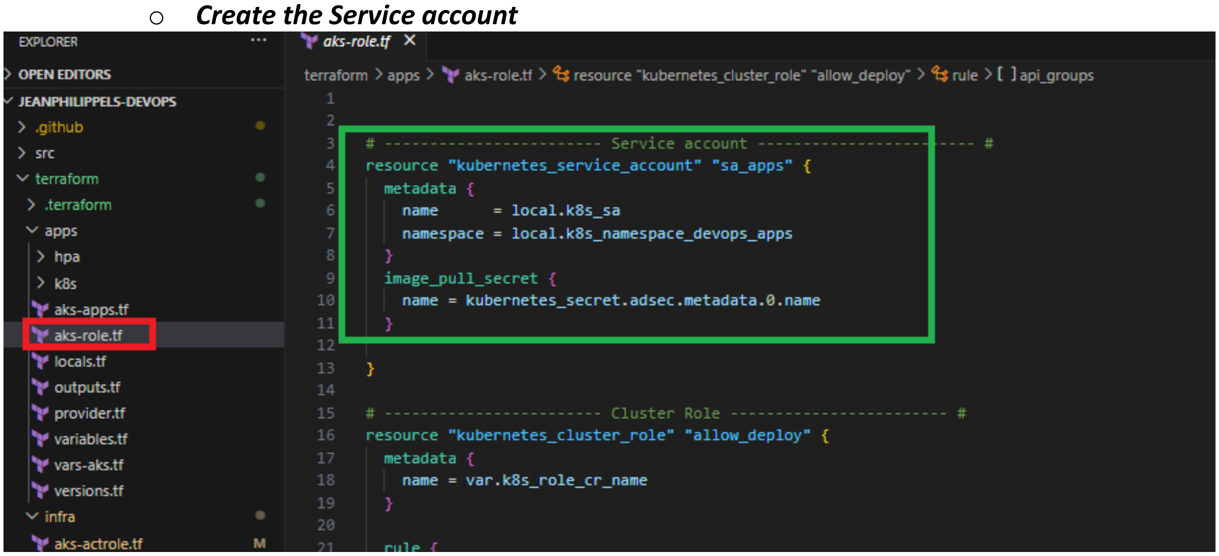




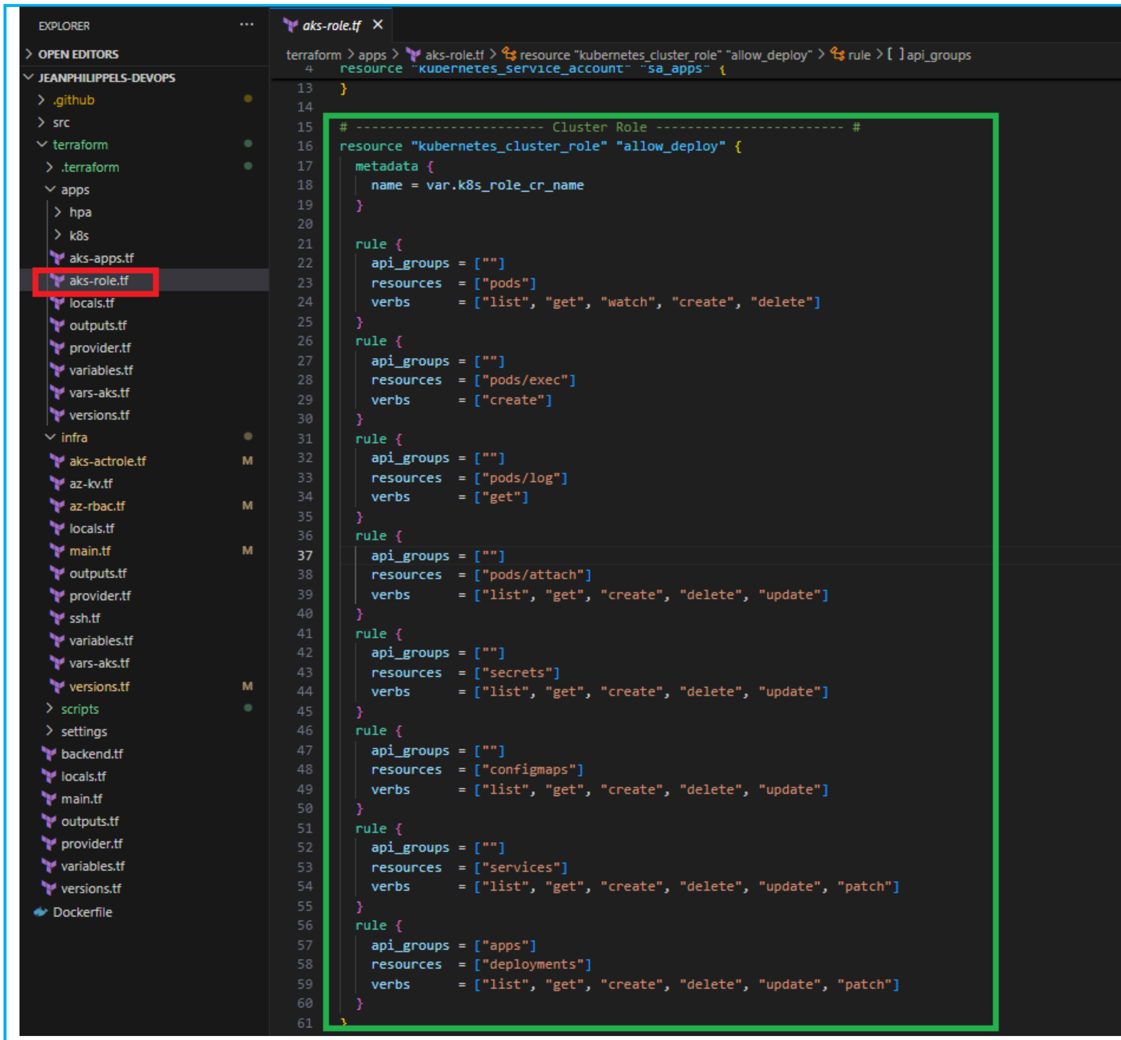
- ***Create the Secrets allowing to use private docker repository***



- *Create the Service account*

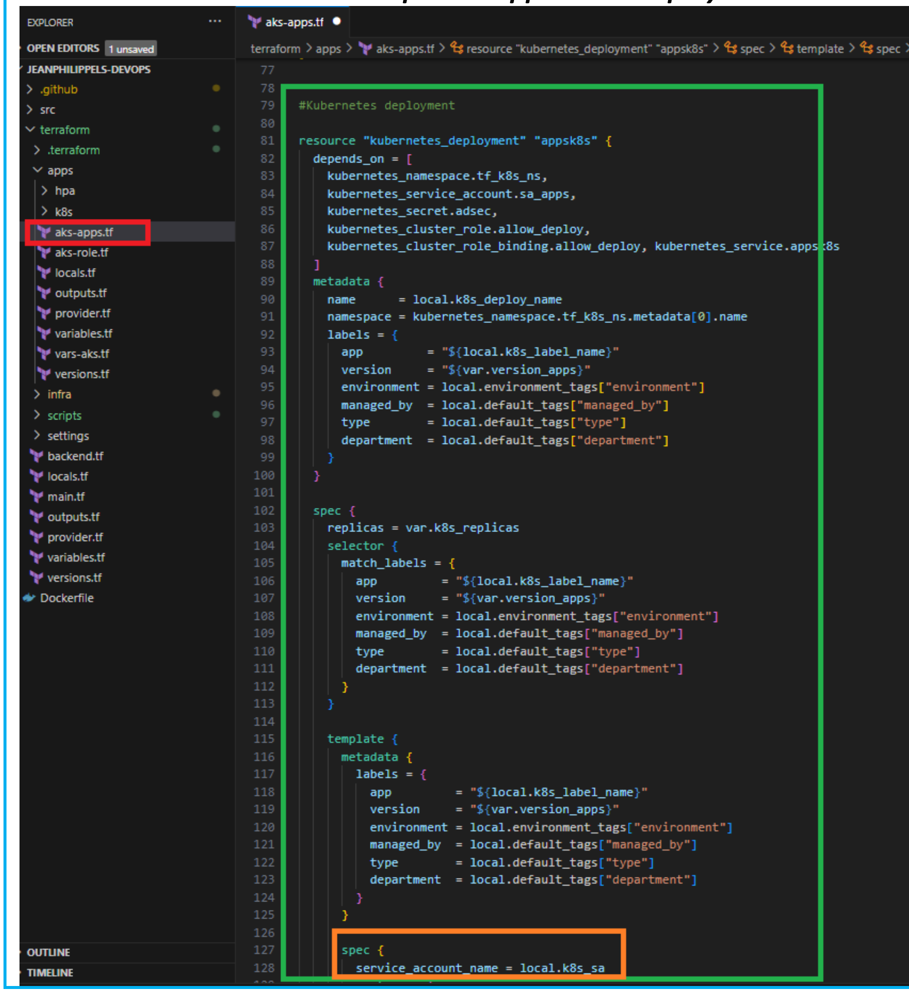


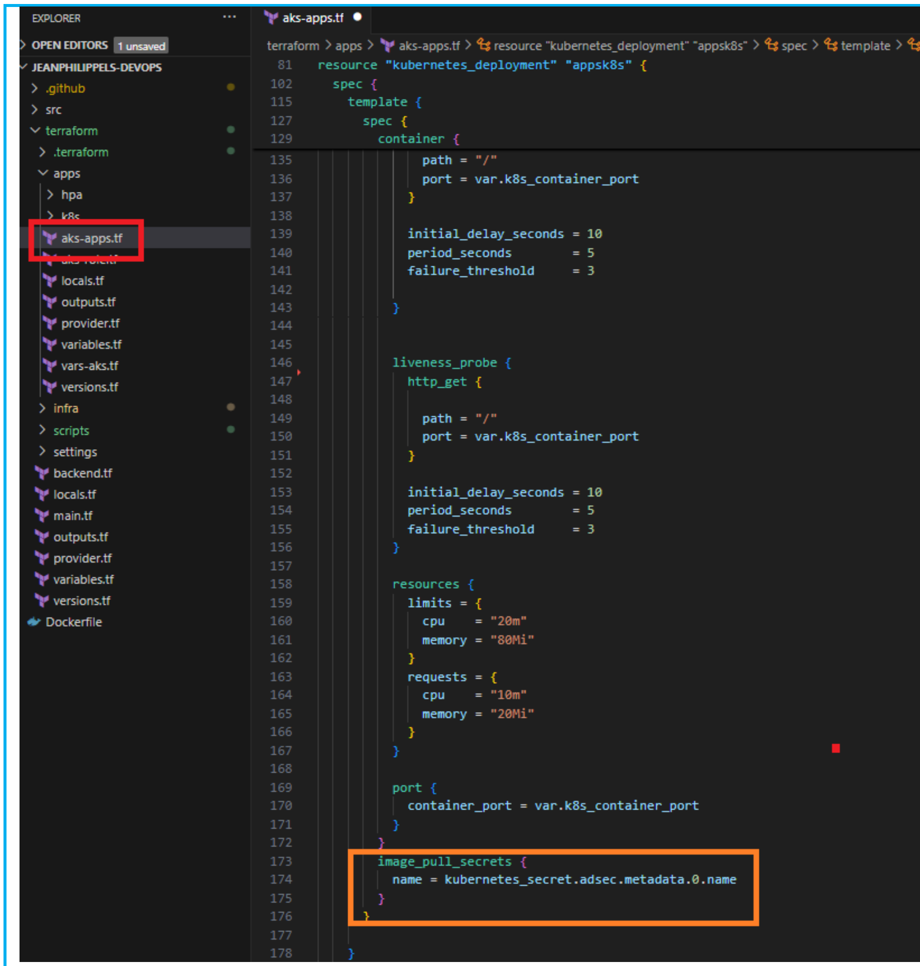
- *Create the role cluster for the service account*



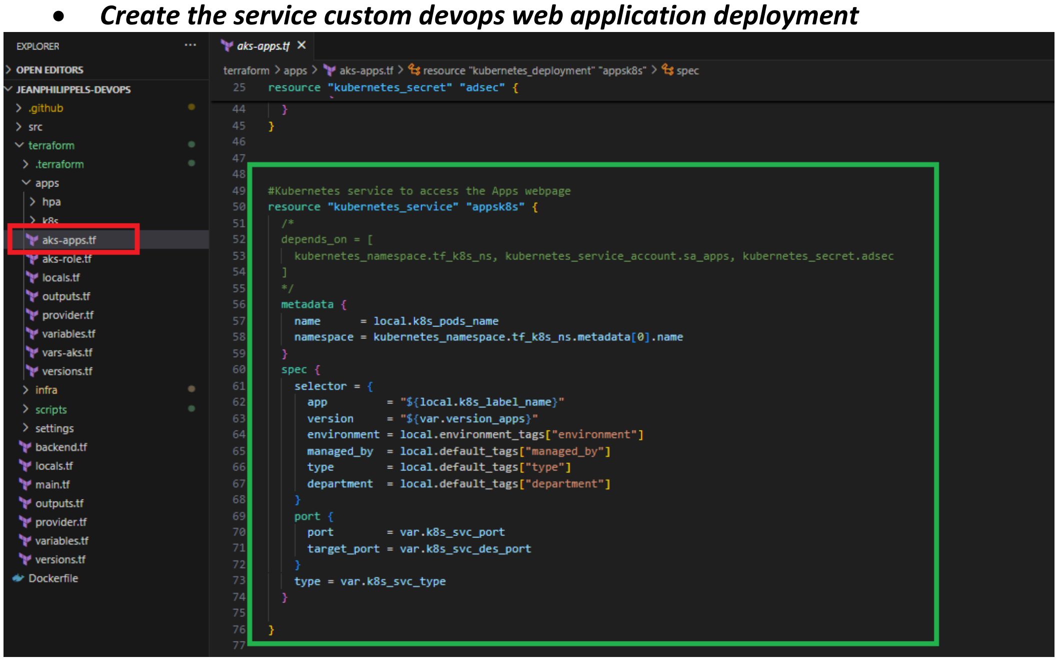
- ***Bind the role cluster to the service account***

- ***Create the custom devops web application deployment with secret from a docker hub repository***





- ***Create the service custom devops web application deployment***



- **Provision the Apps in Azure Kubernetes Services**

```
PS C:\Users\WpcmD2\Documents\GitHub\jeanphilippels-devops\terraform> terraform apply --target=module.application --var-file=settings/dev/variables.tfvars --auto-approve
Acquiring state lock. This may take a few moments...
module.architecture.random_pet.ssh_key_name: Refreshing state... [id=sshclosingoryx]
module.architecture.data.azurerm_client_config.currentsw: Reading...
module.architecture.data.azurerm_client_config.currents: Reading...
module.architecture.azurerm_resource_group.rgaks: Refreshing state... [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-k8s-jpls]
module.architecture.data.azurerm_client_config.currents: Read complete after 0s [id=Y2xpZW50Q29uZmlncy9jbGllbnRJZD00OTY0MjE3Mi03ZTE1LTQ0NDAtYTE1ZS1iZDNmZTg0Y2EyOTM7b2JqZWN0SWQ9YWViNDFkOTEtNDhmYy00ZjhjL
WIzMjAtMzMzNDZkZDM3NDcxO3N1YnNjcmlwdGlvbklkPThkZTg5YzQxLTdhZmMtNDc4OC05NjY2LTdkOTdkNmMwMmNhYjt0ZW5hbnRJZD03YmEyNmE2ZC04ZjQzLTQ0NjctYjQ0Mi02N2RmZjIyYTFmN2Q=]
module.architecture.data.azurerm_client_config.currentsw: Read complete after 0s [id=Y2xpZW50Q29uZmlncy9jbGllbnRJZD00OTY0MjE3Mi03ZTE1LTQ0NDAtYTE1ZS1iZDNmZTg0Y2EyOTM7b2JqZWN0SWQ9YWViNDFkOTEtNDhmYy00Zjhj
LWIzMjAtMzMzNDZkZDM3NDcxO3N1YnNjcmlwdGlvbklkPThkZTg5YzQxLTdhZmMtNDc4OC05NjY2LTdkOTdkNmMwMmNhYjt0ZW5hbnRJZD03YmEyNmE2ZC04ZjQzLTQ0NjctYjQ0Mi02N2RmZjIyYTFmN2Q=]
module.architecture.azuread_group.aks_admin: Refreshing state... [id=/groups/436fa8a1-831a-4ebe-8618-7afa59ab06e3]
module.architecture.azapi_resource.ssh_public_key: Refreshing state... [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-k8s-jpls/providers/Microsoft.Compute/sshPublicKeys/s
shclosingoryx]
module.architecture.azapi_resource_action.ssh_public_key_gen: Refreshing state... [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-k8s-jpls/providers/Microsoft.Compute/sshP
ublicKeys/sshclosingoryx/generateKeyPair]
module.architecture.azurerm_kubernetes_cluster.default: Refreshing state... [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-k8s-jpls/providers/Microsoft.ContainerService/m
anagedClusters/k8s-dev-cluster]
module.application.data.azurerm_kubernetes_cluster.ak8s: Reading...
module.application.data.azurerm_kubernetes_cluster.ak8s: Read complete after 5s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-k8s-jpls/providers/Microsoft.ContainerServi
ce/managedClusters/k8s-dev-cluster]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # module.application.kubernetes_cluster_role.allow_deploy will be created
  + resource "kubernetes_cluster_role" "allow_deploy" {
      + id = (known after apply)

      + metadata {
          + generation       = (known after apply)
          + name             = "CR-DEV-DEPLOY"
          + resource_version = (known after apply)
          + uid              = (known after apply)
        }

      + rule {
          + api_groups = [
              + null,
            ]
          + resources  = [
              + "pods",
```

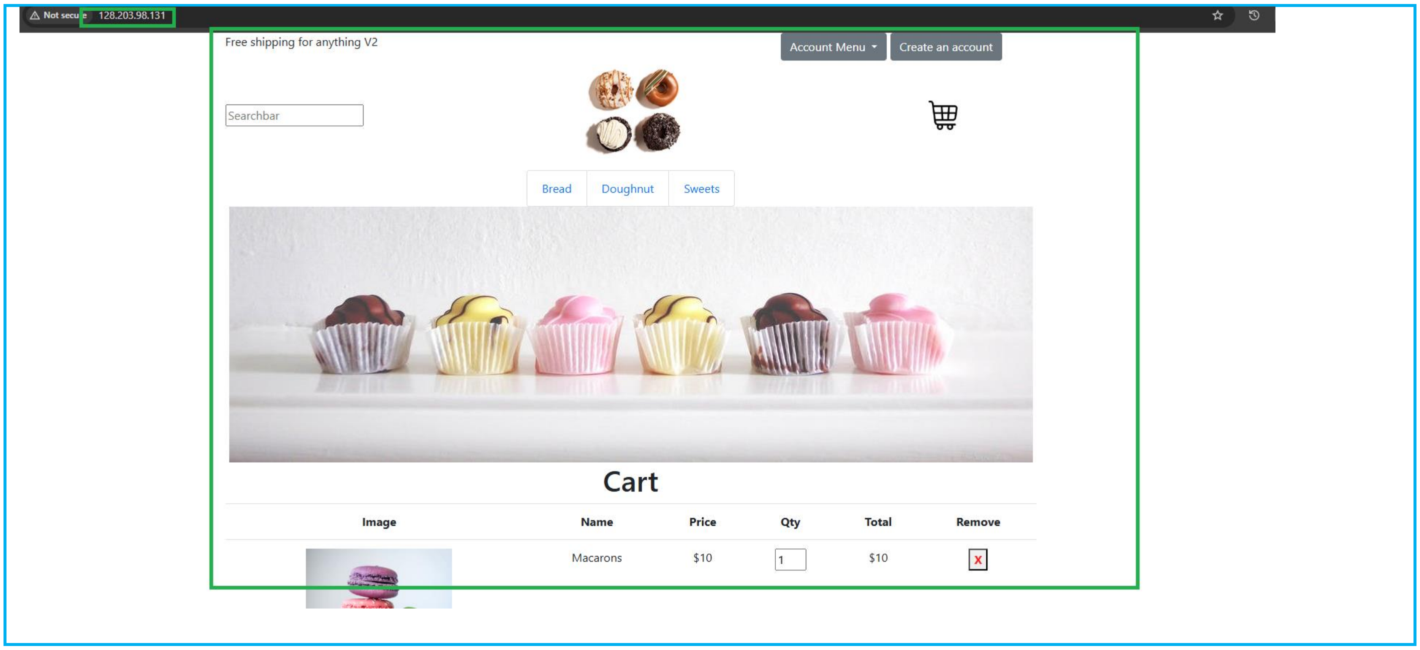
```
  # module.application.kubernetes_deployment.appsk8s will be created
  + resource "kubernetes_deployment" "appsk8s" {
      + id               = (known after apply)
      + wait_for_rollout = false

      + metadata {
          + generation       = (known after apply)
          + labels           = {
              + "app"         = "devops_dev_apps"
              + "department"  = "devops"
              + "environment" = "dev"
              + "managed_by"  = "terraform"
              + "type"        = "apps"
              + "version"     = "V1.0"
            }
          + name             = "devops-deploy-jeanphilippels"
          + namespace        = "dev-apps-jeanphilippels"
          + resource_version = (known after apply)
          + uid              = (known after apply)
        }

      + spec {
          + min_ready_seconds         = 0
          + paused                    = false
          + progress_deadline_seconds = 600
          + replicas                  = "2"
          + revision_history_limit    = 10

          + selector {
              + match_labels = {
                  + "app"         = "devops_dev_apps"
                  + "department"  = "devops"
                  + "environment" = "dev"
                  + "managed_by"  = "terraform"
                  + "type"        = "apps"
                  + "version"     = "V1.0"
                }
            }
```

```
Plan: 8 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + apps_service_ip            = (known after apply)
module.application.kubernetes_namespace.tf_k8s_ns: Creating...
module.application.kubernetes_cluster_role.allow_deploy: Creating...
module.application.kubernetes_cluster_role.allow_deploy: Creation complete after 1s [id=CR-DEV-DEPLOY]
module.application.kubernetes_namespace.tf_k8s_ns: Creation complete after 1s [id=dev-apps-jeanphilippels]
module.application.kubernetes_secret.adsec: Creating...
module.application.kubernetes_service.appsk8s: Creating...
module.application.kubernetes_secret.adsec: Creation complete after 1s [id=dev-apps-jeanphilippels/devops-dev-sec]
module.application.kubernetes_service_account.sa_apps: Creating...
module.application.kubernetes_service_account.sa_apps: Creation complete after 0s [id=dev-apps-jeanphilippels/devopsdevsa]
module.application.kubernetes_cluster_role_binding.allow_deploy: Creating...
module.application.kubernetes_cluster_role_binding.allow_deploy: Creation complete after 1s [id=CR-BIND-dev-DEPLOY]
module.application.kubernetes_service.appsk8s: Creation complete after 9s [id=dev-apps-jeanphilippels/devopsdevpods]
module.application.kubernetes_deployment.appsk8s: Creating...
module.application.kubernetes_deployment.appsk8s: Creation complete after 1s [id=dev-apps-jeanphilippels/devops-deploy-jeanphilippels]
module.application.kubernetes_horizontal_pod_autoscaler.hpajpls: Creating...
module.application.kubernetes_horizontal_pod_autoscaler.hpajpls: Creation complete after 1s [id=dev-apps-jeanphilippels/k8s-hpa-dev-jeanphilippels]

  Warning: Resource targeting is in effect

  You are creating a plan with the -target option, which means that the result of this plan may not represent all of the changes requested by the current configuration.

  The -target option is not for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an
  error message.


  Warning: Applied changes may be incomplete

  The plan was created with the -target option in effect, so some changes requested in the configuration may have been ignored and the output values may not be fully updated. Run the following command
  to verify that no other changes are pending:
      terraform plan

  Note that the -target option is not suitable for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to
  use it as part of an error message.

Releasing state lock. This may take a few moments...

Apply complete! Resources: 8 added, 0 changed, 0 destroyed.
```

- ***Testing the devops web deployment***

```
PS C:\terraform_jpls\Kubernetes_jpls> kubectl get deploy -n dev-apps-jeanphilippels
NAME                           READY   UP-TO-DATE   AVAILABLE   AGE
devops-deploy-jeanphilippels   1/1     1            1           4m19s
PS C:\terraform_jpls\Kubernetes_jpls> kubectl get svc -n dev-apps-jeanphilippels
NAME            TYPE           CLUSTER-IP     EXTERNAL-IP      PORT(S)        AGE
devopsdevpods   LoadBalancer   10.0.121.108   128.203.98.131   80:32748/TCP   29m
PS C:\terraform_jpls\Kubernetes_jpls> 
```

```
PS C:\terraform_jpls\Kubernetes_jpls> kubectl describe deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels
Name:                   devops-deploy-jeanphilippels
Namespace:              dev-apps-jeanphilippels
CreationTimestamp:      Sun, 16 Mar 2025 15:18:35 +1300
Labels:                 app=devops_dev_apps
                        department=devops
                        environment=dev
                        managed_by=terraform
                        version=V1.0
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=devops_dev_apps,department=devops,environment=dev,managed_by=terraform,version=V1.0
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:           app=devops_dev_apps
                    department=devops
                    environment=dev
                    managed_by=terraform
                    version=V1.0
  Service Account:  devopsdevsa
  Containers:
   devops-dev-apps:
    Image:      dockerlspf123/devops-jeanphilippels:V1.0
    Port:       80/TCP
    Host Port:  0/TCP
    Limits:
      cpu:     20m
      memory:  80Mi
    Requests:
      cpu:        10m
      memory:     20Mi
    Liveness:     http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3
    Readiness:    http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
  Node-Selectors: <none>
  Tolerations:    <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   devops-deploy-jeanphilippels-8657f6847d (1/1 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  4m56s deployment-controller  Scaled up replica set devops-deploy-jeanphilippels-8657f6847d from 0 to 1
PS C:\terraform_jpls\Kubernetes_jpls>
```


Not secure 128.203.98.131
Free shipping for anything V2
Account Menu
Create an account
Searchbar
Bread
Doughnut
Sweets
Cart
Image
Name
Price
Qty
Total
Remove
Macarons
$10
1
$10
X

SETTING UP CI/CD PIPELINE

GitHub Actions | Workflow Push Docker image to private repository

- **Building DevOps image with Dockerfile**
- **Building container with the DevOps image**
- **Testing container with the DevOps image**
- **Pushing DevOps image to private docker repository**

GitHub Actions is a platform that allows to do Continuous Integration and Continuous Delivery (CI/ CD) by automating build, test and deployment in pipeline

- **Building DevOps image with Dockerfile**

.github > workflows > tf-docker.yml

```
name: 2-Build and Push docker image
description: 'Build and push docker image'
on:
  workflow_run:
    workflows: ["1-Build and Test"]
    types: [completed]
    branches:
      - 'dev'
  #workflow_dispatch:

env:
  #ENV: "${{ vars.ENV }}"
  ENV: "dev"
  version: "V1.0"
  container_name: "mycaketest-jpls"

jobs:
  release-docker:
    name: Buid | Push Images to docker hub
    runs-on: ubuntu-latest
    steps:
      - name: Check out the repo
        uses: actions/checkout@v3
        with:
          ref: 'dev'

      - name: Build Docker Image
        run: docker build -t ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }} .
```

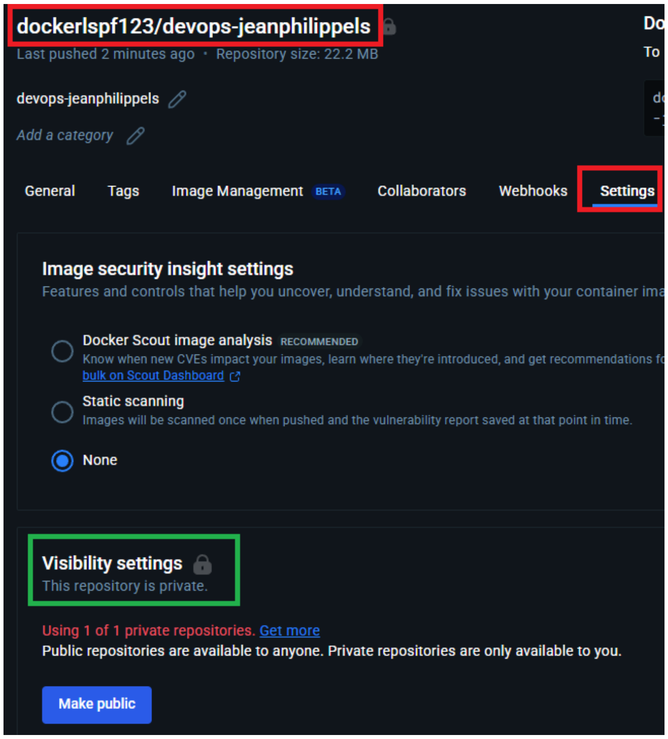
- **Building container with the DevOps image**

```
      - name: Run Docker Container
        run: docker run -d -p 80:80 --name $container_name ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}
```

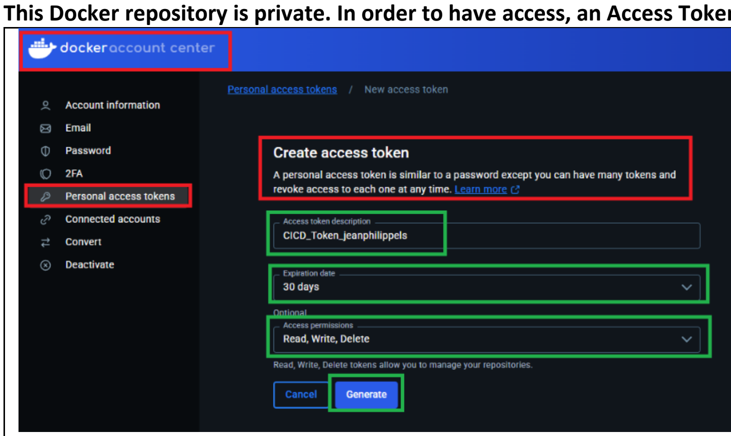
- **Testing container with the DevOps image**

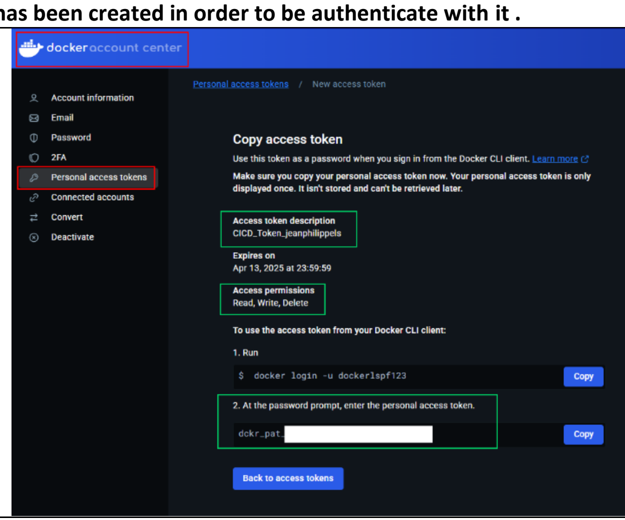
```
      - name: Test Index Page
        run: |
          sleep 10  # Allow container to start
          curl --fail http://localhost:80/
```

- Pushing images to private docker repository

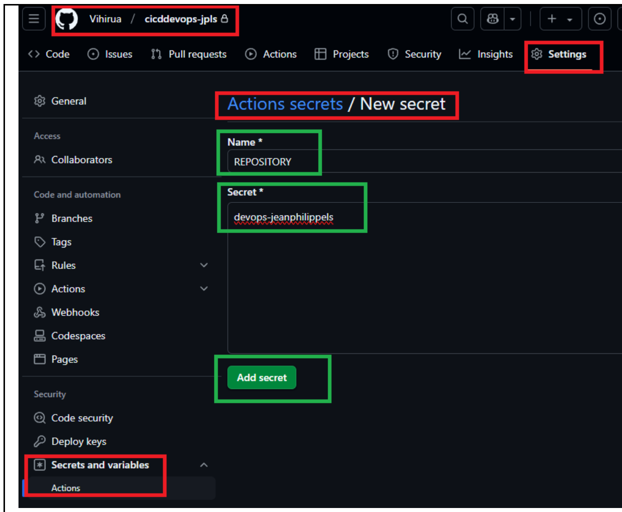


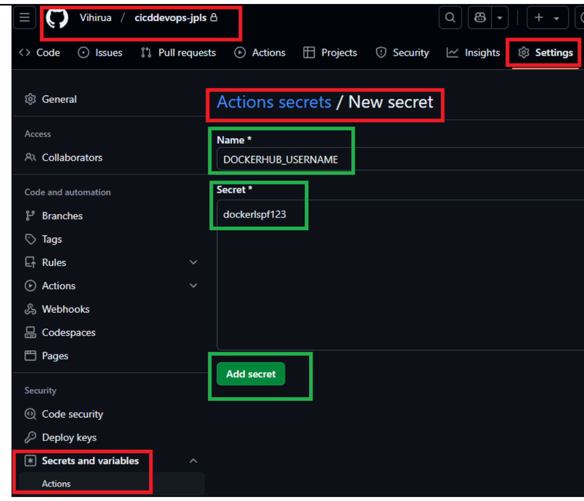
This Docker repository is private. In order to have access, an Access Token has been created in order to be authenticate with it .

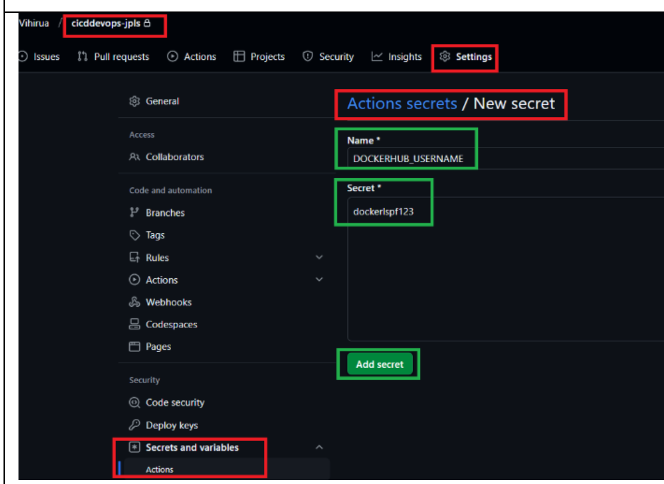


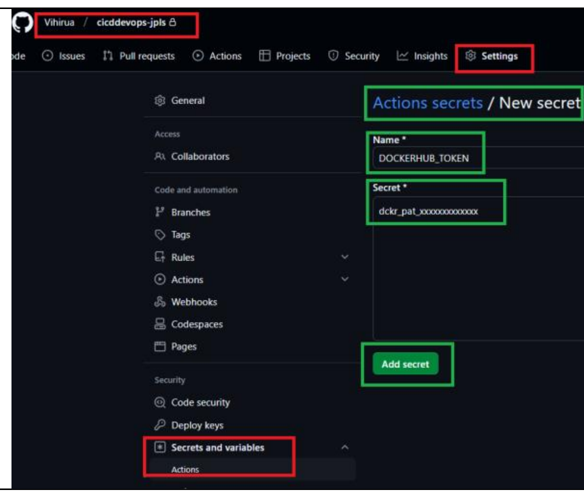


Secrets have been created in GitHub to use for GitHub actions:







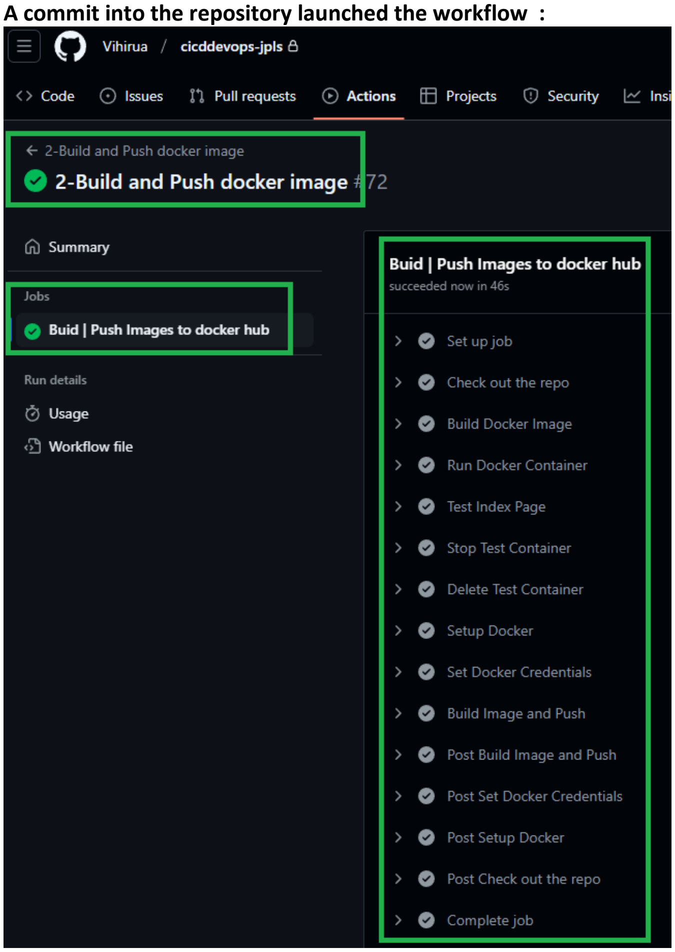


Workflow on GitHub for Building and Pushing Docker image into the private docker hub registry :

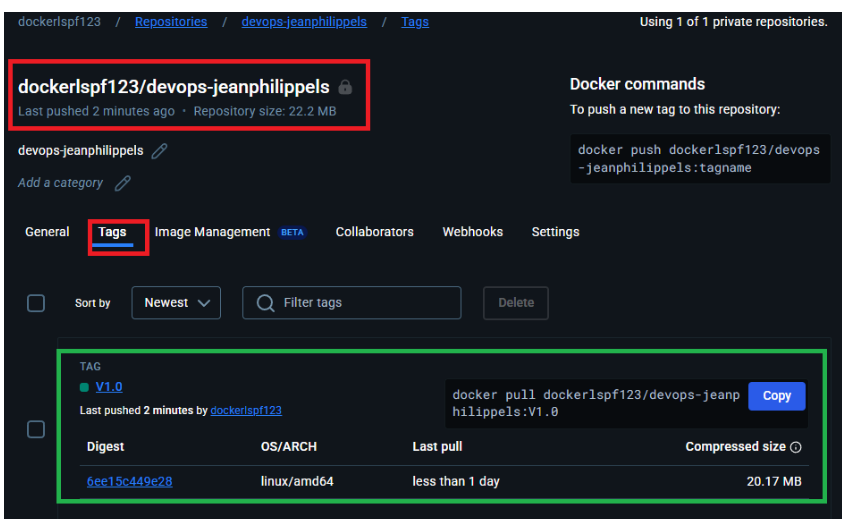
.github > workflows > tf-docker.yml

```yaml
name: 2-Build and Push docker image
description: 'Build and push docker image'
on:
  workflow_run:
    workflows: ["1-Build and Test"]
    types: [completed]
    branches:
      - 'dev'
  #workflow_dispatch:

env:
  #ENV: "${{ vars.ENV }}"
  ENV: "dev"
  version: "V1.0"
  container_name: "mycaketest-jpls"

jobs:
  release-docker:
    name: Buid | Push Images to docker hub
    runs-on: ubuntu-latest
    steps:
      - name: Check out the repo
        uses: actions/checkout@v3
        with:
          ref: 'dev'

      - name: Build Docker Image
        run: docker build -t ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }} .

      - name: Run Docker Container
        run: docker run -d -p 80:80 --name $container_name ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}

      - name: Test Index Page
        run: |
          sleep 10  # Allow container to start
          curl --fail http://localhost:80/
      - name: Stop Test Container
        run: |
          sleep 5  # Allow container to stop
          docker stop $(docker ps -q)
      - name: Delete Test Container
        run: |
          sleep 5  # Allow container to be deleted
          docker rm $(docker ps -a -q)
      - name: Setup Docker
        uses: docker/setup-buildx-action@v2

      - name: Set Docker Credentials
        uses: docker/login-action@v2
        with:
          username: ${{ secrets.DOCKERHUB_USERNAME }}
          password: ${{ secrets.DOCKERHUB_TOKEN }}
      - name: Build Image and Push
        id: build-push-api
        uses: docker/build-push-action@v3
        with:
          context: .
          file: Dockerfile
          push: true
          tags: ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}

```

A commit into the repository launched the workflow :



After the workflow has been successfully ran, the new image has been push into the private docker repository :

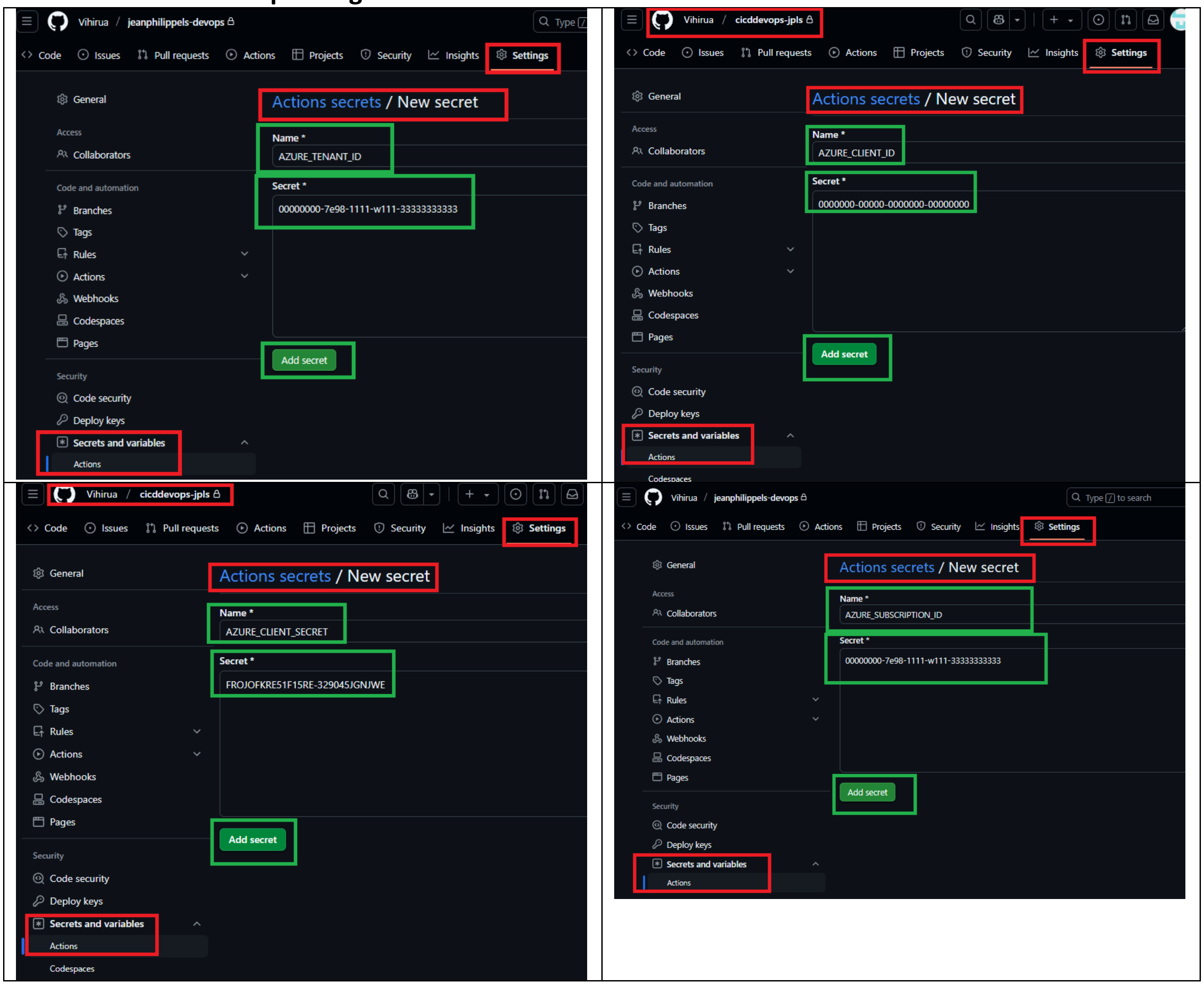


GitHub Actions | Workflow Terraform

Workflow Unit Test

When pushing code to the branch, this workflow will analyze if the scripts are well formed and respond to the format.

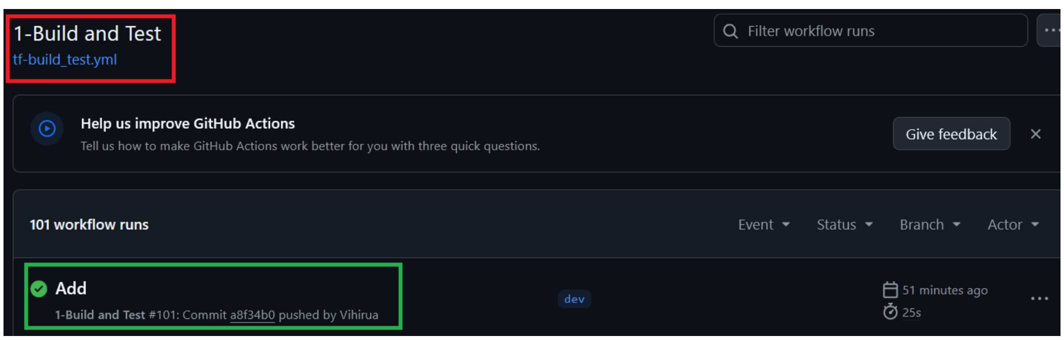
- **Test unit when pushing into branches**

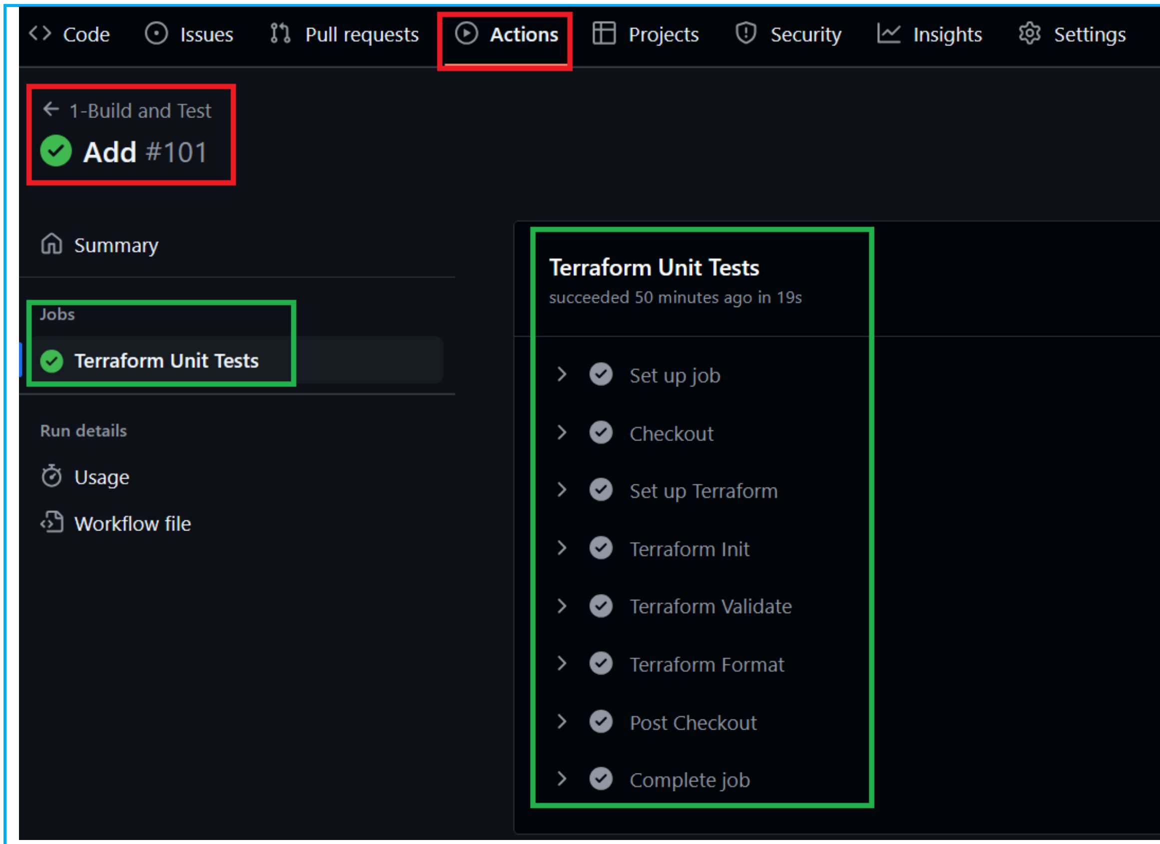


.github > workflows > tf-build_test.yml

```yaml
name: 1-UnitTest
description: 'Build and Test code'
on:
  push:
    branches:
      - 'dev'
    paths:
      - 'terraform/**'
      - '.github/workflows/tf-build_test.yml'
concurrency:
  group: ${{ github.workflow }}-${{ github.ref }}  # Grouping to ensure only one job per branch
  cancel-in-progress: true
env:
  ARM_CLIENT_ID: ${{ secrets.AZURE_CLIENT_ID }}
  ARM_CLIENT_SECRET: ${{ secrets.AZURE_CLIENT_SECRET }}
  ARM_SUBSCRIPTION_ID: ${{ secrets.AZURE_SUBSCRIPTION_ID }}
  ARM_TENANT_ID: ${{ secrets.AZURE_TENANT_ID }}
  #ENV: "${{ vars.ENV }}"
  ENV: "dev"
  CONF_FILE: "backend-unittest.conf"
  BACKEND_SCRIPT: "backend-unit-test"
  BACKEND_RM_SCRIPT: "backend-unit-test-rm"
  WORKING_DIR: "./terraform"
  WORKING_SCRIPTS_DIR: "scripts"
jobs:
  terraform-unit-tests:
    name: "Terraform Unit Tests"
    runs-on: ubuntu-latest
    concurrency:
      group: dev_environment
      cancel-in-progress: true
    environment: development
    defaults:
      run:
        shell: bash
        working-directory: ${{ env.WORKING_DIR}}
    steps:
      # Checkout the repository
      - name: Checkout
        uses: actions/checkout@v3
      - name: Initialization of the Backend on Azure
        id: init-backend
        shell: bash
        run: |
          chmod +x ./${{ env.WORKING_SCRIPTS_DIR}}/${{ env.BACKEND_SCRIPT}}.sh
          ./${{ env.WORKING_SCRIPTS_DIR}}/${{ env.BACKEND_SCRIPT}}.sh
        env:
          AZURE_CLIENT_ID: ${{ secrets.AZURE_CLIENT_ID }}
          AZURE_CLIENT_SECRET: ${{ secrets.AZURE_CLIENT_SECRET }}
          AZURE_TENANT_ID: ${{ secrets.AZURE_TENANT_ID }}
     # Install Terraform - the latest version
```

```yaml
      # Install Terraform - the latest version
      - name: Set up Terraform
        uses: hashicorp/setup-terraform@v3
        with:
          terraform_version: 1.6.0 # Specify  Terraform version
      # Initialize Terraform
      - name: Terraform Init
        id: tf-init
        run: terraform init -reconfigure -backend-config=settings/${ENV}/${CONF_FILE}
      - name: Destroy  Backend on Azure
        id: destroy-backend
        shell: bash
        run: |
          chmod +x ./${{ env.WORKING_SCRIPTS_DIR}}/${{ env.BACKEND_RM_SCRIPT}}.sh
          ./${{ env.WORKING_SCRIPTS_DIR}}/${{ env.BACKEND_RM_SCRIPT}}.sh
        env:
          AZURE_CLIENT_ID: ${{ secrets.AZURE_CLIENT_ID }}
          AZURE_CLIENT_SECRET: ${{ secrets.AZURE_CLIENT_SECRET }}
          AZURE_TENANT_ID: ${{ secrets.AZURE_TENANT_ID }}
      # Validate terraform files
      - name: Terraform Validate
        run: terraform validate

      # Checks that all Terraform configuration files adhere to a canonical format
      - name: Terraform Format
        run: terraform fmt -check -recursive

```

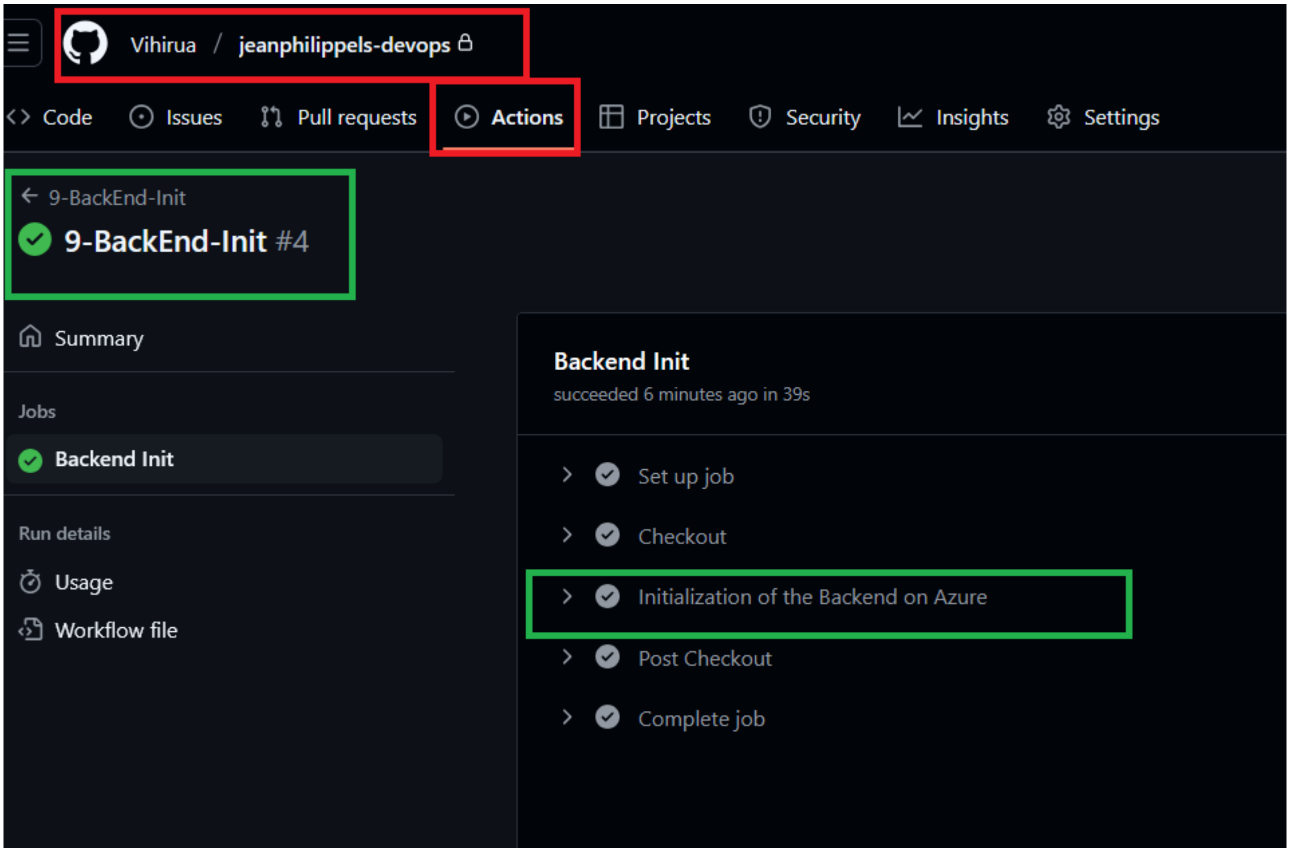


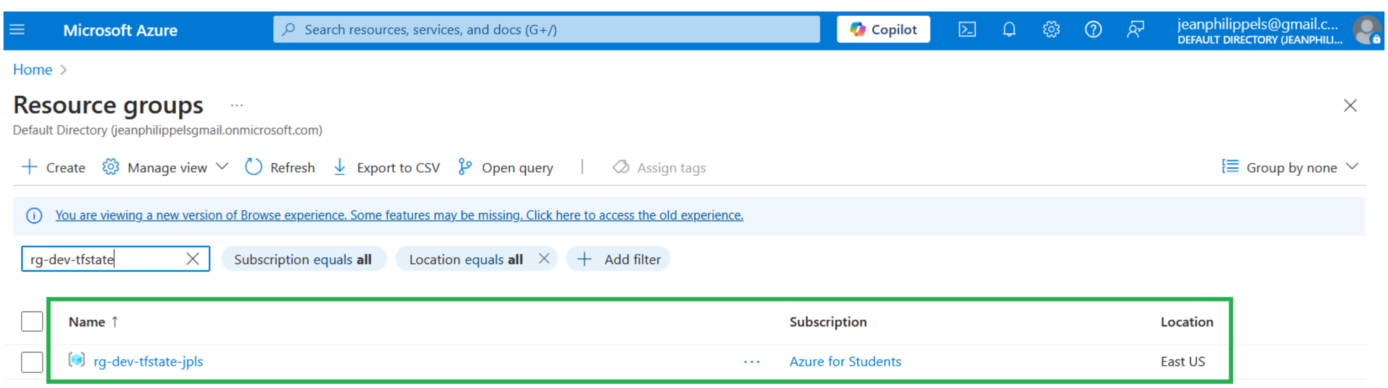

Code
Issues
Pull requests
Actions
Projects
Security
Insights
Settings
1-Build and Test
Add #101
Summary
Jobs
Terraform Unit Tests
Run details
Usage
Workflow file
Terraform Unit Tests
succeeded 50 minutes ago in 19s
Set up job
Checkout
Set up Terraform
Terraform Init
Terraform Validate
Terraform Format
Post Checkout
Complete job

Workflow Init Backend

This workflow provisions the backend to use on Azure allowing to test manually module with terraform if teams want to .

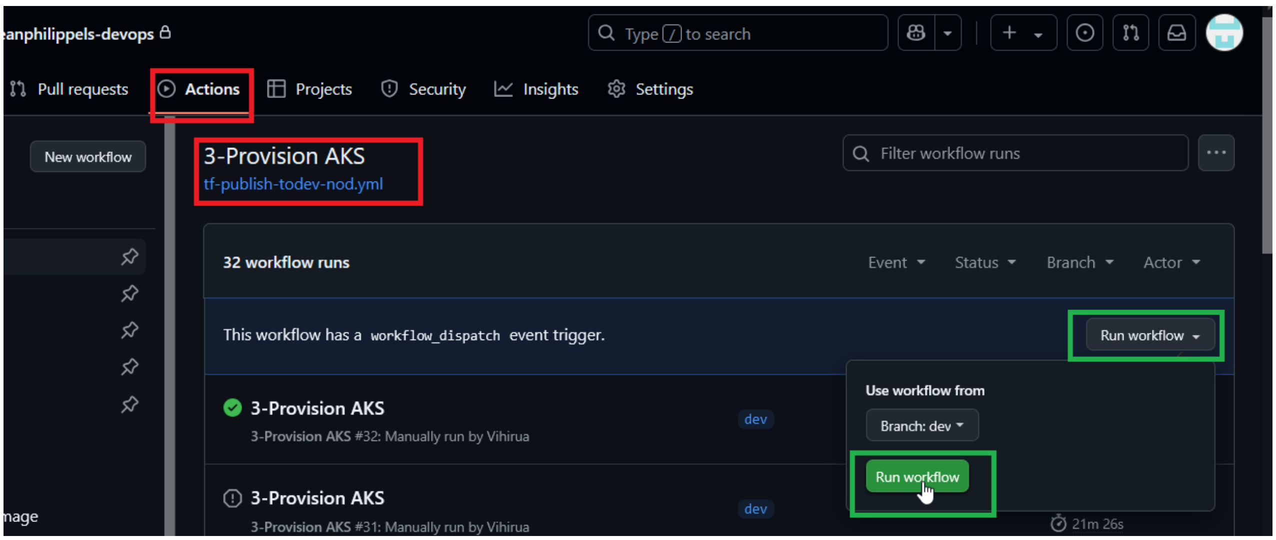
- ***Shell to create resource group and container for the backend***

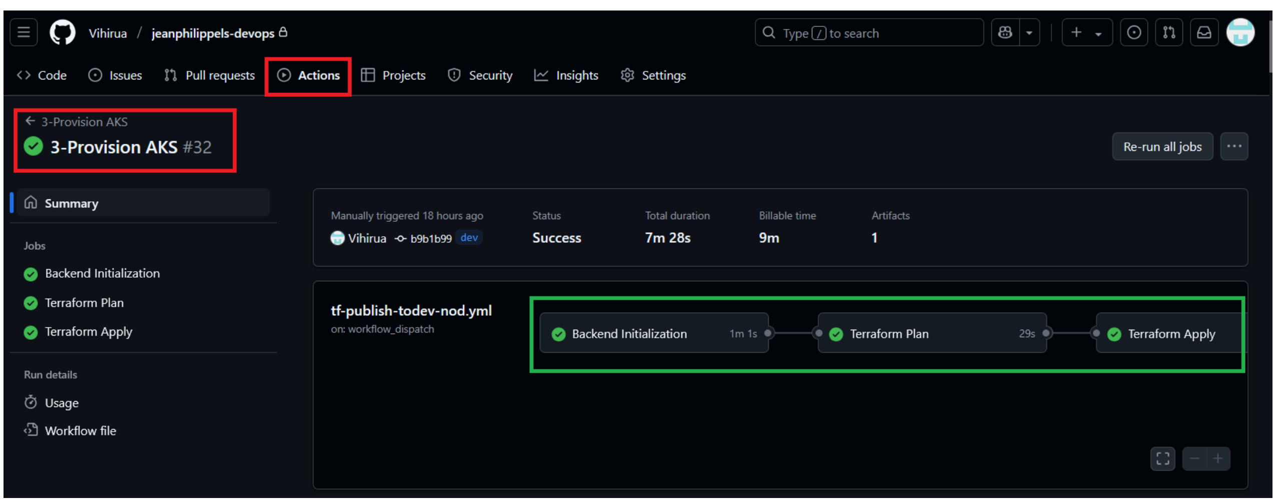


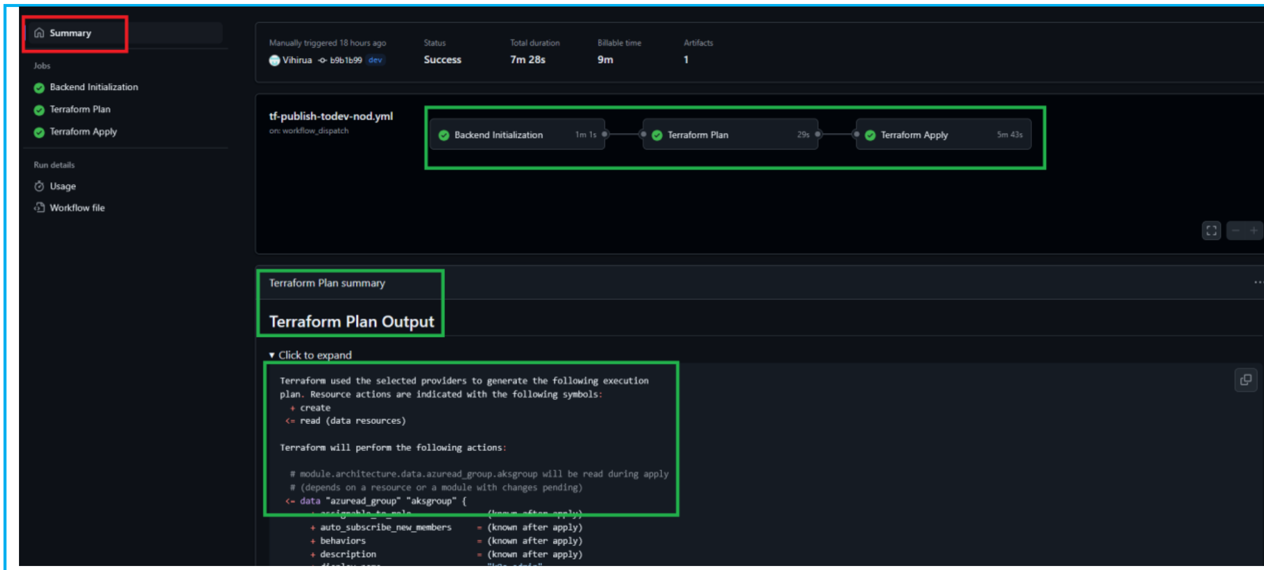


Workflow Azure Kubernetes Services

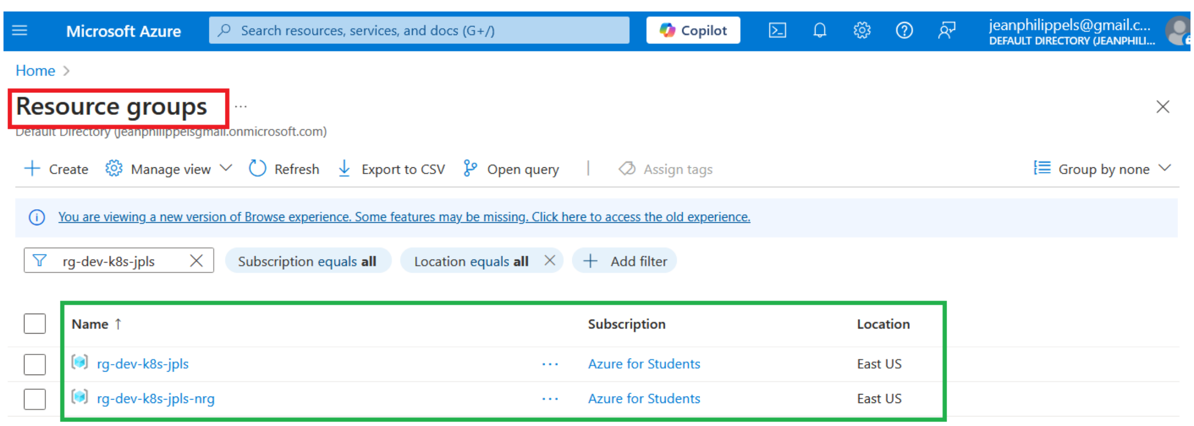
This workflow provisions Azure Kubernetes Services with the backend and the Azure Kubernetes Services

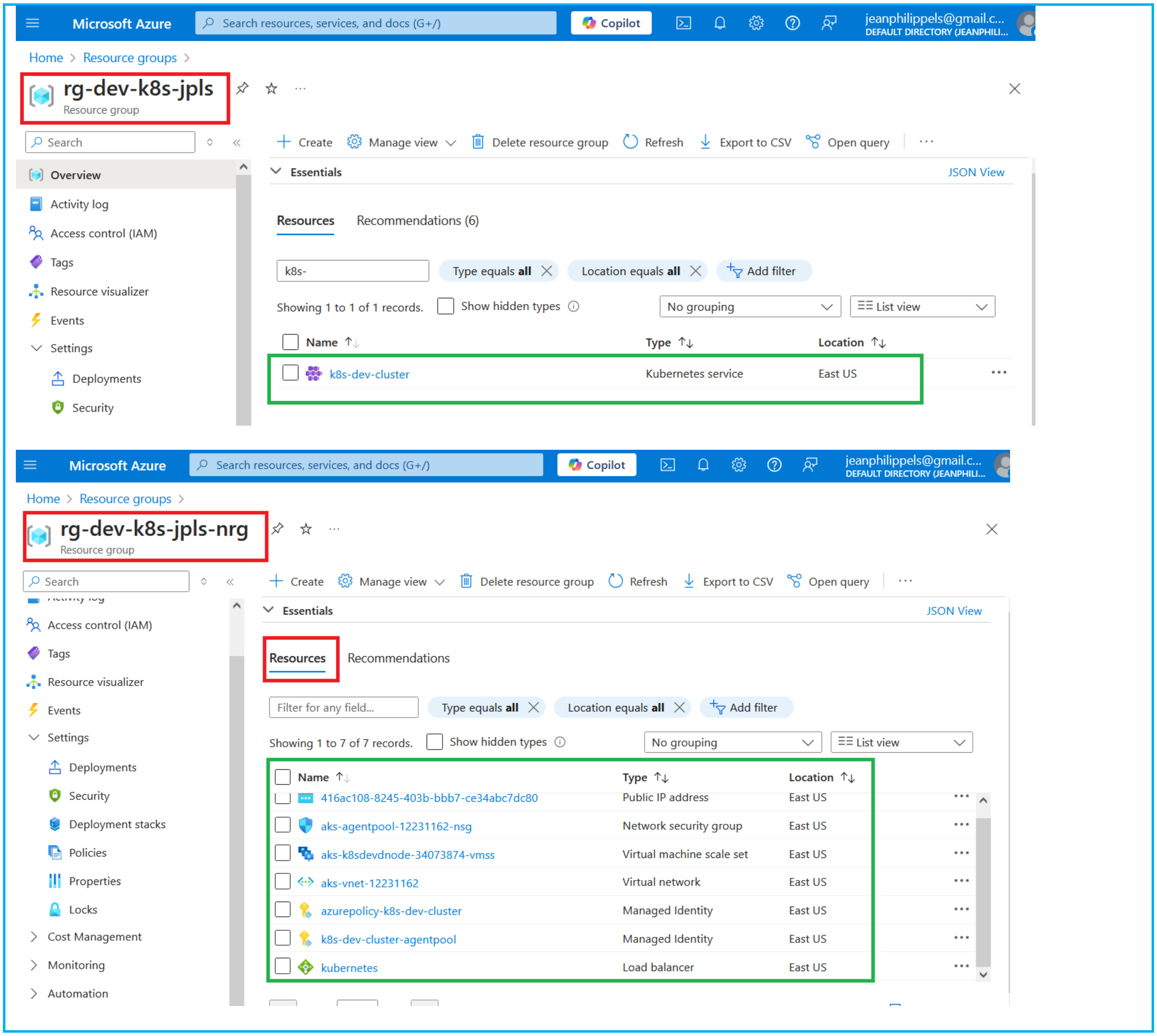





Summary
Jobs
Backend Initialization
Terraform Plan
Terraform Apply
Run details
Usage
Workflow file
Manually triggered 18 hours ago
Status
Success
Total duration
7m 28s
Billable time
9m
Artifacts
1
tf-publish-todev-nod.yml
Backend Initialization
Terraform Plan
Terraform Apply
Terraform Plan summary
Terraform Plan Output
Click to expand
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
<= read (data resources)
Terraform will perform the following actions:

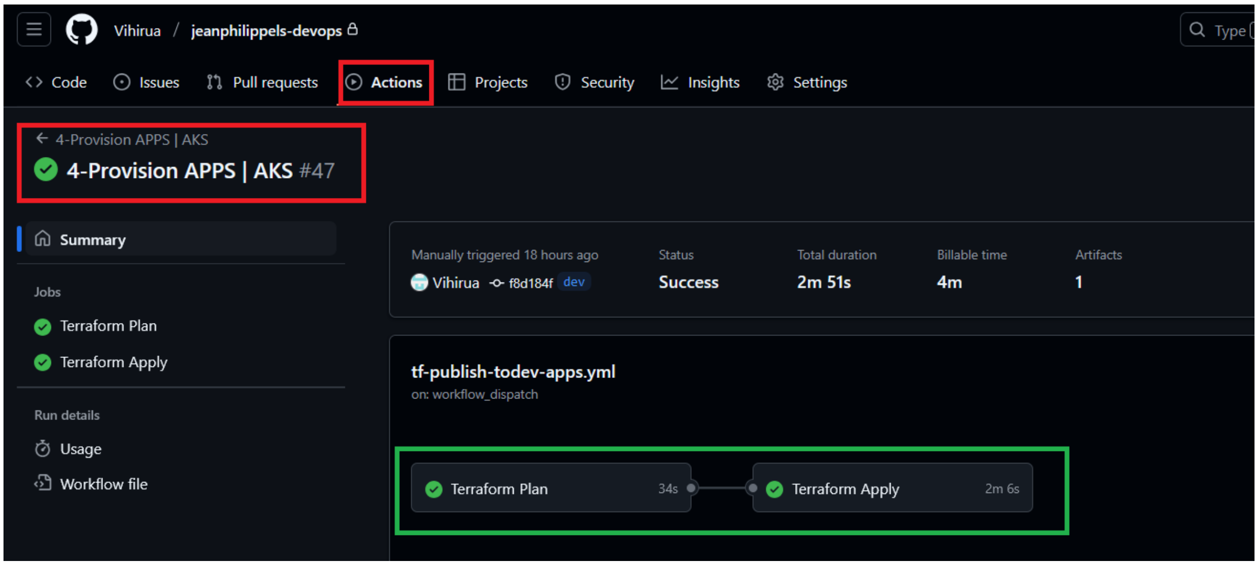

Summary
Backend Initialization
Terraform Plan
Terraform Apply
Usage
Workflow file
# module.architecture.local_file.kubeconfig will be created
+ resource "local_file" "kubeconfig" {
# module.architecture.random_pet.ssh_key_name will be created
+ resource "random_pet" "ssh_key_name" {
Plan: 16 to add, 0 to change, 0 to destroy.
Changes to Outputs:

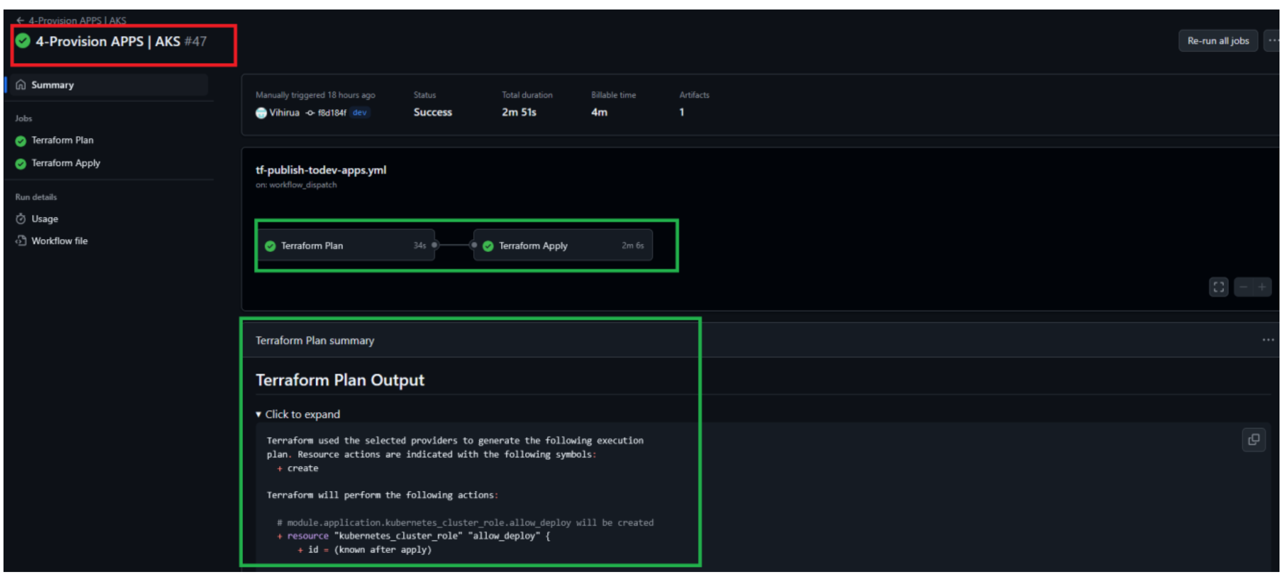

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Home
Resource groups
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Group by none
You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.
rg-dev-k8s-jpls
Subscription equals all
Location equals all
Add filter
Name
Subscription
Location
rg-dev-k8s-jpls
Azure for Students
East US
rg-dev-k8s-jpls-nrg
Azure for Students
East US

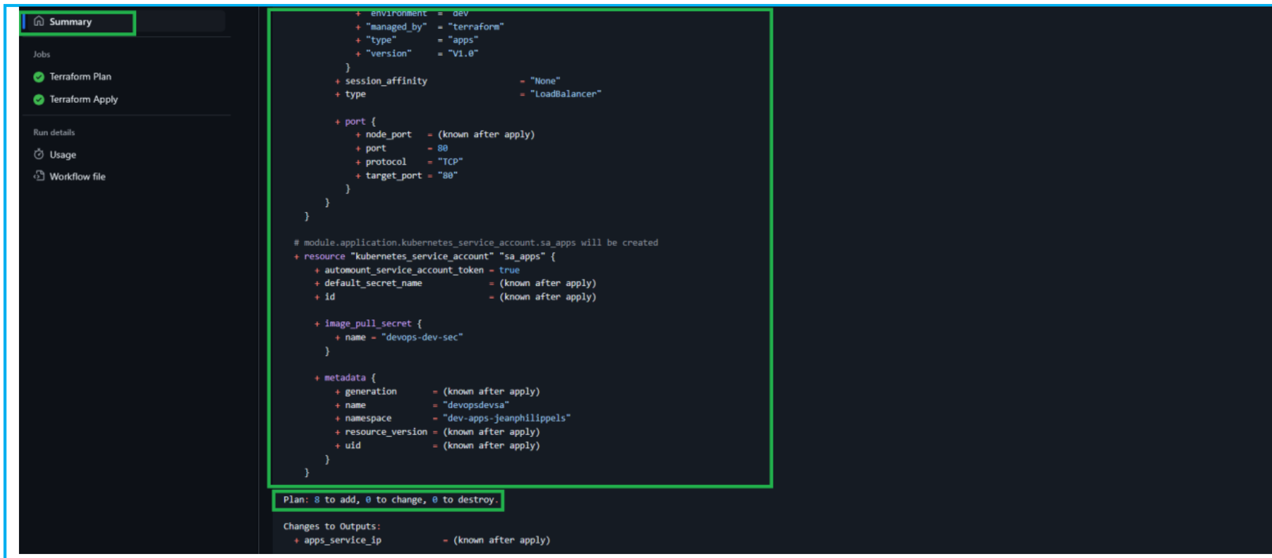

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
jeanphilippels@gmail.c...
DEFAULT DIRECTORY (JEANPHILI...
Home > Resource groups >
rg-dev-k8s-jpls
Resource group
Search
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Deployments
Security
Create
Manage view
Delete resource group
Refresh
Export to CSV
Open query
Essentials
JSON View
Resources
Recommendations (6)
k8s-
Type equals all
Location equals all
Add filter
Showing 1 to 1 of 1 records.
Show hidden types
No grouping
List view
Name
Type
Location
k8s-dev-cluster
Kubernetes service
East US
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
jeanphilippels@gmail.c...
DEFAULT DIRECTORY (JEANPHILI...
Home > Resource groups >
rg-dev-k8s-jpls-nrg
Resource group
Search
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Deployments
Security
Deployment stacks
Policies
Properties
Locks
Cost Management
Monitoring
Automation
Create
Manage view
Delete resource group
Refresh
Export to CSV
Open query
Essentials
JSON View
Resources
Recommendations
Filter for any field...
Type equals all
Location equals all
Add filter
Showing 1 to 7 of 7 records.
Show hidden types
No grouping
List view
Name
Type
Location
416ac108-8245-403b-bbb7-ce34abc7dc80
Public IP address
East US
aks-agentpool-12231162-nsg
Network security group
East US
aks-k8sdevdnode-34073874-vmss
Virtual machine scale set
East US
aks-vnet-12231162
Virtual network
East US
azurepolicy-k8s-dev-cluster
Managed Identity
East US
k8s-dev-cluster-agentpool
Managed Identity
East US
kubernetes
Load balancer
East US

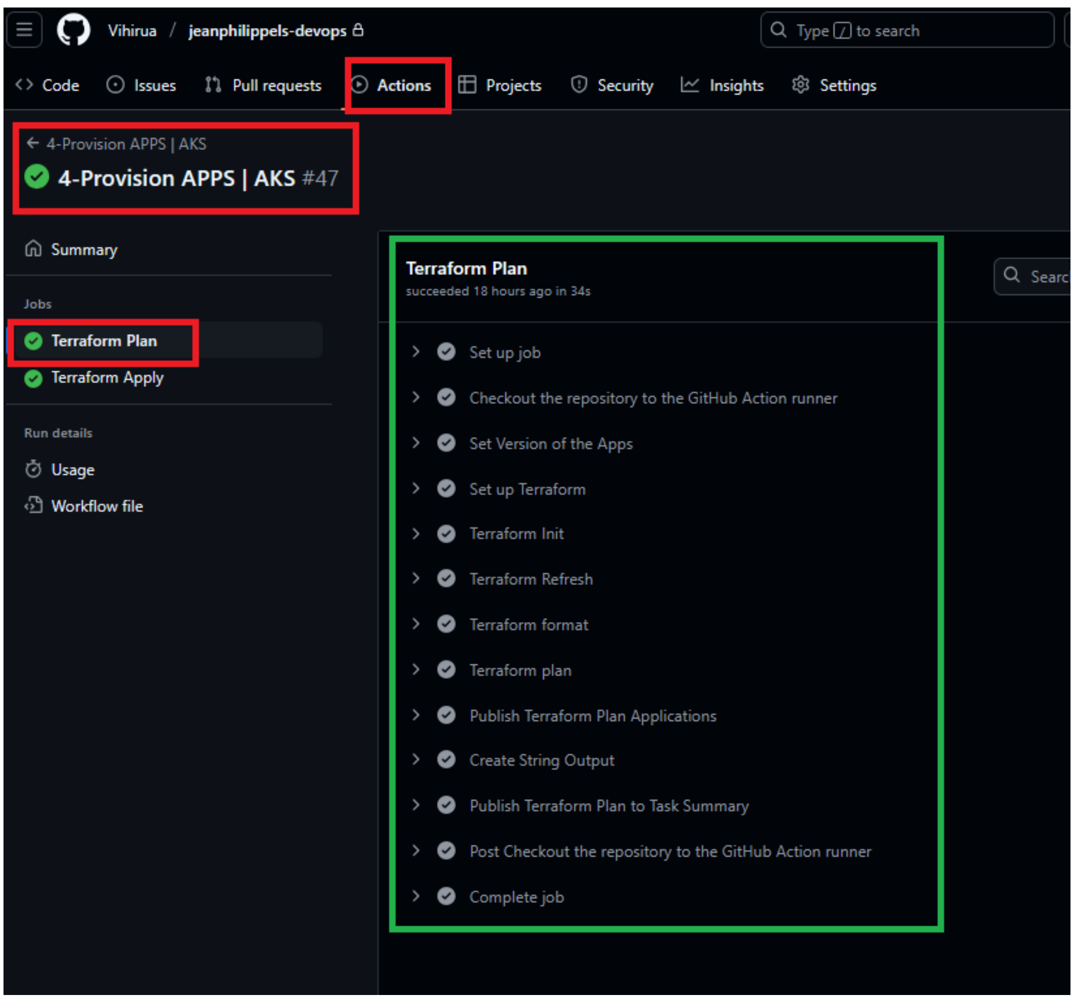
Workflow Applications to Azure Kubernetes Services

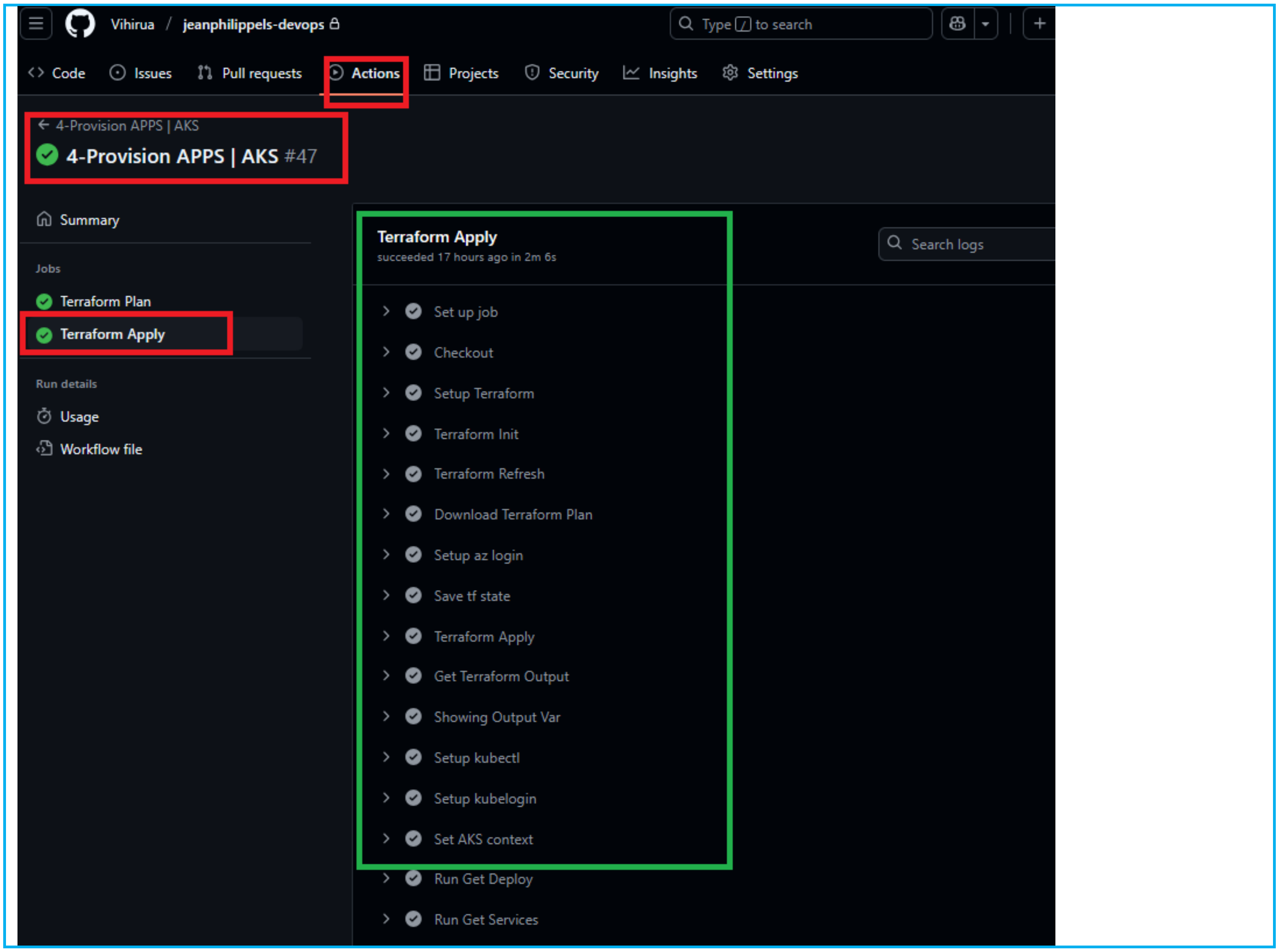
This workflow provisions the applications into the Azure Kubernetes Services





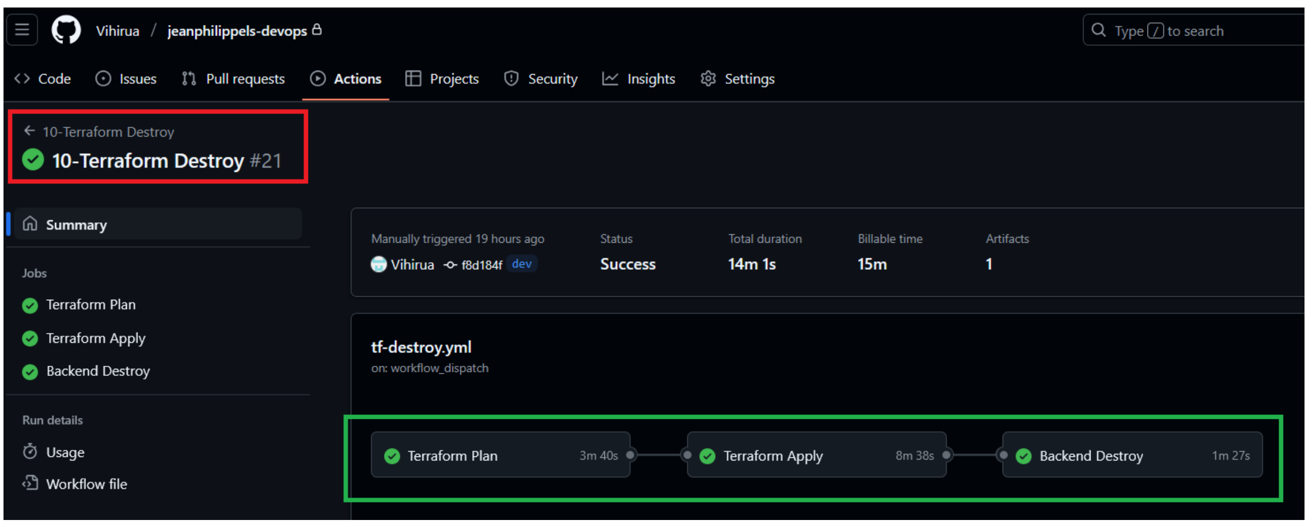

Summary
Jobs
Terraform Plan
Terraform Apply
Run details
Usage
Workflow file
+ "managed_by" = "terraform"
+ "type" = "apps"
+ "version" = "V1.0"
}
+ session_affinity = "None"
+ type = "LoadBalancer"
+ port {
+ node_port = (known after apply)
+ port = 80
+ protocol = "TCP"
+ target_port = "80"
}
}
}
# module.application.kubernetes_service_account.sa_apps will be created
+ resource "kubernetes_service_account" "sa_apps" {
+ automount_service_account_token = true
+ default_secret_name = (known after apply)
+ id = (known after apply)
+ image_pull_secret {
+ name = "devops-dev-sec"
}
+ metadata {
+ generation = (known after apply)
+ name = "devopsdevsa"
+ namespace = "dev-apps-jeanphilippels"
+ resource_version = (known after apply)
+ uid = (known after apply)
}
}
Plan: 8 to add, 0 to change, 0 to destroy.
Changes to Outputs:
+ apps_service_ip = (known after apply)

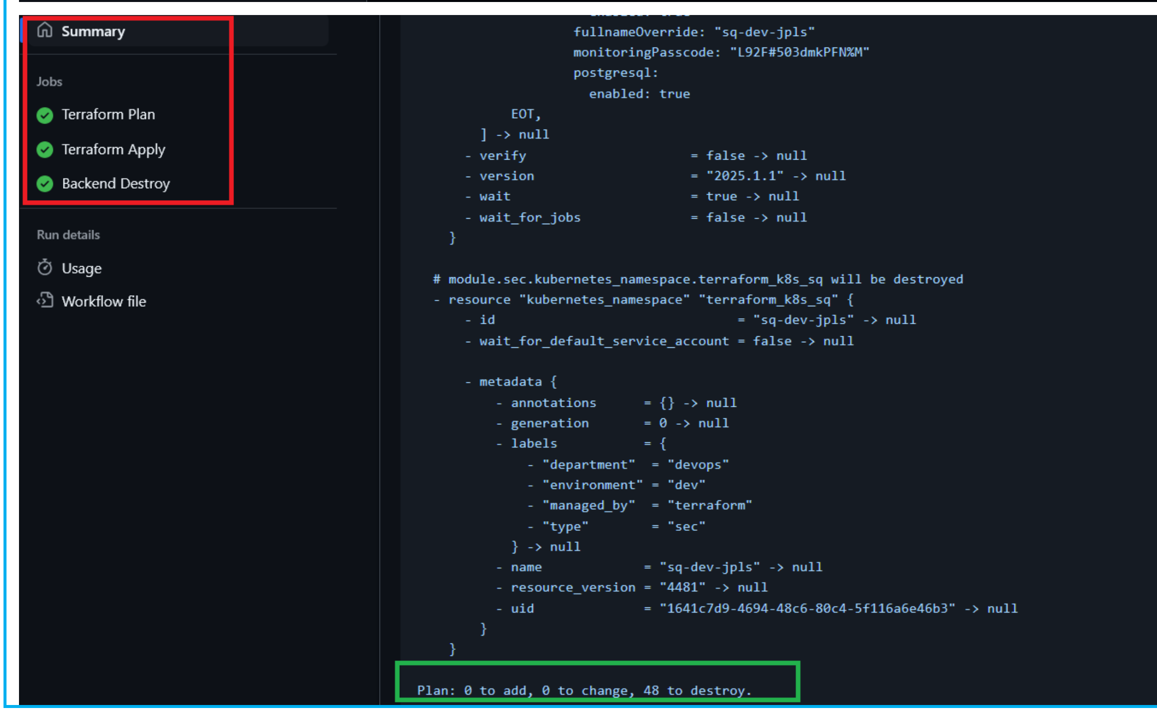

Vihirua / jeanphilippels-devops
Type / to search
Code
Issues
Pull requests
Actions
Projects
Security
Insights
Settings
4-Provision APPS | AKS
4-Provision APPS | AKS #47
Summary
Jobs
Terraform Plan
Terraform Apply
Run details
Usage
Workflow file
Terraform Plan
succeeded 18 hours ago in 34s
Set up job
Checkout the repository to the GitHub Action runner
Set Version of the Apps
Set up Terraform
Terraform Init
Terraform Refresh
Terraform format
Terraform plan
Publish Terraform Plan Applications
Create String Output
Publish Terraform Plan to Task Summary
Post Checkout the repository to the GitHub Action runner
Complete job

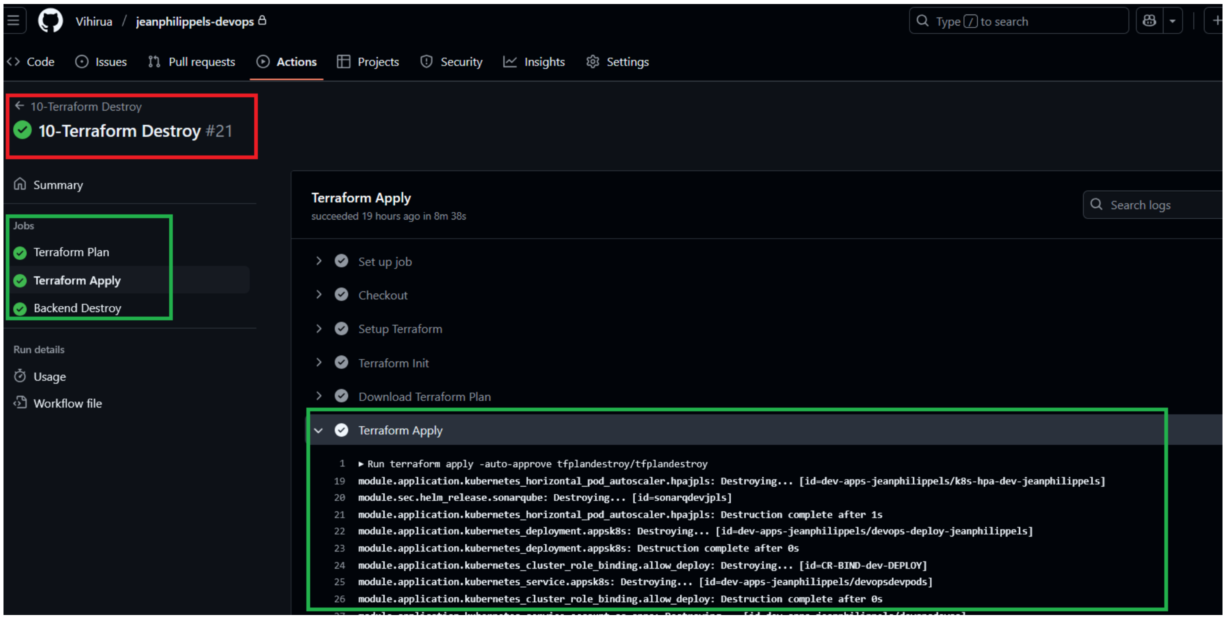

Vihirua / jeanphilippels-devops
Type / to search
Code
Issues
Pull requests
Actions
Projects
Security
Insights
Settings
4-Provision APPS | AKS
4-Provision APPS | AKS #47
Summary
Jobs
Terraform Plan
Terraform Apply
Run details
Usage
Workflow file
Terraform Apply
succeeded 17 hours ago in 2m 6s
Search logs
Set up job
Checkout
Setup Terraform
Terraform Init
Terraform Refresh
Download Terraform Plan
Setup az login
Save tf state
Terraform Apply
Get Terraform Output
Showing Output Var
Setup kubectl
Setup kubelogin
Set AKS context
Run Get Deploy
Run Get Services

Workflow Destroy

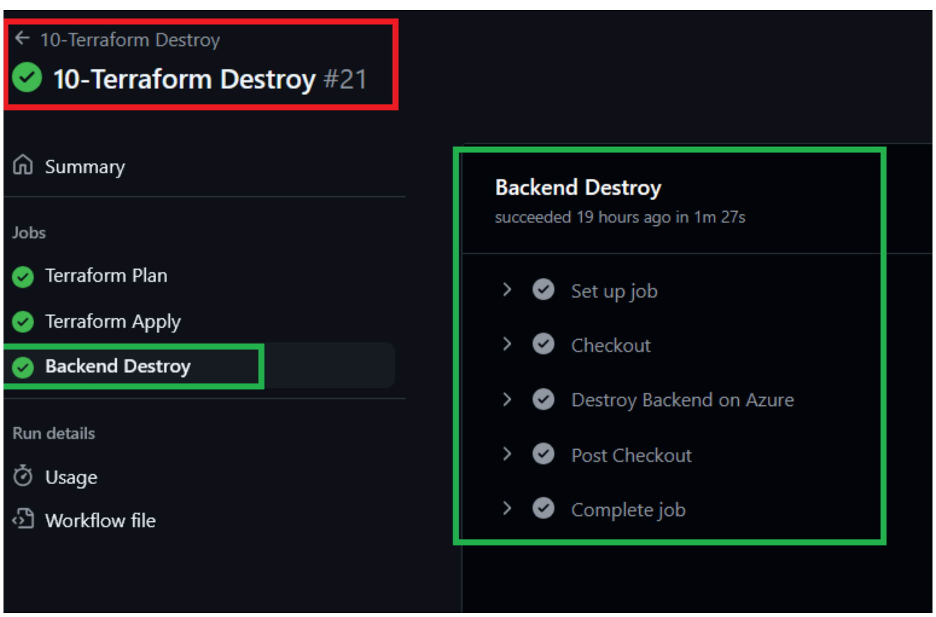
This workflow destroys provisioned azure resources







Vihirua / jeanphilippels-devops
Type / to search
Code
Issues
Pull requests
Actions
Projects
Security
Insights
Settings
← 10-Terraform Destroy
10-Terraform Destroy #21
Summary
Jobs
Terraform Plan
Terraform Apply
Backend Destroy
Run details
Usage
Workflow file
Terraform Apply
succeeded 19 hours ago in 8m 38s
Search logs
Set up job
Checkout
Setup Terraform
Terraform Init
Download Terraform Plan
Terraform Apply
1 ▸ Run terraform apply -auto-approve tfplandestroy/tfplandestroy
19 module.application.kubernetes_horizontal_pod_autoscaler.hpajpls: Destroying... [id=dev-apps-jeanphilippels/k8s-hpa-dev-jeanphilippels]
20 module.sec.helm_release.sonarqube: Destroying... [id=sonarqdevjpls]
21 module.application.kubernetes_horizontal_pod_autoscaler.hpajpls: Destruction complete after 1s
22 module.application.kubernetes_deployment.appsk8s: Destroying... [id=dev-apps-jeanphilippels/devops-deploy-jeanphilippels]
23 module.application.kubernetes_deployment.appsk8s: Destruction complete after 0s
24 module.application.kubernetes_cluster_role_binding.allow_deploy: Destroying... [id=CR-BIND-dev-DEPLOY]
25 module.application.kubernetes_service.appsk8s: Destroying... [id=dev-apps-jeanphilippels/devopsdevpods]
26 module.application.kubernetes_cluster_role_binding.allow_deploy: Destruction complete after 0s



← 10-Terraform Destroy
10-Terraform Destroy #21
Summary
Jobs
Terraform Plan
Terraform Apply
Backend Destroy
Run details
Usage
Workflow file
Backend Destroy
succeeded 19 hours ago in 1m 27s
Set up job
Checkout
Destroy Backend on Azure
Post Checkout
Complete job